(Re-)Building a Personality Science of the Whole Person



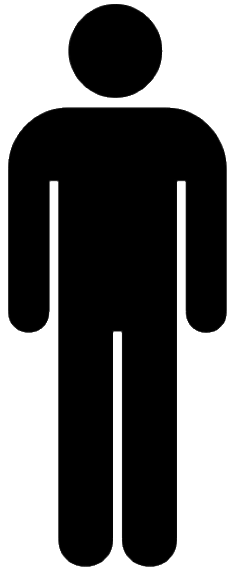
Emorie D. Beck, Ph.D.
University of California, Davis
beck-lab.ucdavis.edu

UC DAVIS
UNIVERSITY OF CALIFORNIA

Description

Prediction

Explanation



Thoughts

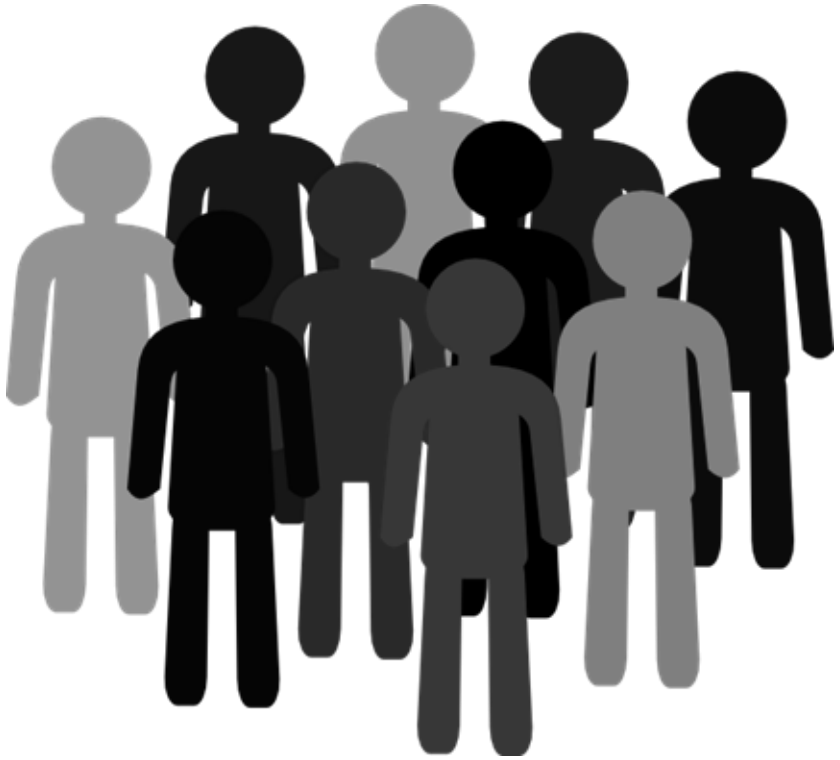
Feelings

Behaviors

Description

Prediction

Explanation



Thoughts

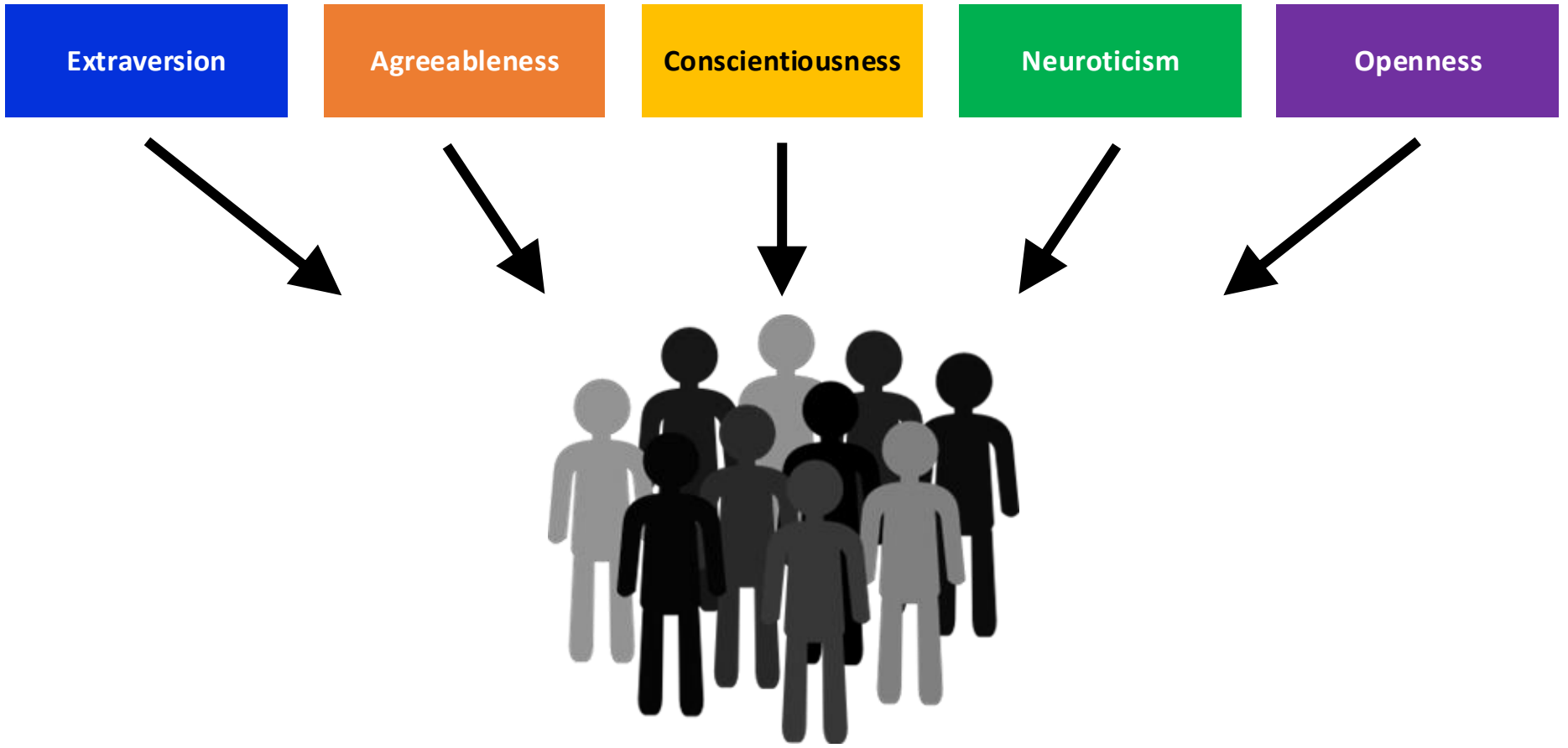
Feelings

Behaviors

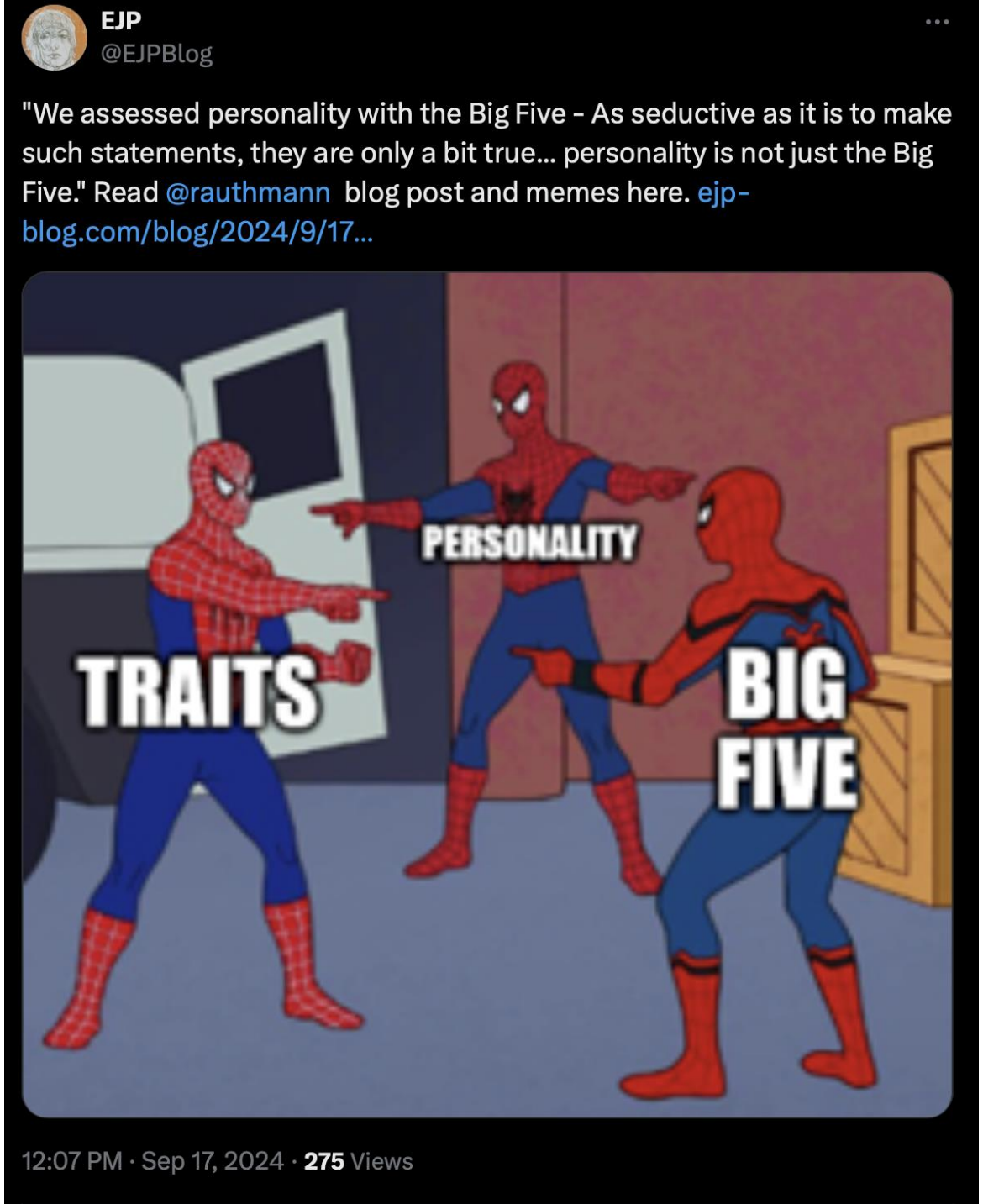
What is personality?

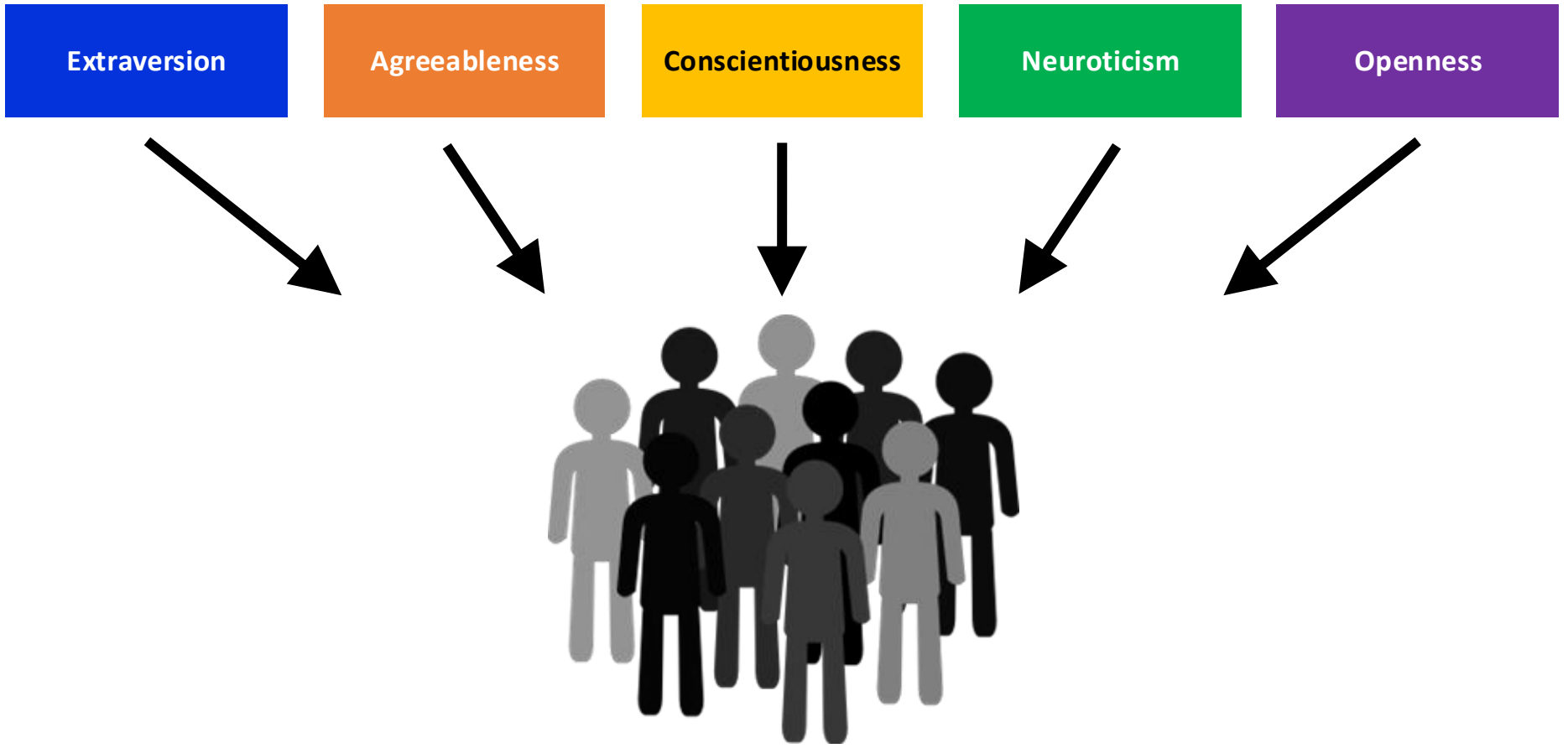
"Personality refers to those **characteristics** of the person that account for **consistent patterns of feelings, thinking, and behaving**."

(Pervin, Cervone & John, 2005, p. 6)



Nomothetic
Between-Person
Variable Centered


EJP
@EJPBlog
"We assessed personality with the Big Five - As seductive as it is to make such statements, they are only a bit true... personality is not just the Big Five." Read @rauthmann blog post and memes here. ejp-blog.com/blog/2024/9/17...
PERSONALITY
TRAITS
BIG FIVE
12:07 PM · Sep 17, 2024 · 275 Views

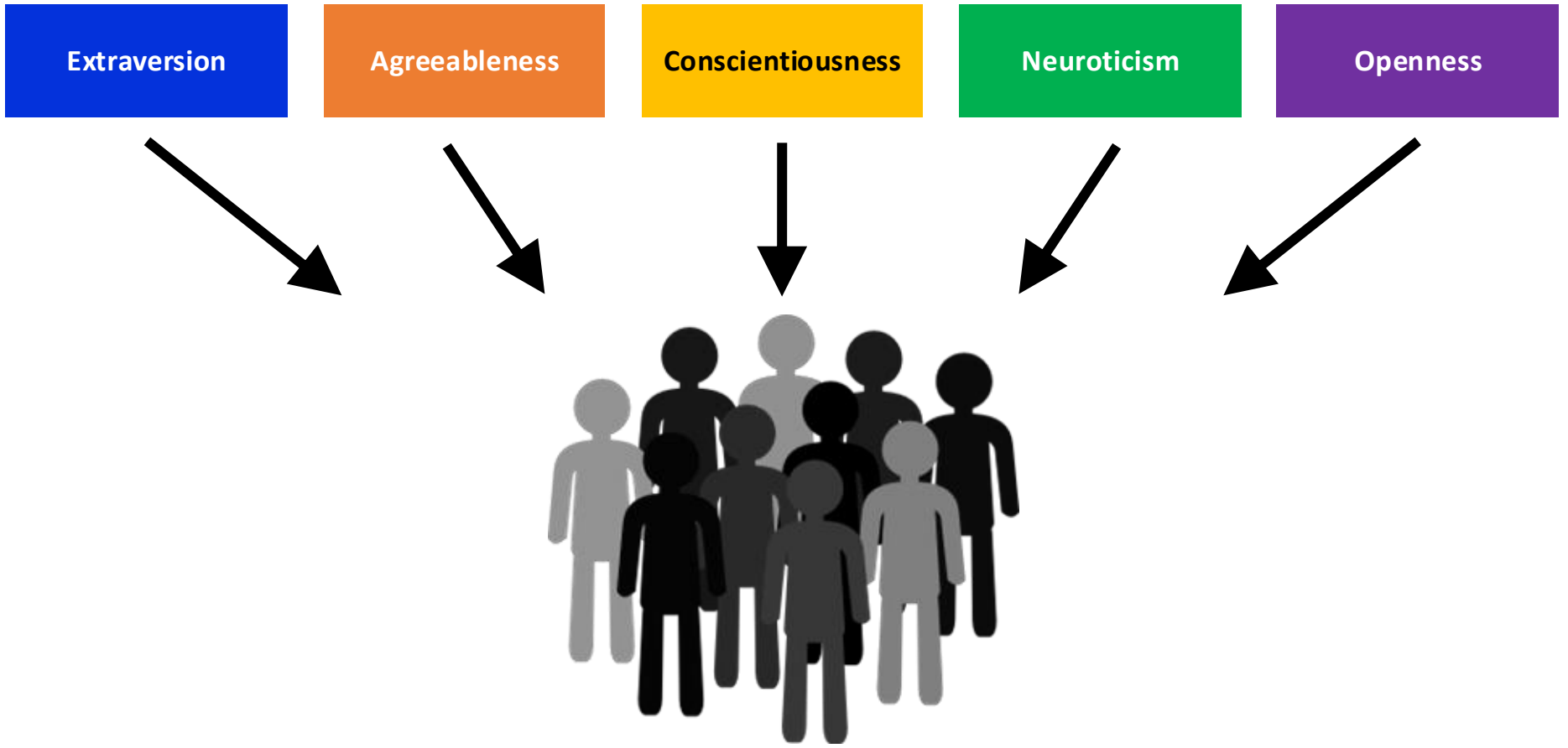


Nomothetic
Between-Person
Variable Centered

What is personality?

Personality is what personality tests test.

(Jack Wright, personal communication, 2013)



Nomothetic
Between-Person
Variable Centered

What is personality?

"Personality is the **dynamic organization** within the individual of those **psychophysical systems** that determine his **unique adjustments to the environment**."

(Allport, 1937, p. 32)



e.g., Allport, 1937, 1961, 1968

What is personality?

Personality is what personality tests test.

(Jack Wright, personal communication, 2013)

The Whole Person

Contexts

Dynamics & Systems

e.g., Beck & Jackson, 2020a, *JPSP*; 2020b, *CDPS*; Lee & Beck, 2025, *JPSP*; Nissen & Beck, 2025a,b, *JPSP*, *SPPC*

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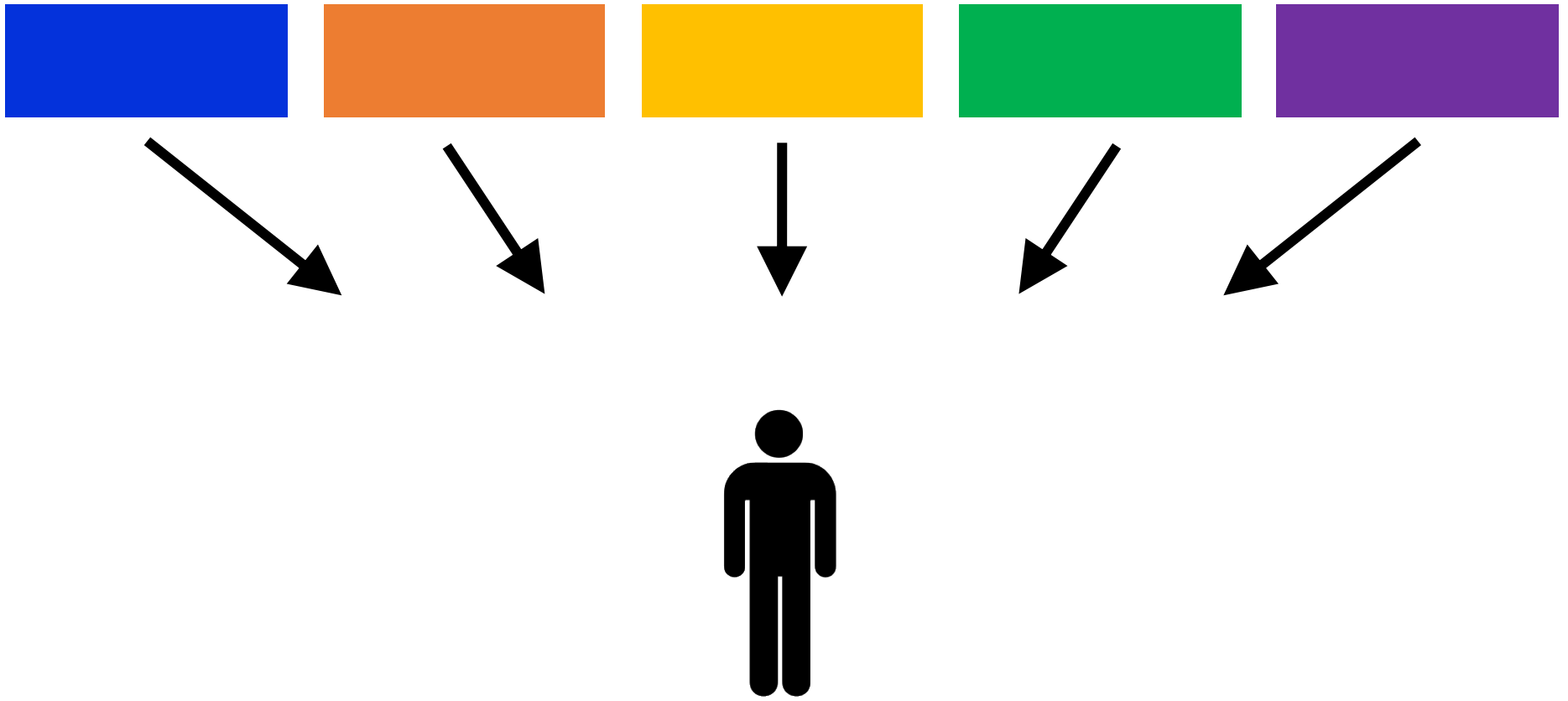
(Jack Wright, personal communication, 2013)

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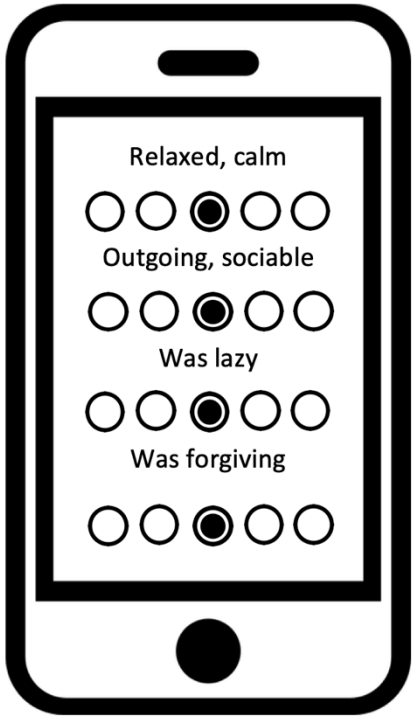
Idiographic
Person-Specific
N = 1

e.g., Beck & Jackson, 2020a, *JPSP;* 2020b, *CDPS*

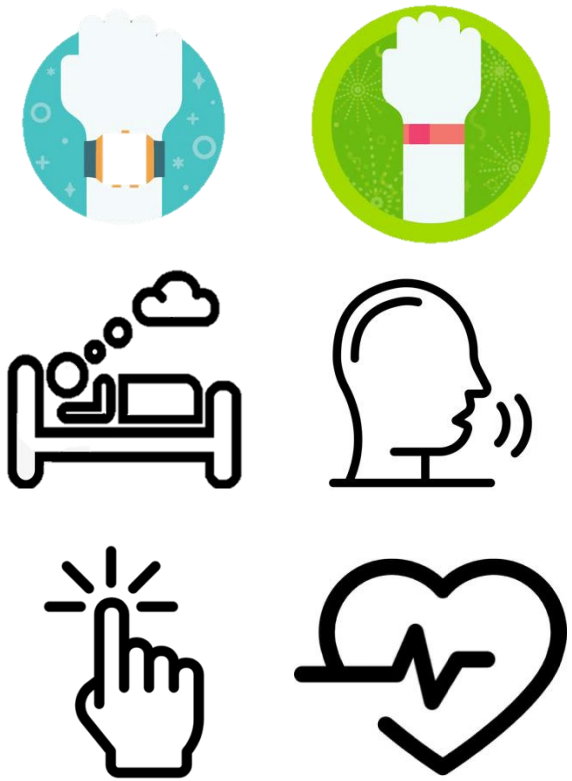


Modern Opportunities for *N* = 1 Research

ESM / EMA



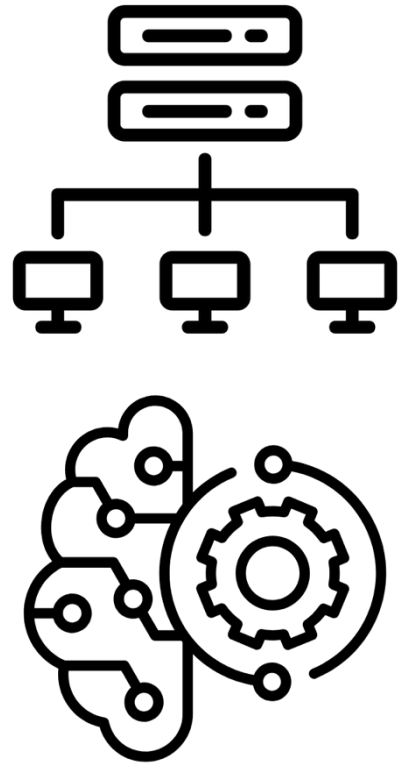
Mobile Sensing



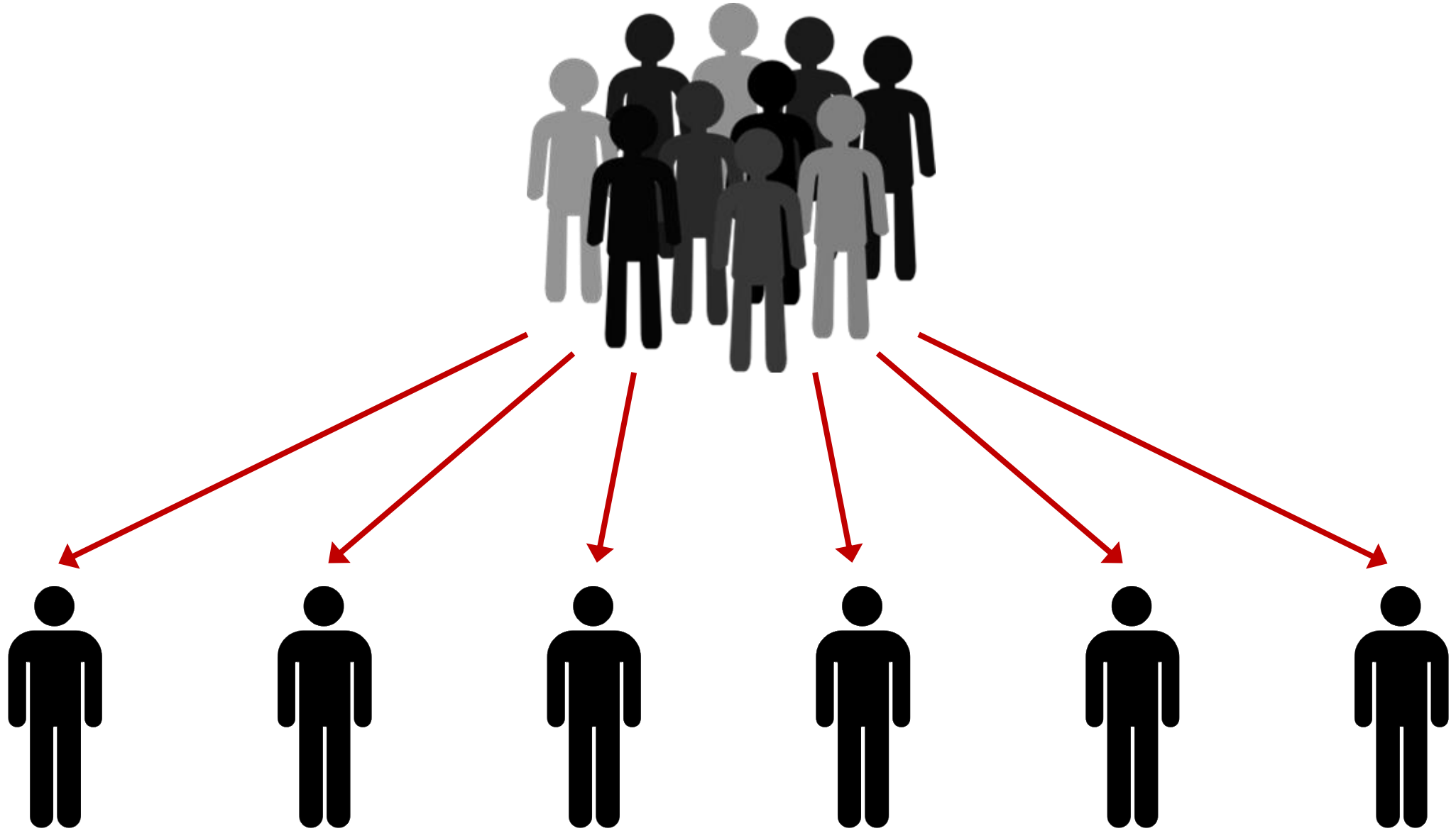
Open-Ended Responses



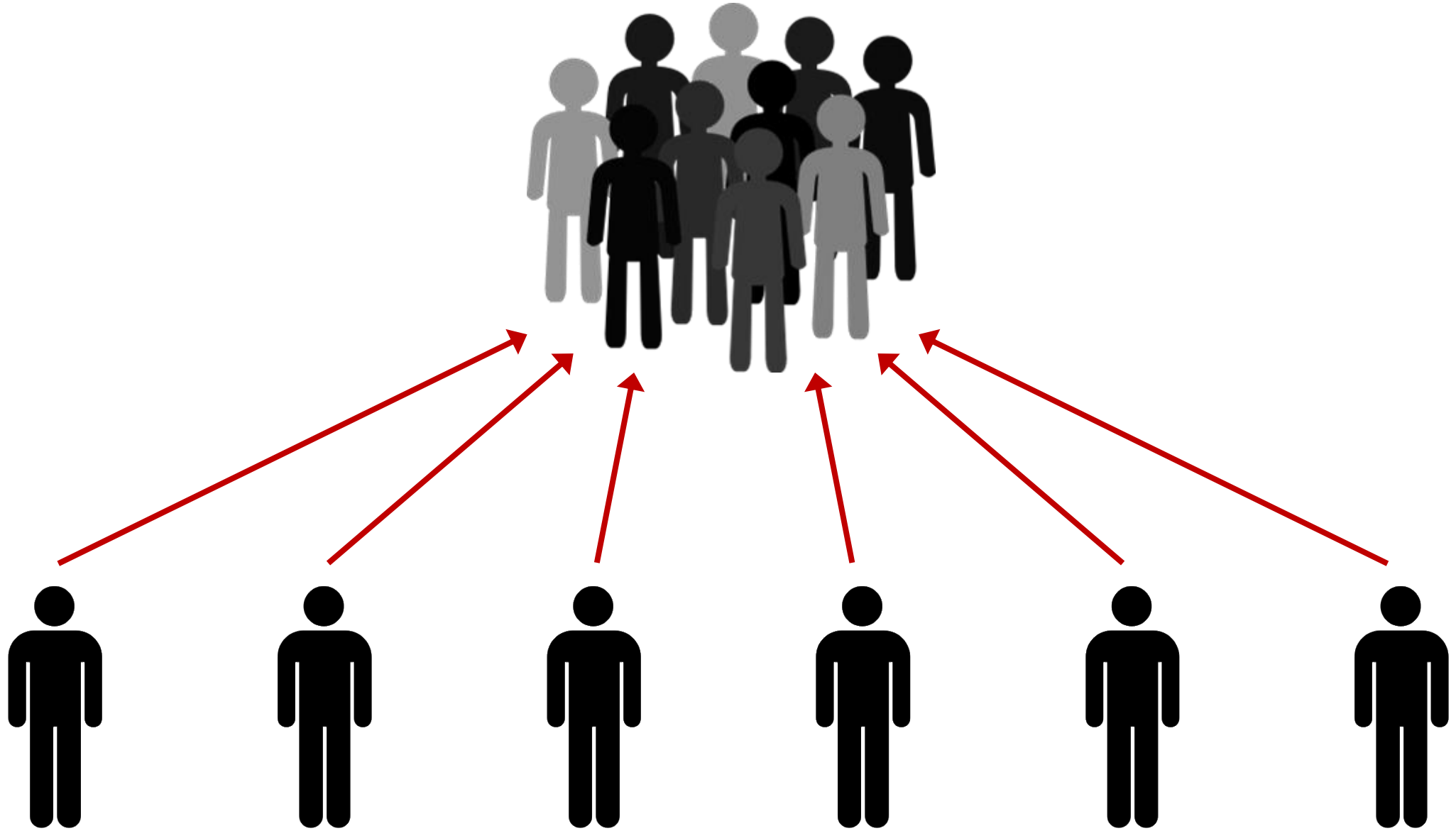
Computational Tools



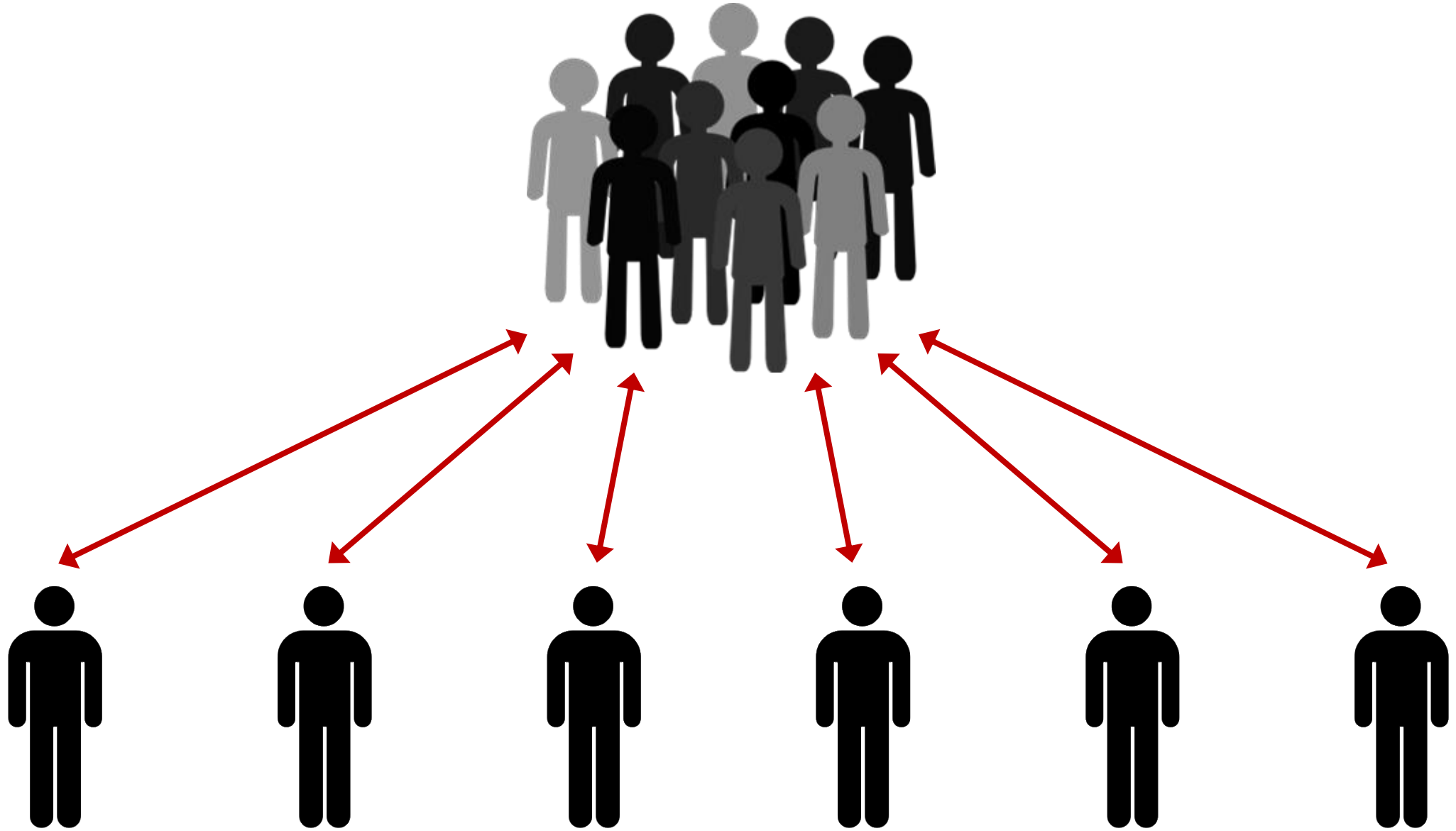
Top-Down Psychological Science



Bottom-Up Psychological Science



Bottom-Up Psychological Science



What is personality?

Personality is what personality tests test.

(Jack Wright, personal communication, 2013)

The Whole Person

Contexts

Dynamics & Systems

e.g., Beck & Jackson, 2020a, *JPSP*; 2020b, *CDPS*; Lee & Beck, 2025, *JPSP*; Nissen & Beck, 2025a,b, *JPSP*, *SPPC*

What is personality?







if **...** ***then***

Context
Situation perception
Experience
Etc.

Behavior
Experience
Etc.

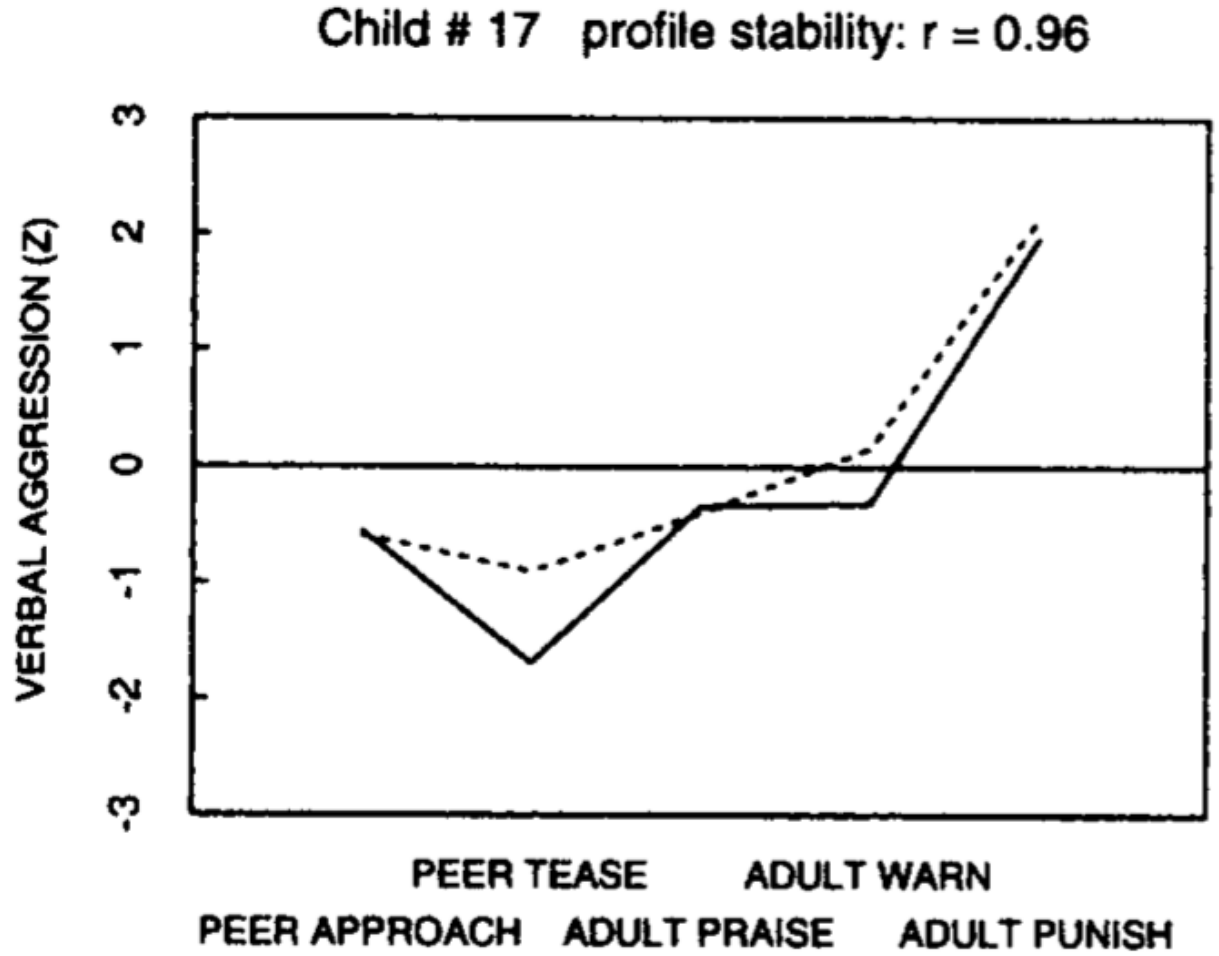
e.g., Mischel & Shoda, 2005; Wright & Mischel, 1987; Shoda, Mischel, & Wright, 1993; Mischel, 1973







What is personality?



e.g., Mischel & Shoda, 2005; Wright & Mischel, 1987; Shoda, Mischel, & Wright, 1993; Mischel, 1973

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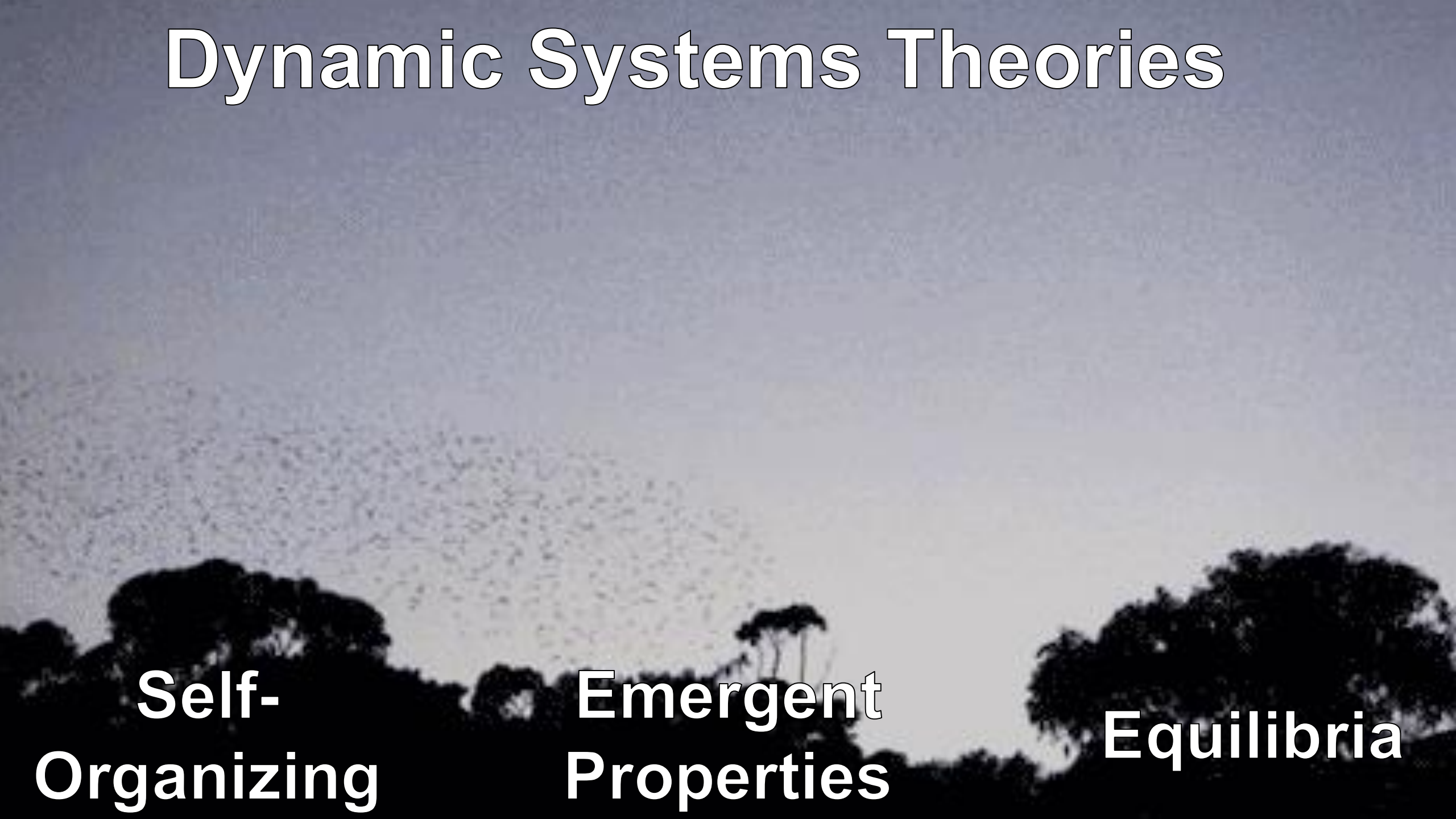
The Whole Person

Contexts

Dynamics & Systems

e.g., Beck & Jackson, 2020a, *JPSP*; 2020b, *CDPS*; Lee & Beck, 2025, *JPSP*; Nissen & Beck, 2025a,b, *JPSP*, *SPPC*

Dynamic Systems Theories



Self-Organizing

Emergent Properties

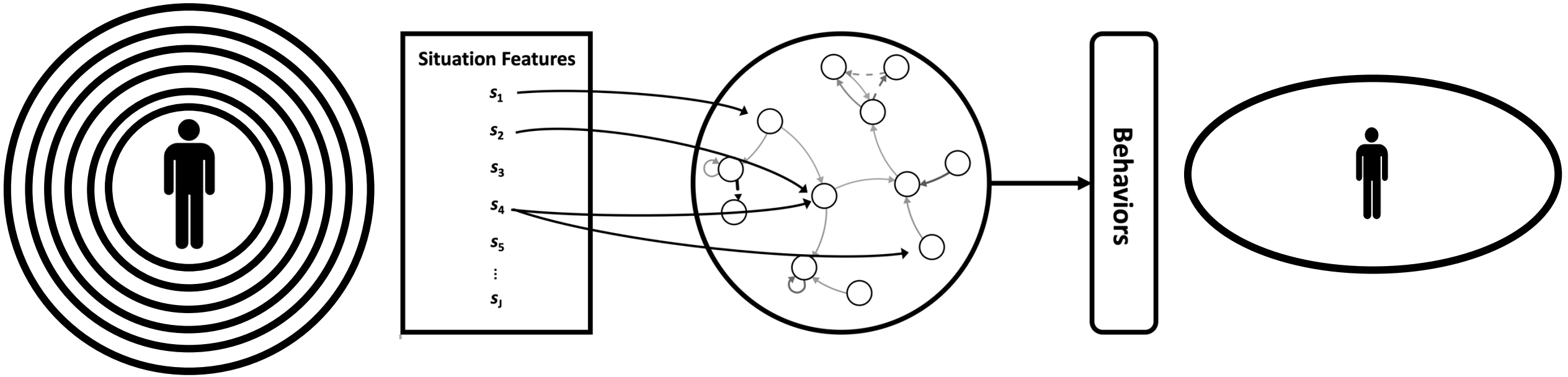
Equilibria

Persons in Context: Systems Theories

Ecological Systems Theory
(e.g., Bronfenbrenner, 1979)

Cognitive Affective Personality System
(e.g., Mischel & Shoda, 1995)

Field Theory
(e.g., Lewin, 1936)



e.g., Lewin, 1936

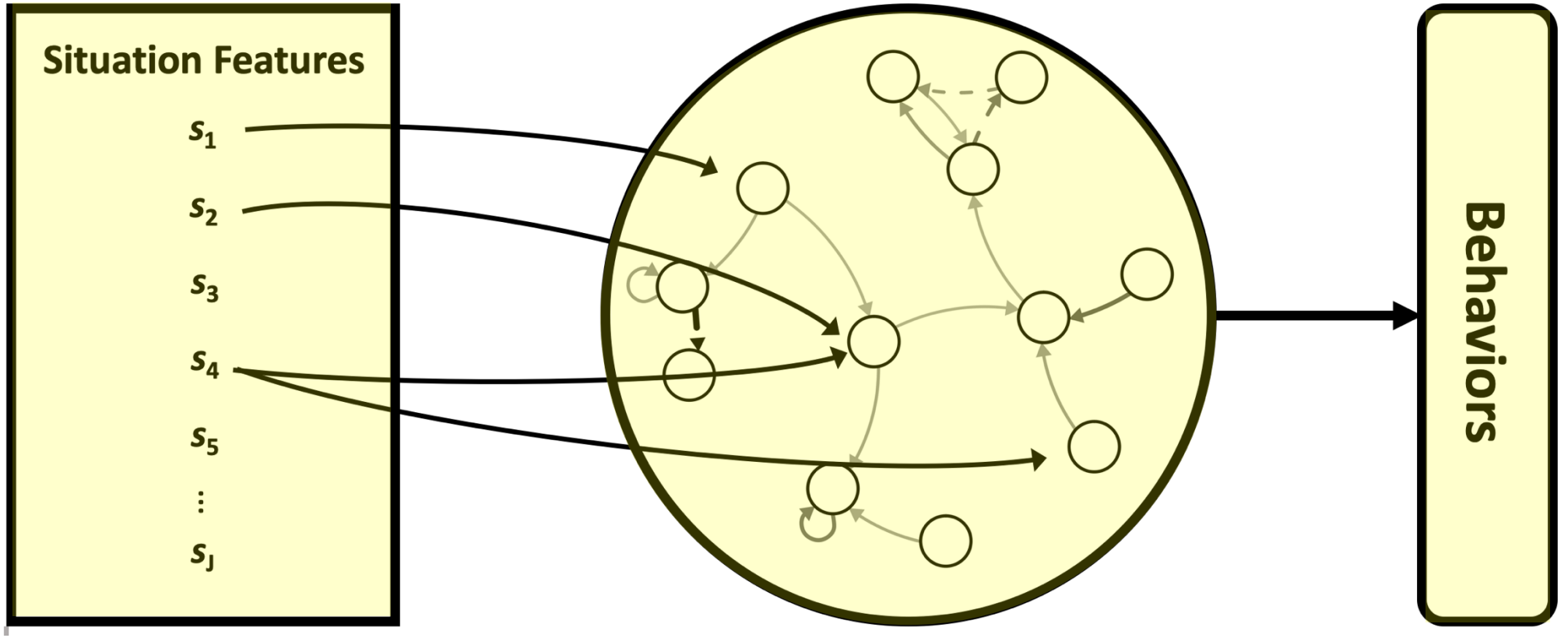




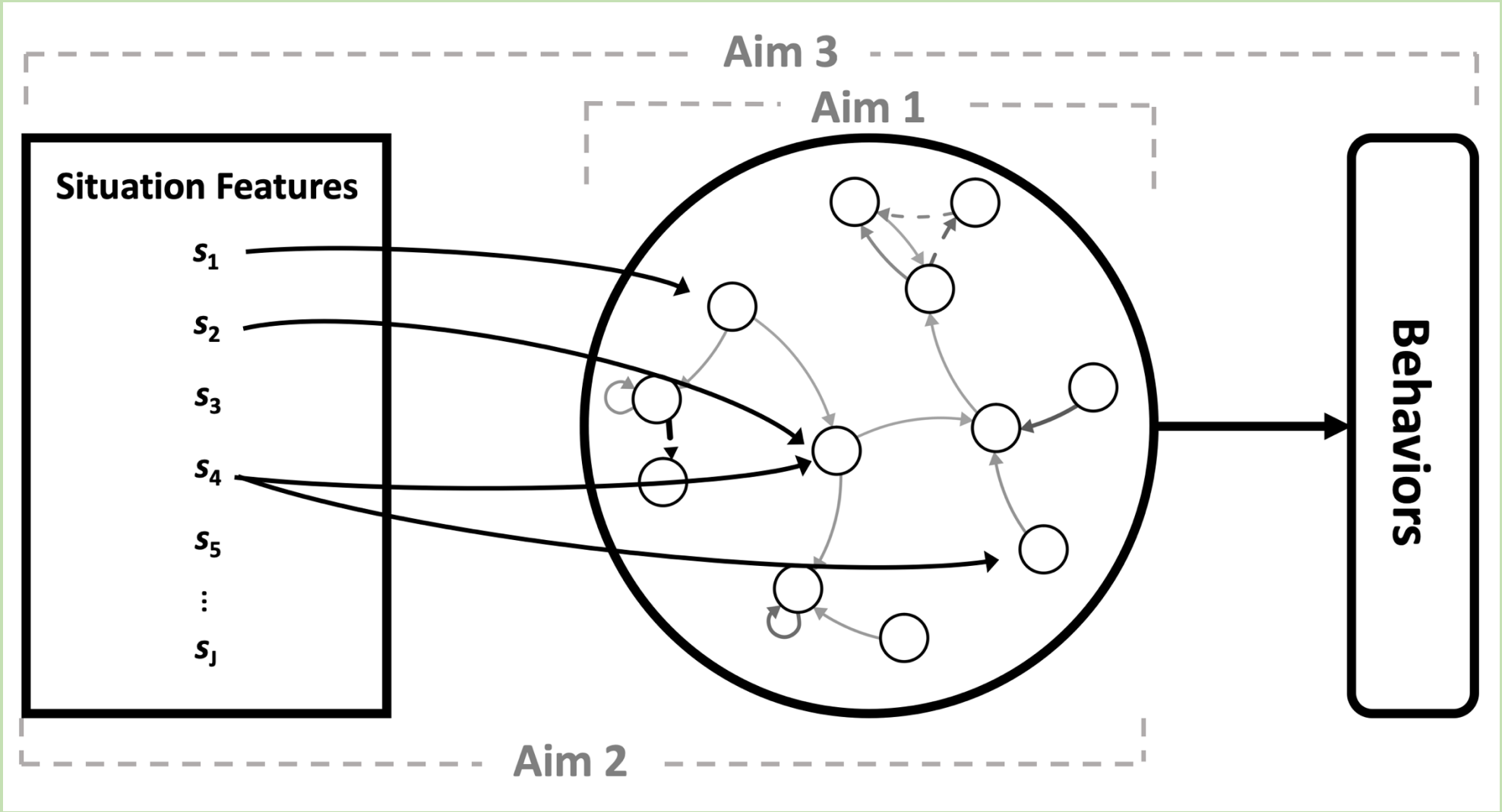


What is personality?

A Cognitive Affective Processing System (CAPS)

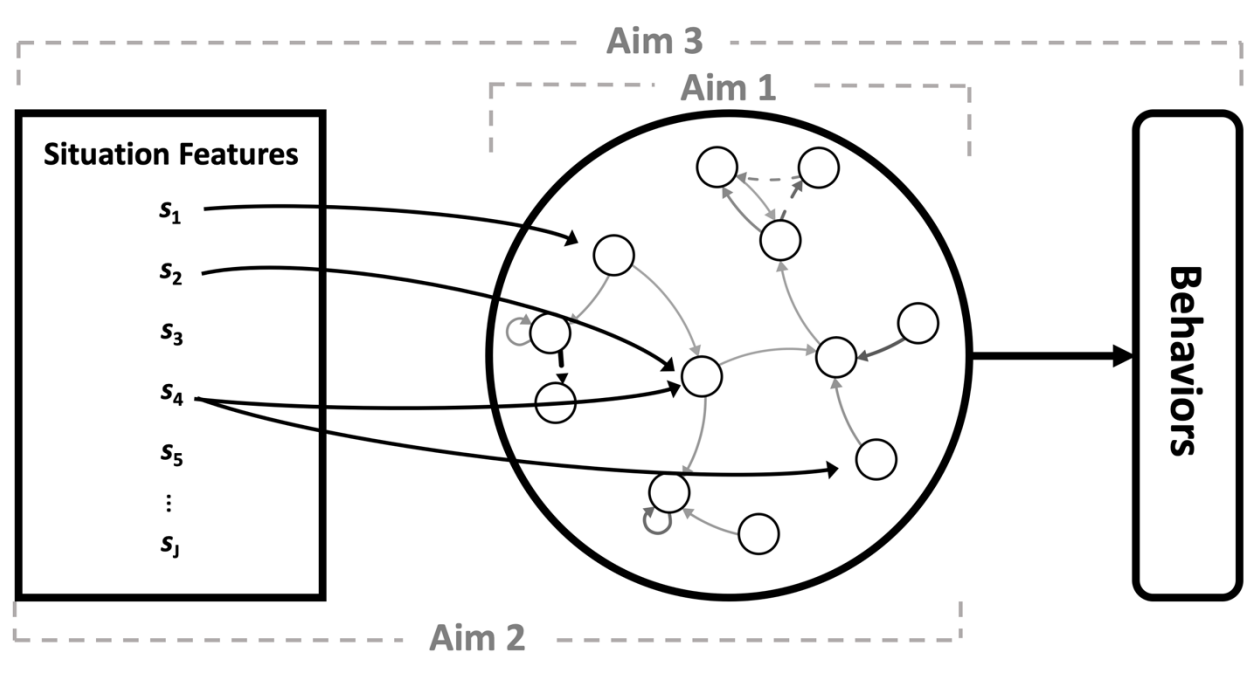


e.g., Mischel & Shoda, 2005; Wright & Mischel, 1987; Shoda, Mischel, & Wright, 1993; Mischel, 1973


Aim 3
Aim 1
Situation Features
s_1
s_2
s_3
s_4
s_5
s_J
Behaviors
Aim 2

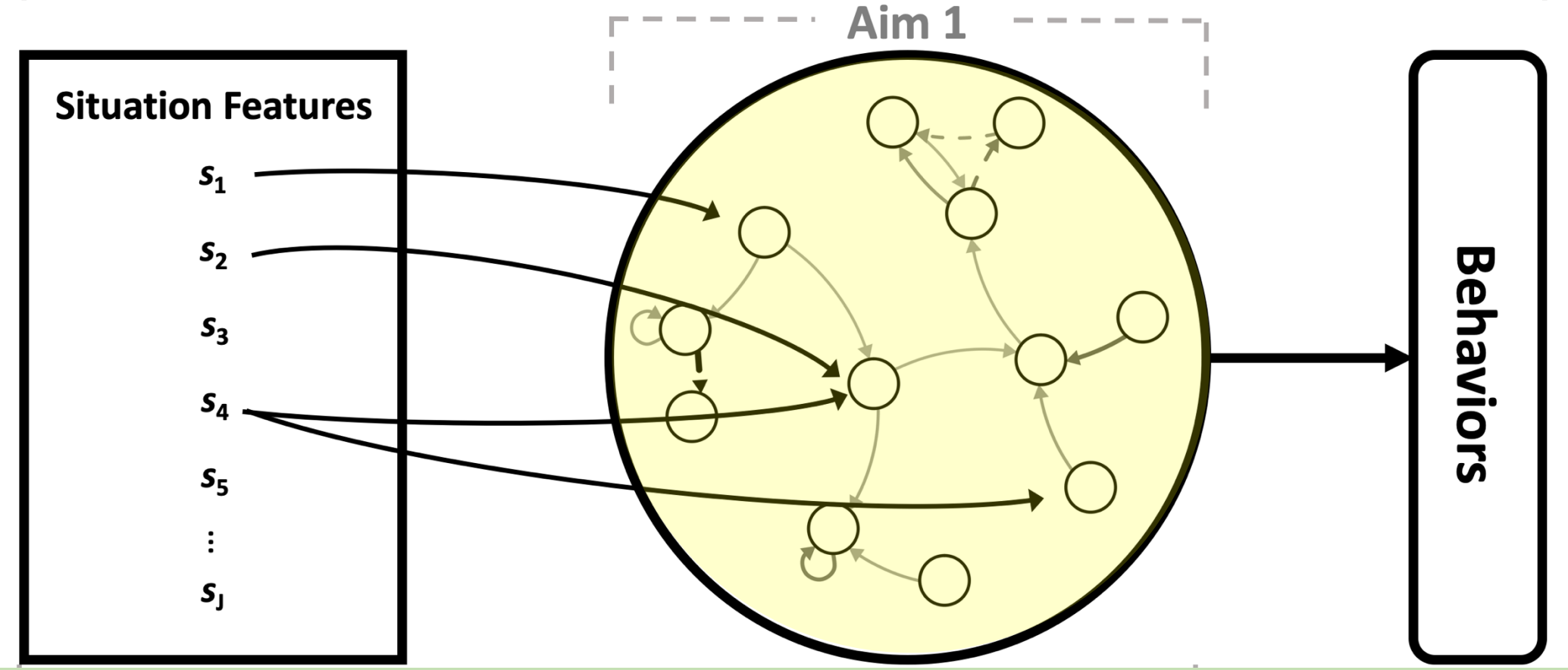
Aim 1: Bringing the individual back in the study of personality structure and change.	**Beck & Jackson** (2020, *JPSP*; 2021b; *EJP*)
Aim 2: Using longitudinal data to understand how well-being unfolds across contexts.	**Beck, et al.** (2025, *Nature Human Behaviour*)
Aim 3: Predicting behavior using machine learning.	**Beck & Jackson** (2023, *Psych Science*)

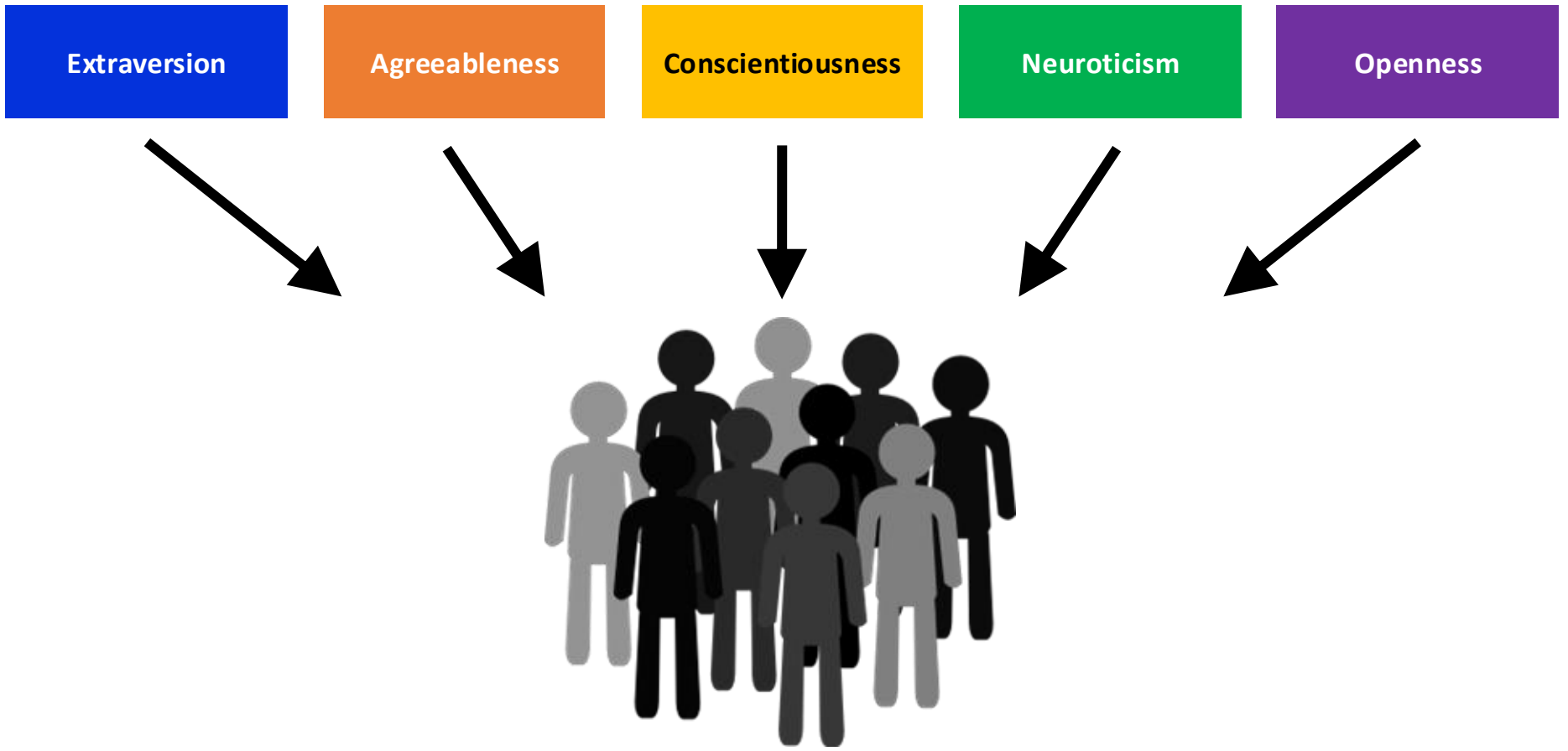
Ongoing Directions



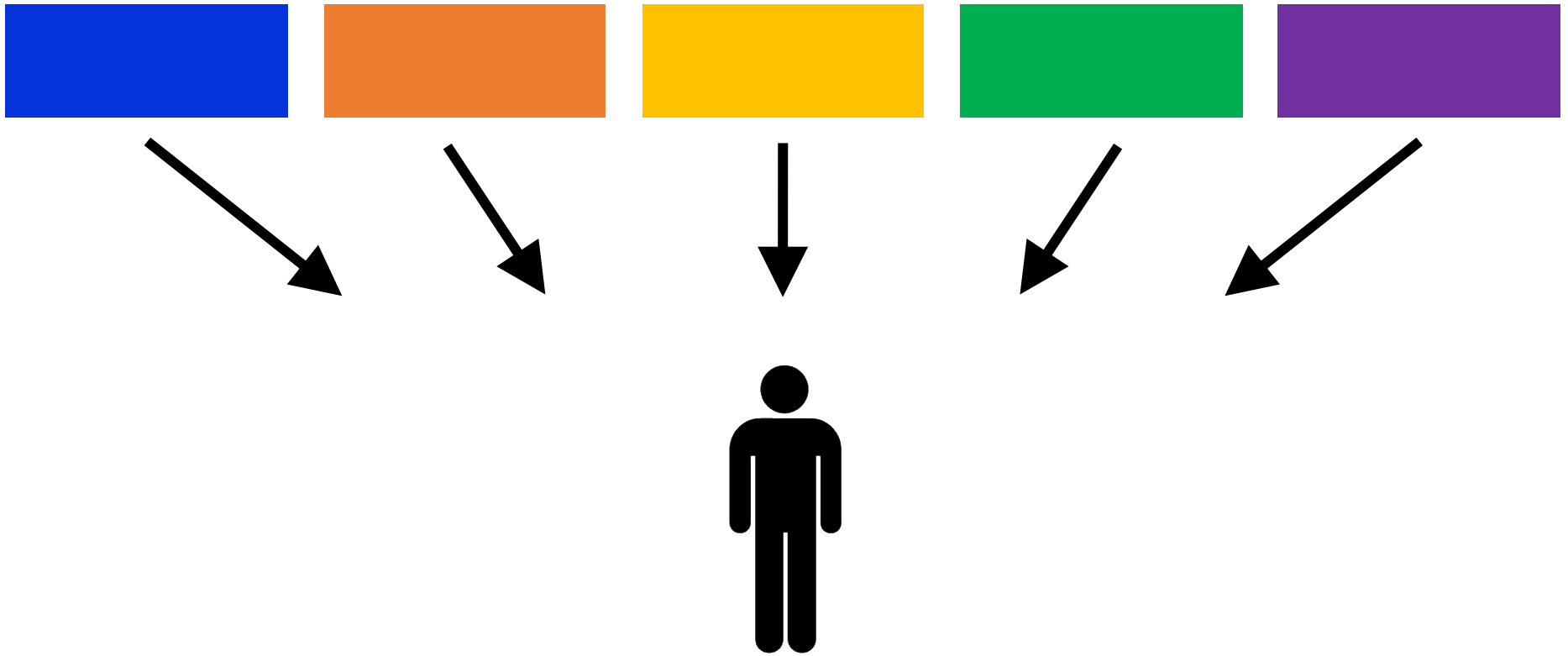
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Beck & Jackson
(2020, *JPSP*; 2021b; *EJP*)





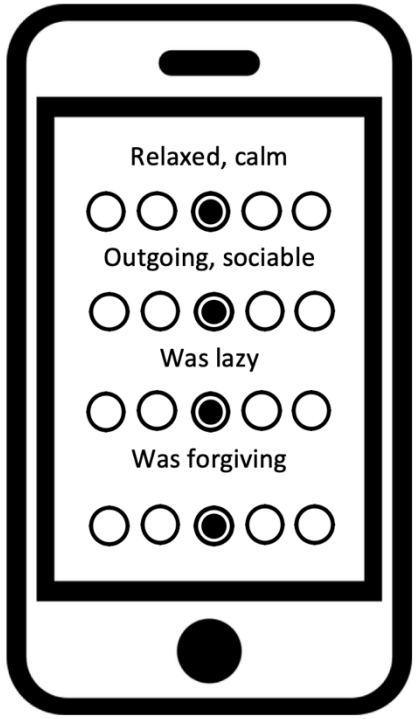
Nomothetic
Variable-Centered



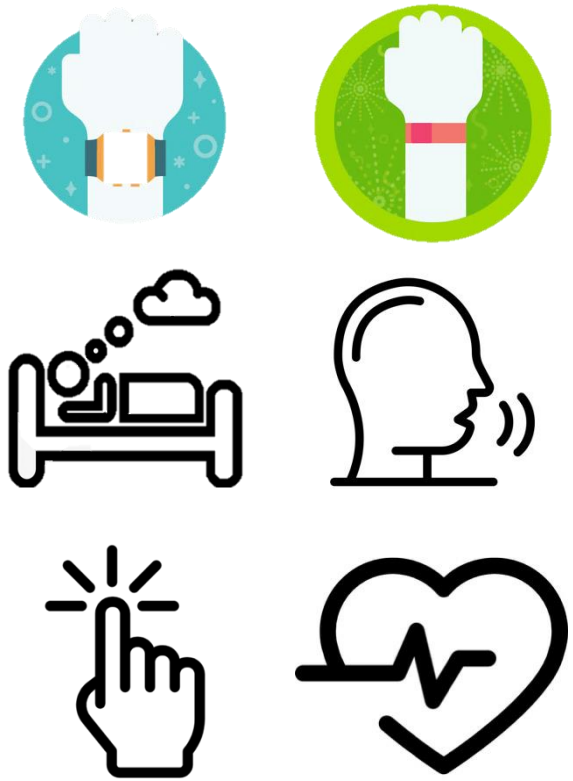
Idiographic
Person-Specific

e.g., Beck & Jackson, 2020a, *JPSP;* 2020b, *CDPS*

ESM / EMA



Mobile Sensing



Methods

2 years of ESM responses from the Personality and Interpersonal Roles Study (PAIRS)

N = 372 Wash U undergrads, total assessments N = 17,715

Measures

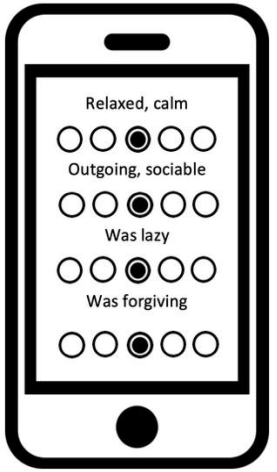
9 items from the Big Five Inventory (BFI)

Procedure

4 assessments / day for 15 days

Modeling

Graphical vector autoregression



Beck & Jackson (2020a, *JPSP*; 2021b, *EJP*)



time	E	A	C	N	O
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

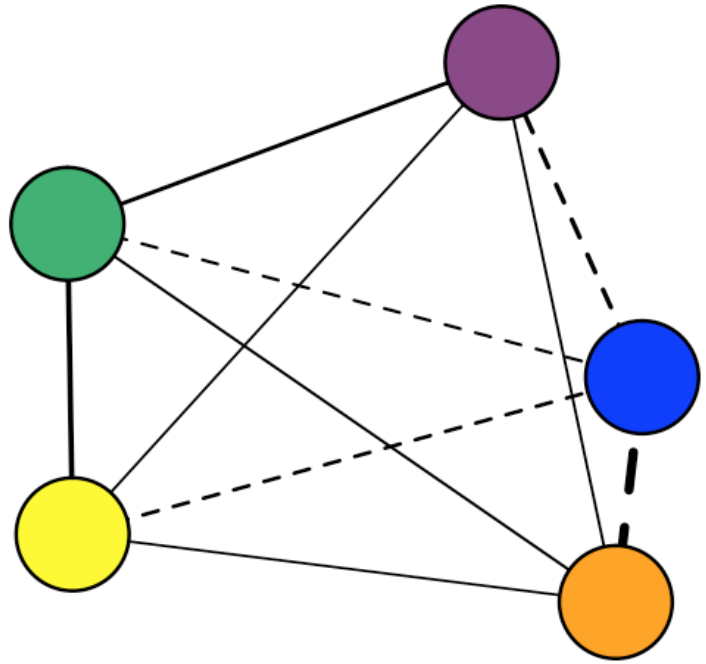
	E	A	C	N	O
E					
A	0.02				
C	-0.05	0.08			
N	0.64	-0.12	-0.10		
O	0.26	-0.27	0.43	0.07	

Contemporaneous: Within Time-Points

$X_{it} \leftrightarrow X_{jt}$

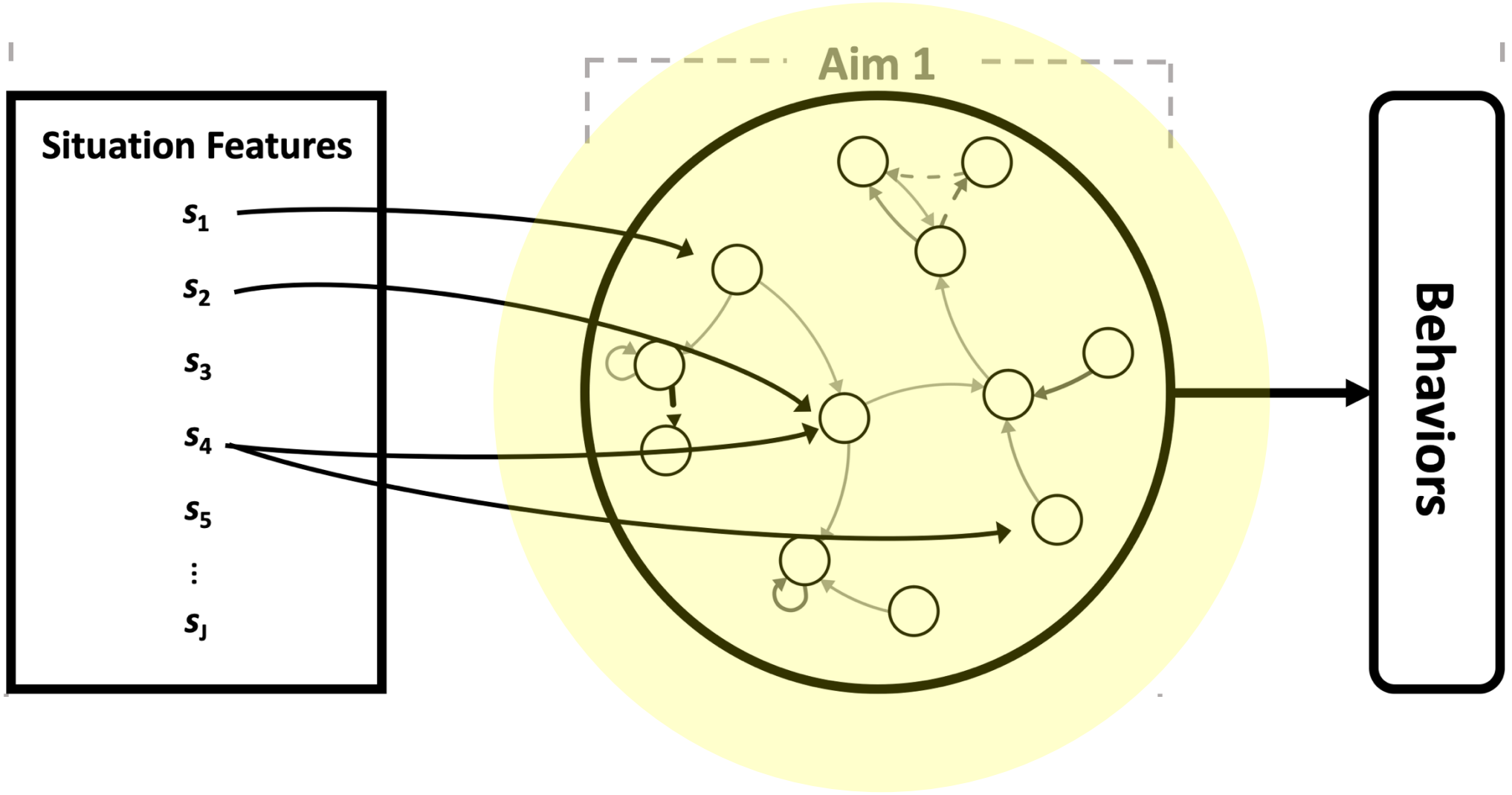


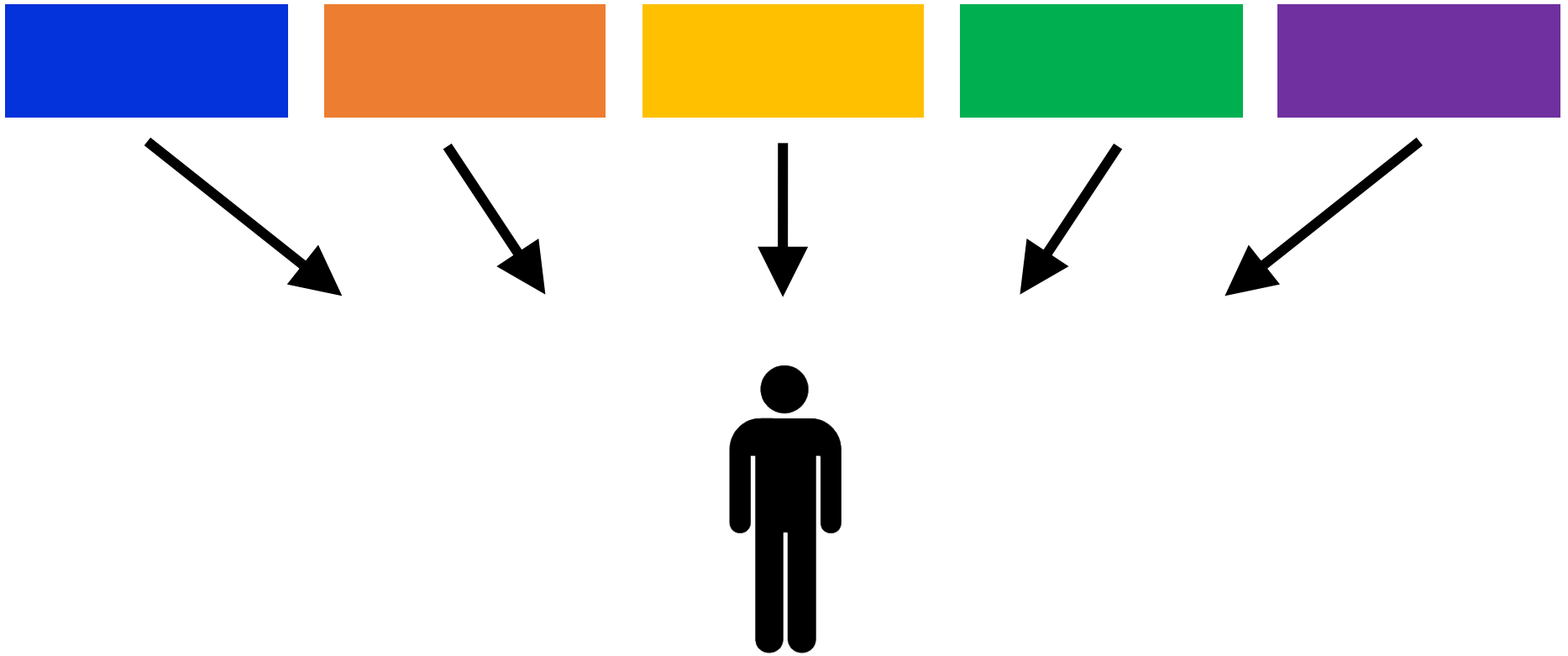
time	E	A	C	N	O
1	4	2	3	2	4
2	2	3	4	1	4
3	3	3	2	2	4
4	3	1	3	2	3
5	4	4	3	2	2
6	4	1	2	3	4
7	2	2	2	2	1
8	4	2	3	3	4
9	3	2	2	1	2
10	4	2	3	3	3
11	3	1	3	2	5



Contemporaneous: Within Time-Points

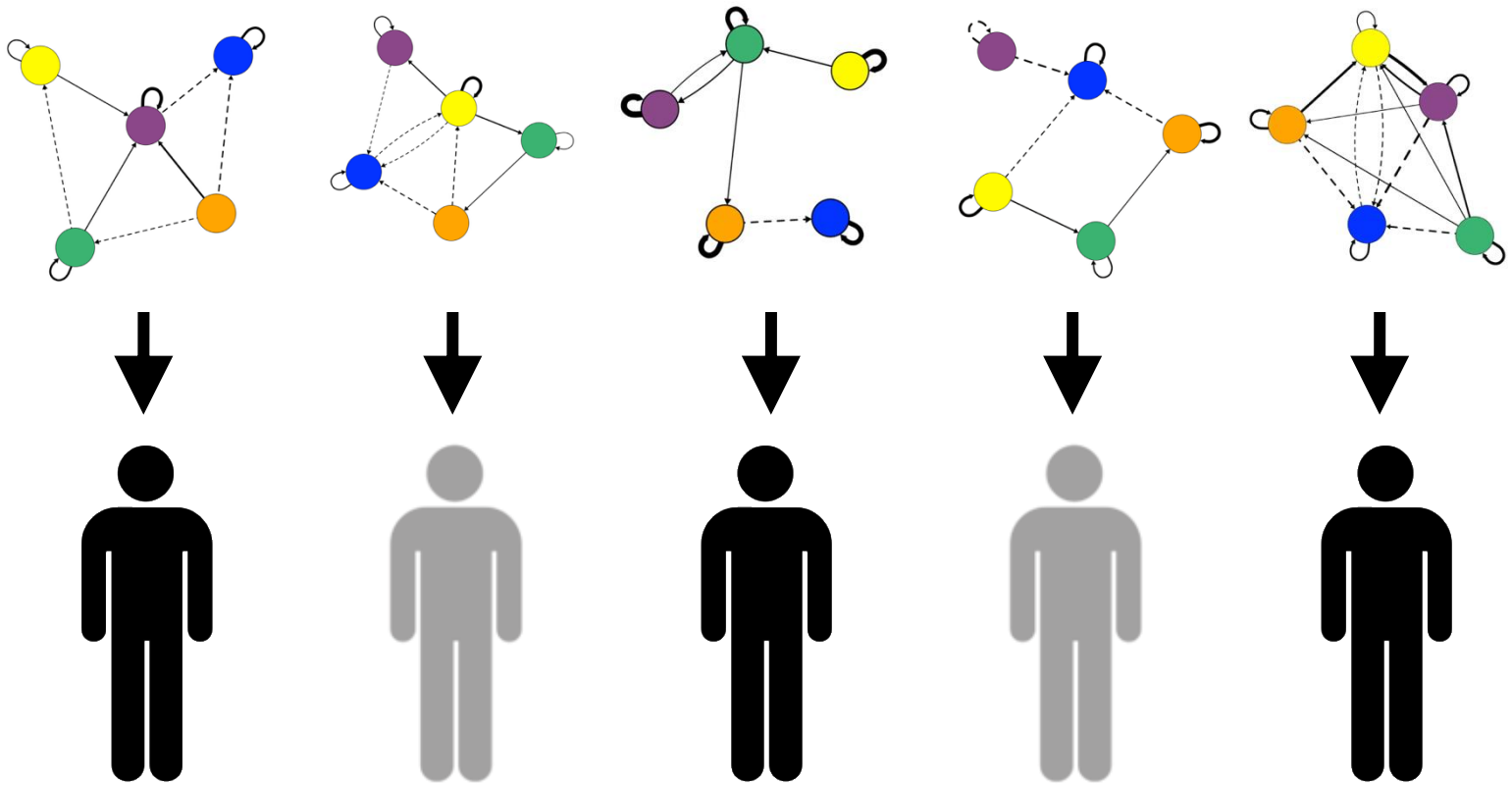
$X_{it} \leftrightarrow X_{jt}$


Aim 1
Situation Features
s_1
s_2
s_3
s_4
s_5
⋮
s_J
Behaviors



Idiographic
Person-Specific

e.g., Beck & Jackson, 2020a, *JPSP;* 2020b, *CDPS*

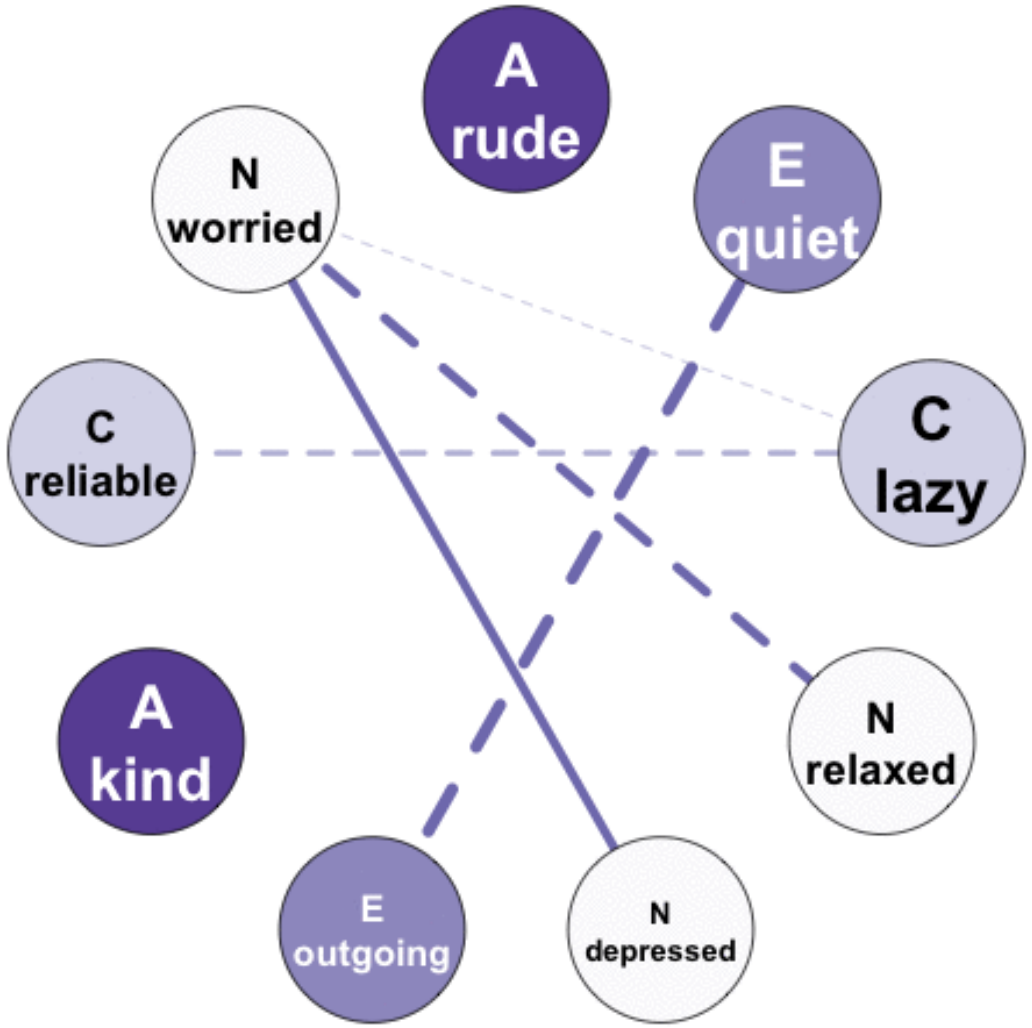


Idiographic
Person-Specific

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RESULTS

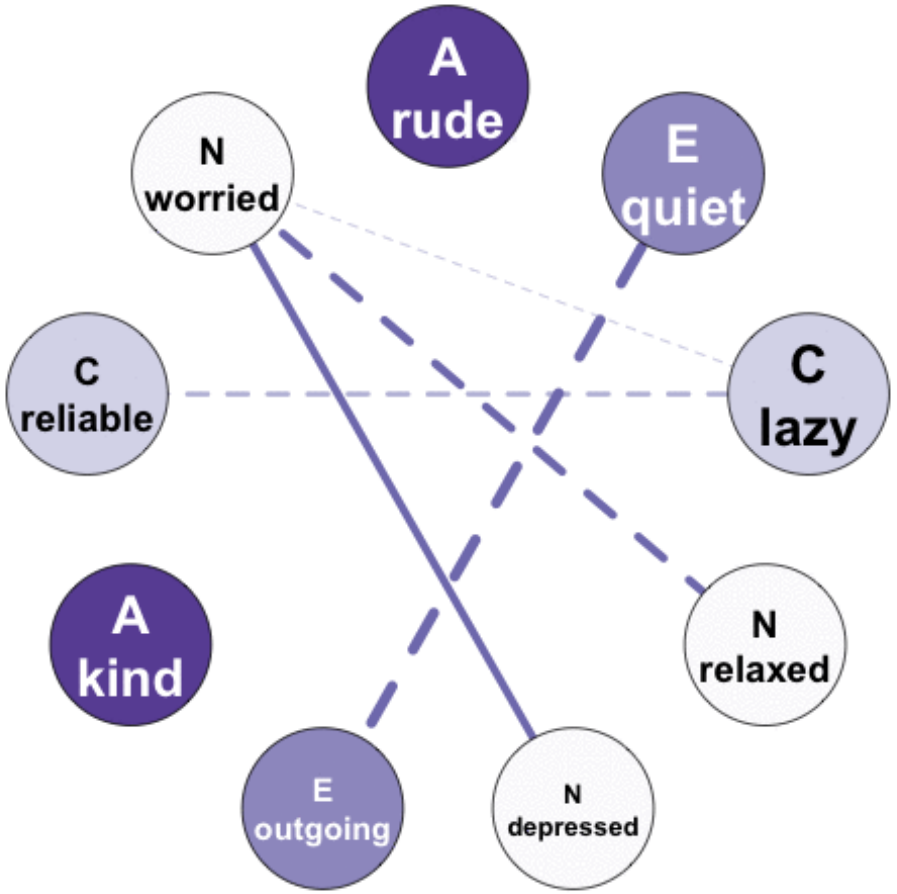
Aim 1

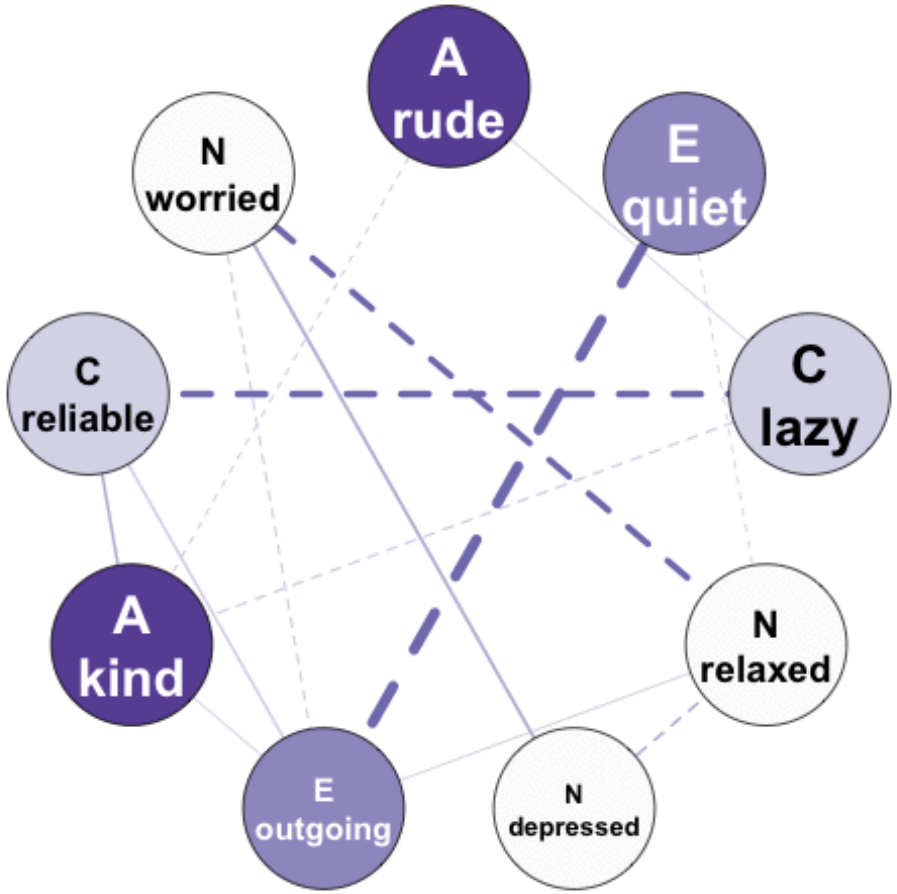


The structures differ across people.

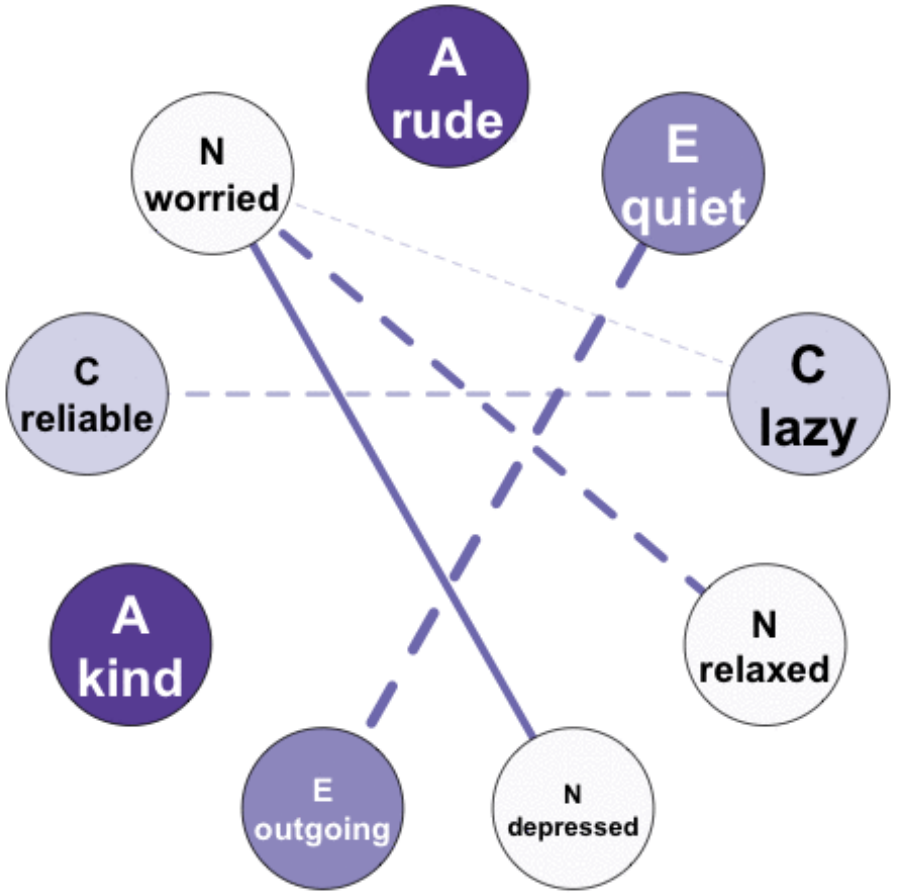
But do they show expected longitudinal consistency?

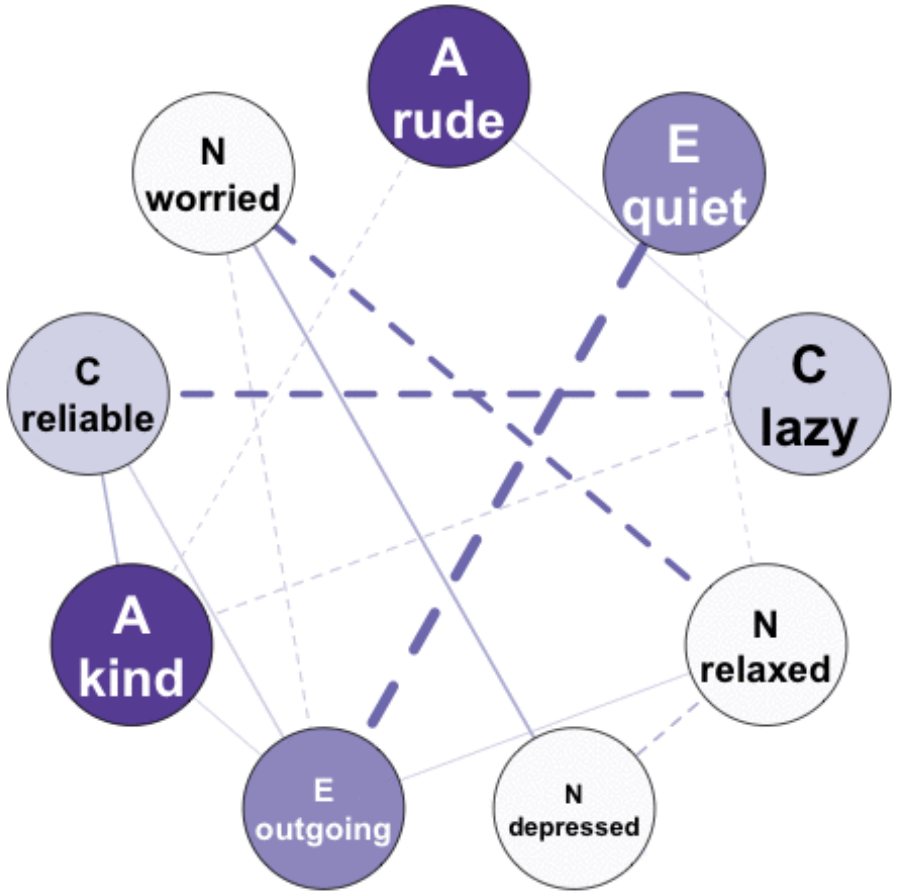
Beck & Jackson (2020a, *JPSP*)



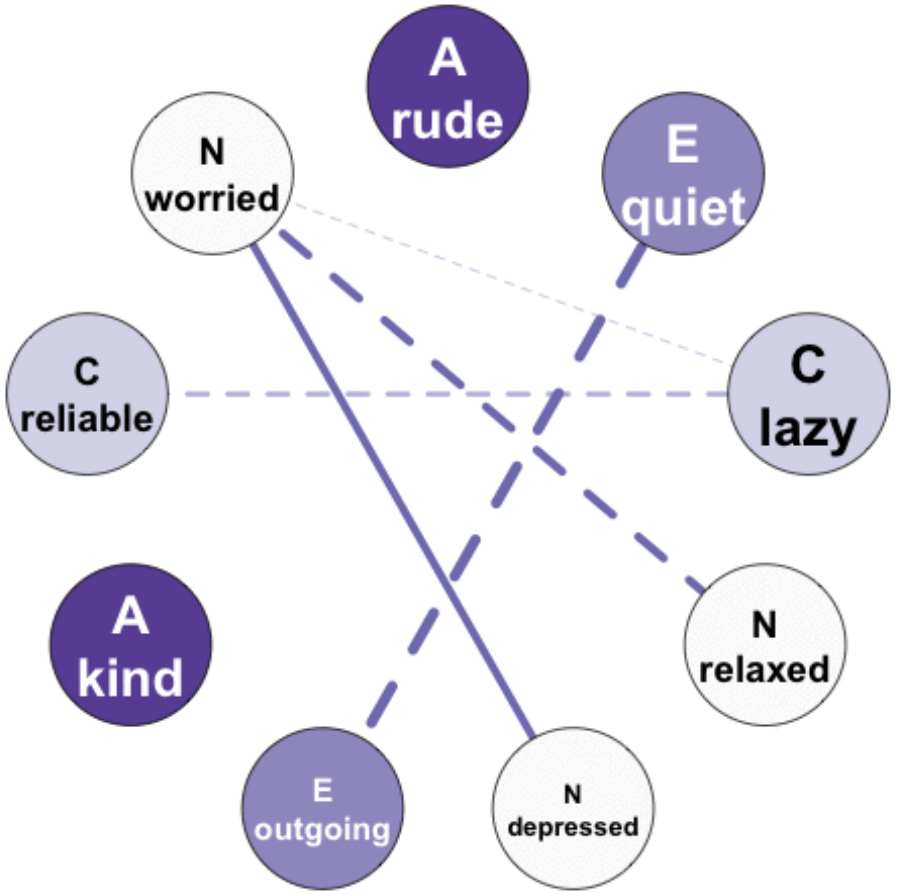


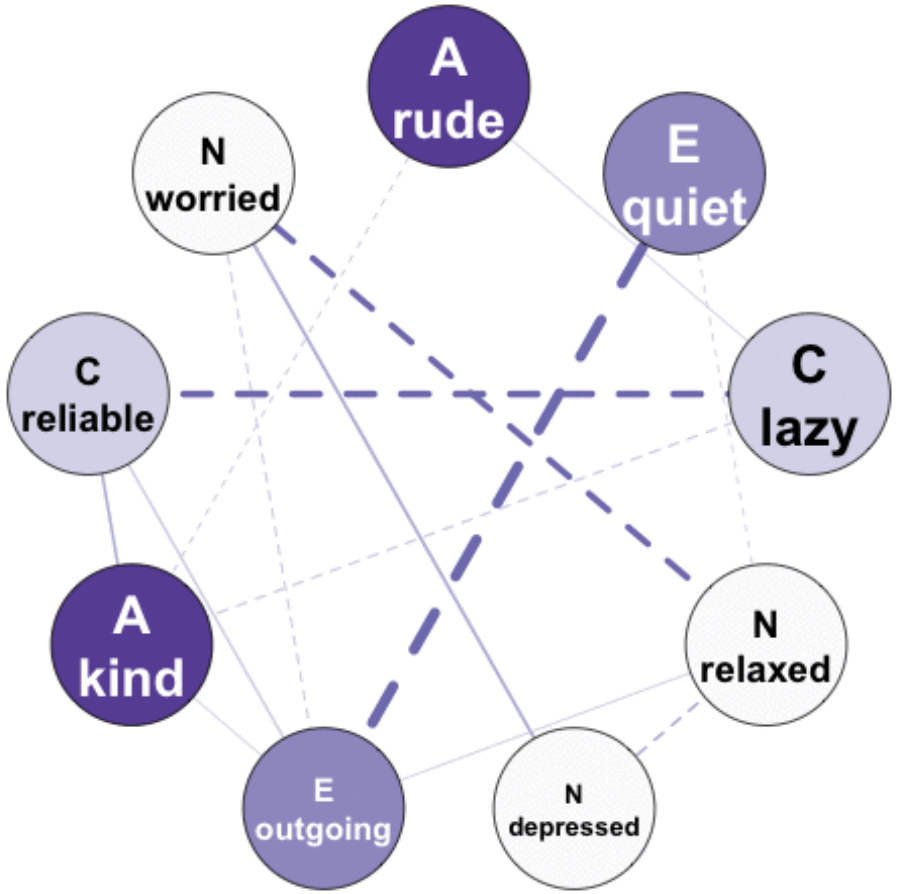
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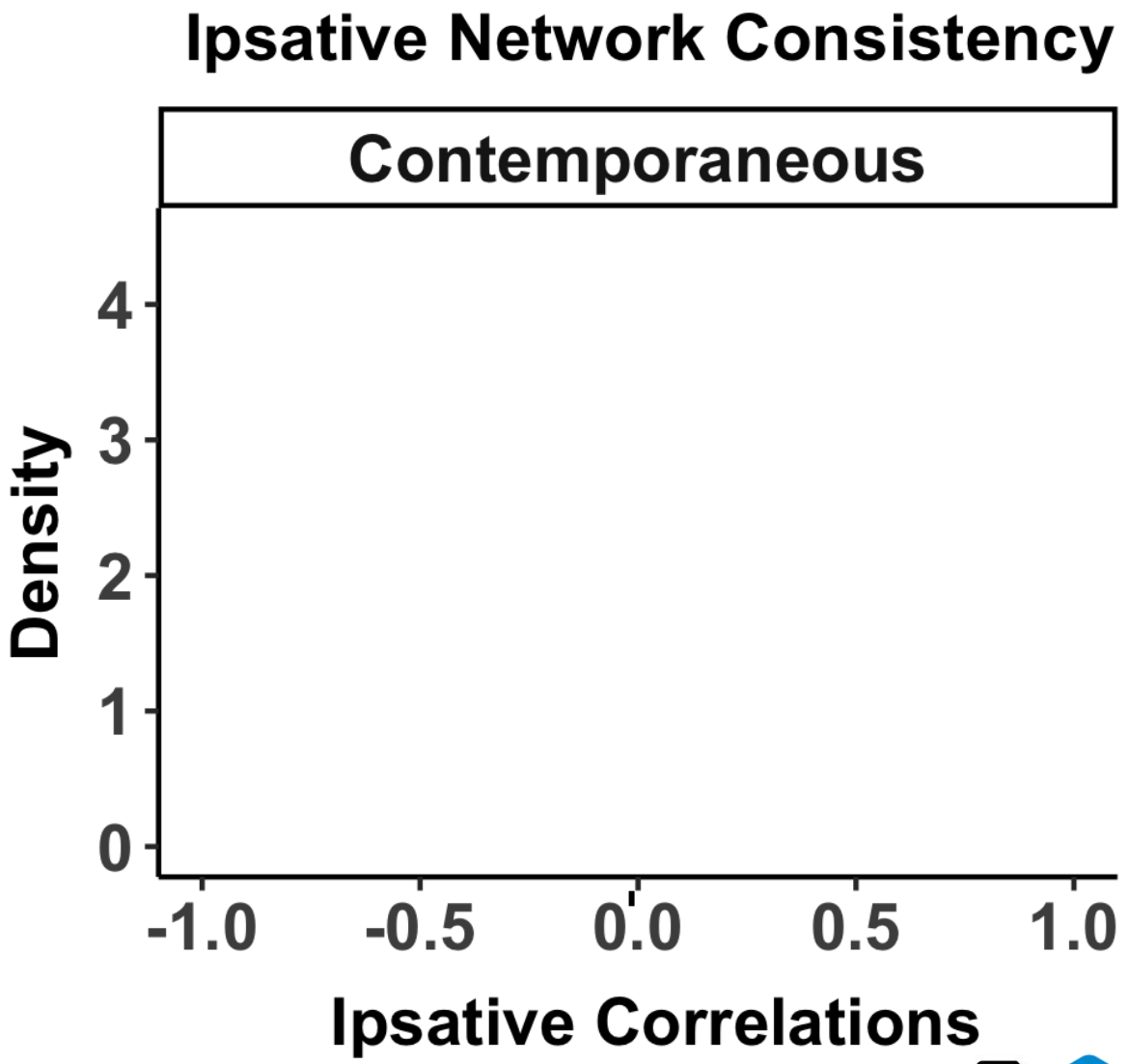
Beck & Jackson (2020a, *JPSP*)





Beck & Jackson (2020a, *JPSP*)

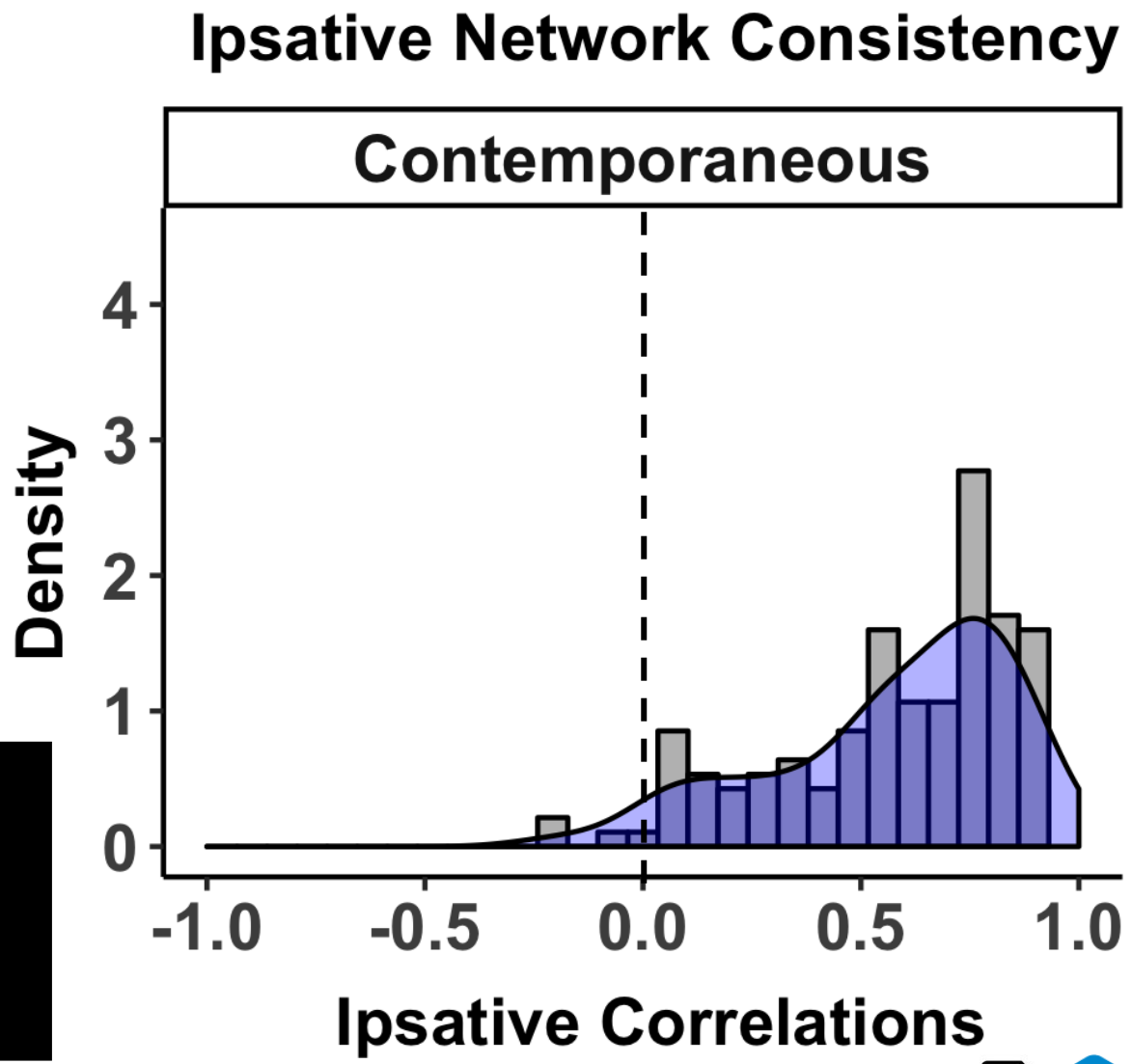
How consistent is idiographic personality across two years?



Beck & Jackson (2020a, *JPSP*)

How consistent is idiographic personality across two years?

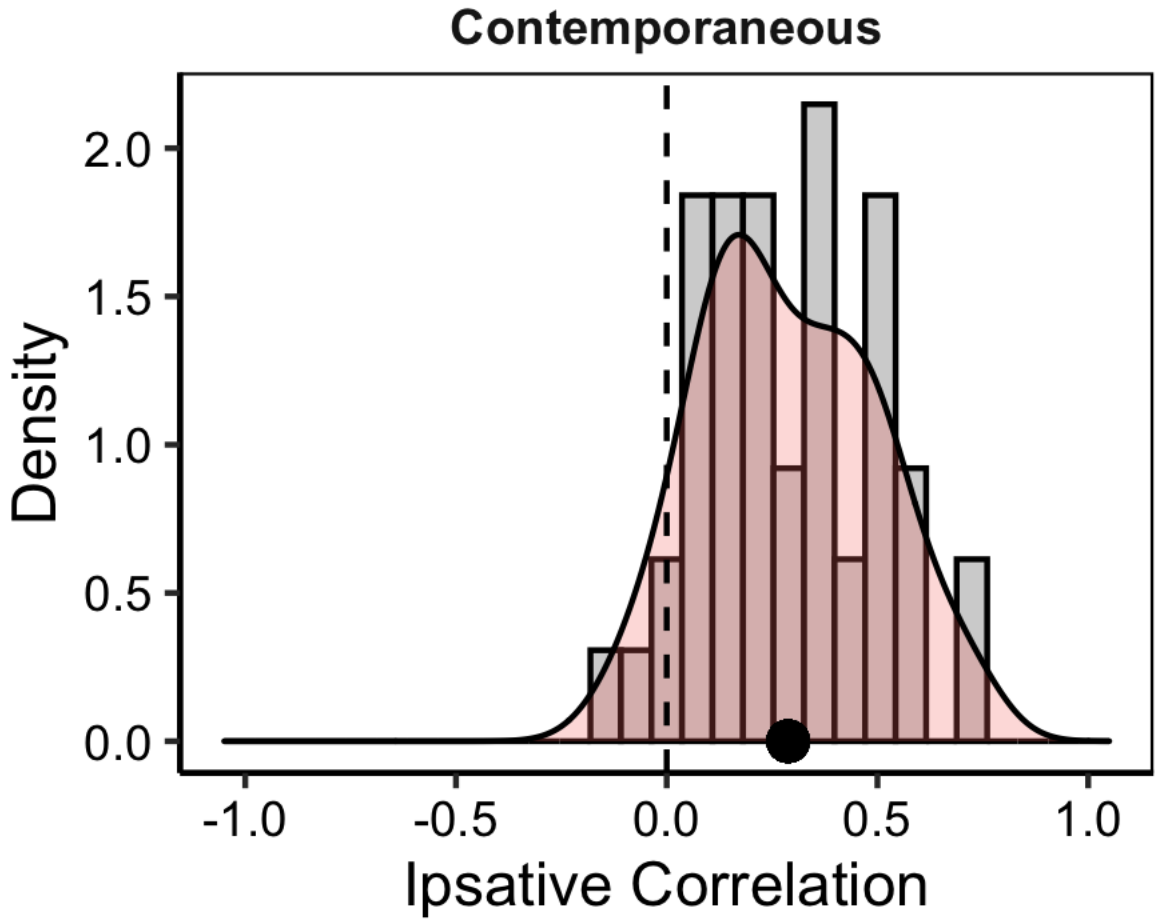
Idiographic Personality is consistent over two years.



Beck & Jackson (2020a, *JPSP*)

How consistent is idiographic personality across two years?

Idiographic Personality is consistent over two years AND global events.



Beck & Jackson (2021b, *EJP*)

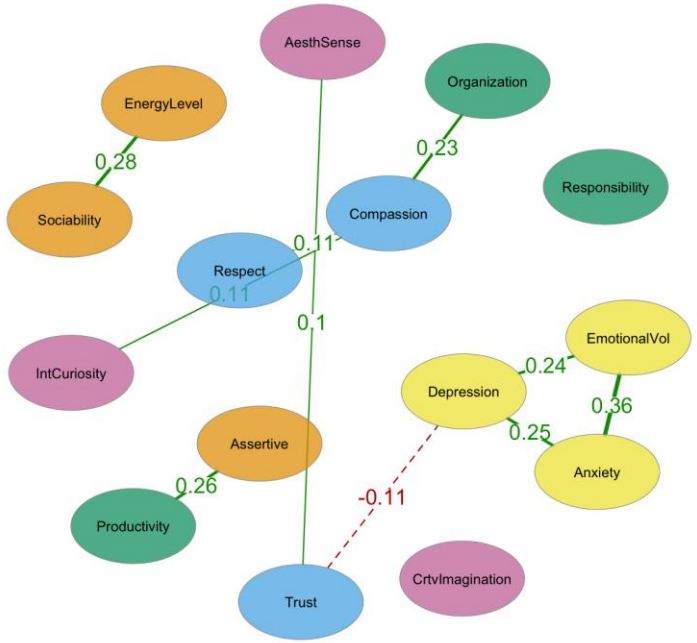
Ongoing Work in This Area

Colin J Lee

Extension to Time-Varying Networks

Lee & Beck (under review)

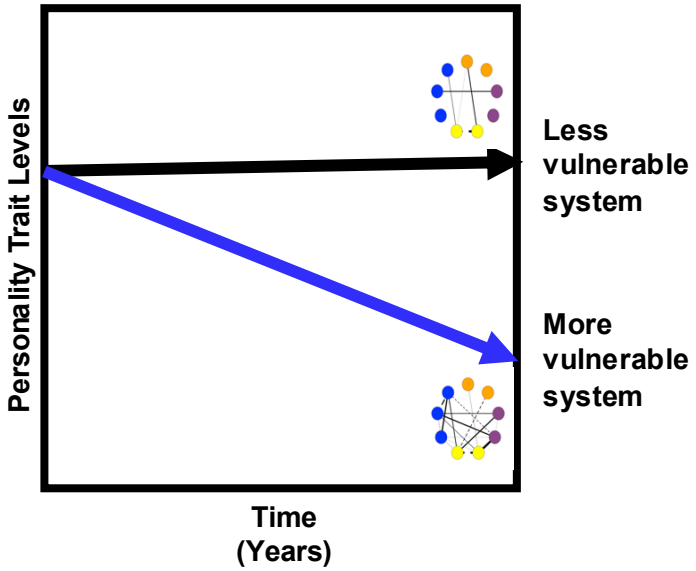
Estimation Point 1



Adam T Nissen

Network properties predict trait change and life outcomes

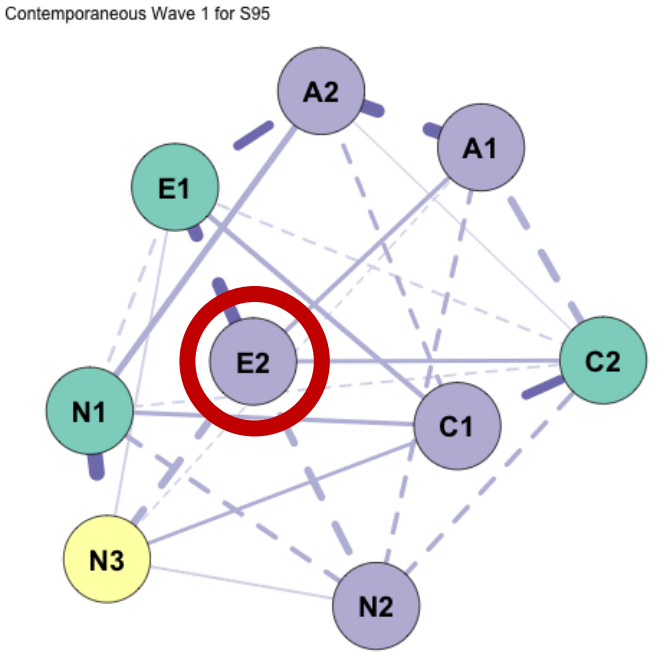
Nissen & Beck (2025),
Nissen et al. (under review)



Elizabeth U Long

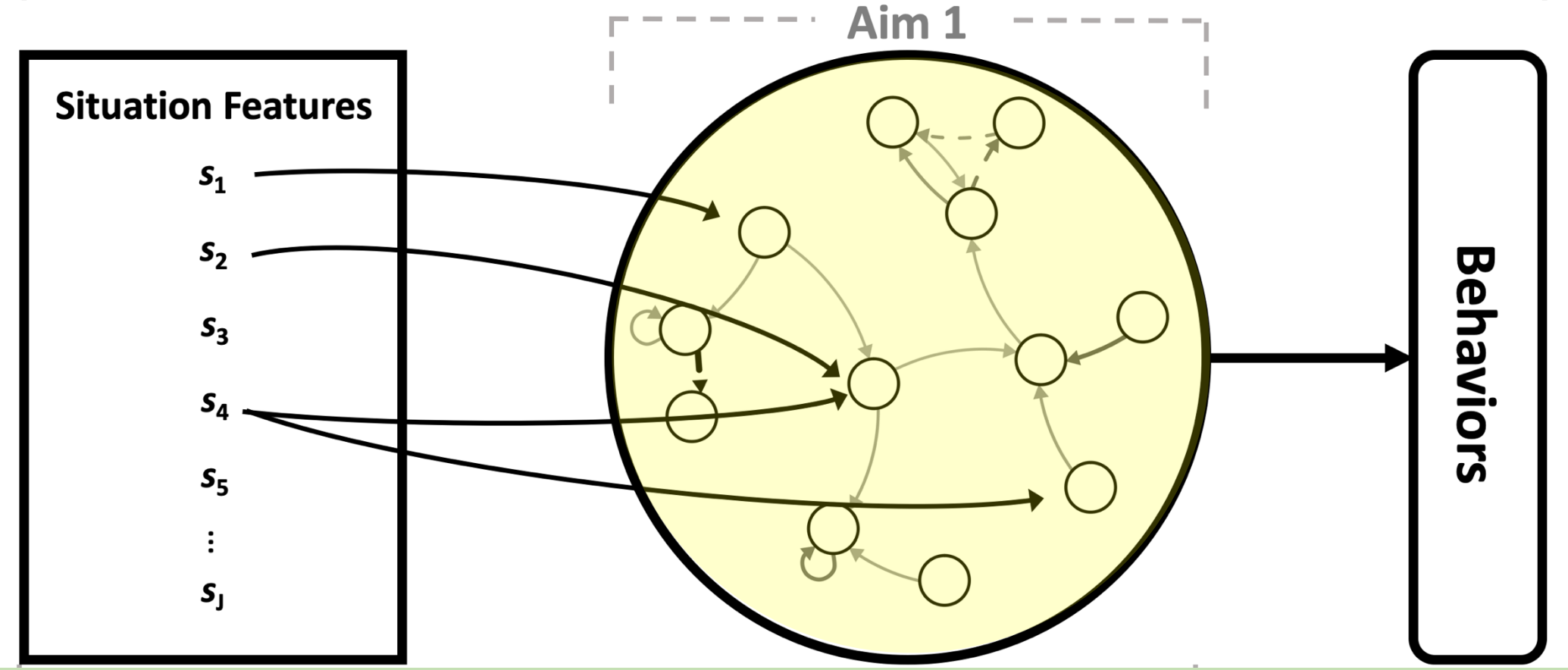
Network Centrality and Trait Centrality

Long & Beck (in prep)



Aim 1: Bringing the individual back in the study of personality structure and change.

Beck & Jackson
(2020, *JPSP*; 2021b; *EJP*)



<u>Aim 1</u>: Bringing the individual back in the study of personality structure and change.

Beck & Jackson
(2020, *JPSP*; 2021b; *EJP*)

Idiographic personality structures are:

Unique (i.e. differ across people).

Relatively consistent over time.

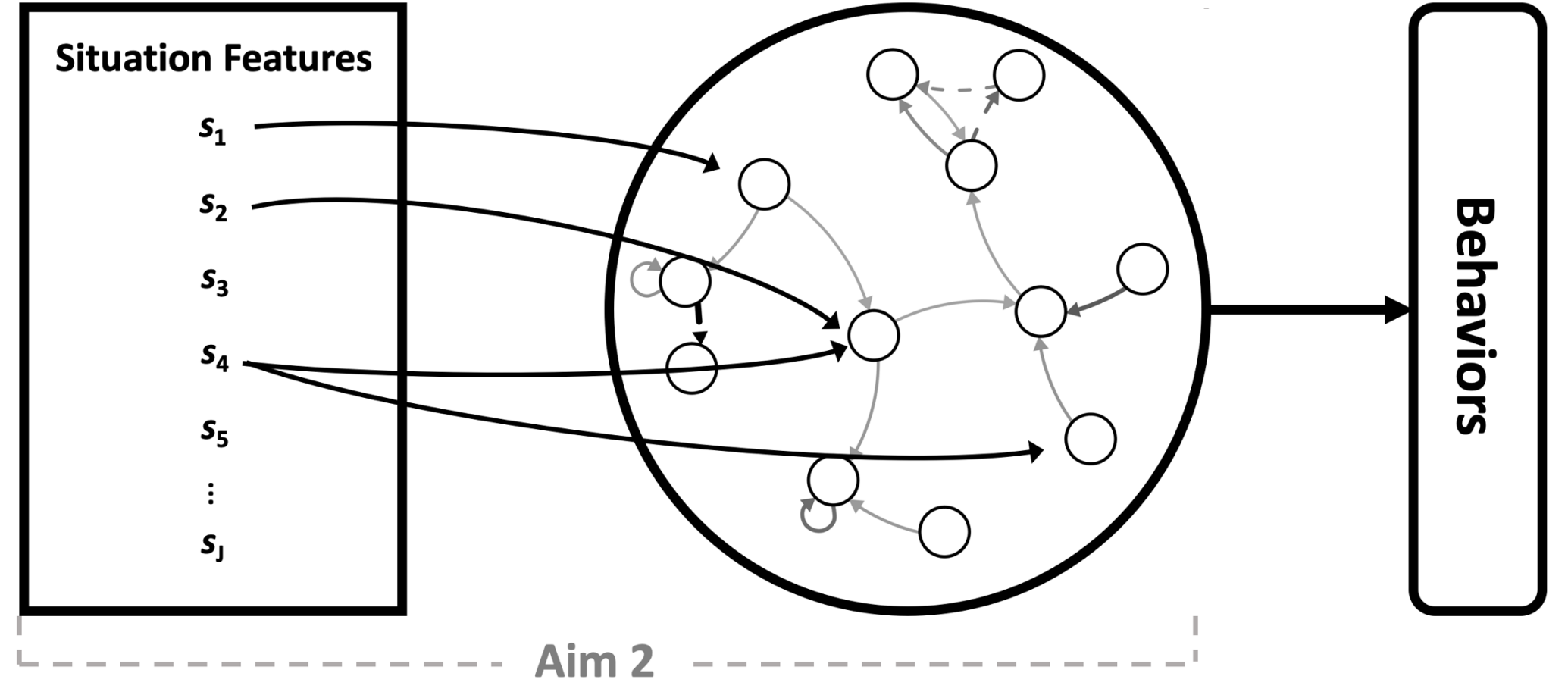
Relatively consistent across global events.

Aim 1: Individuals show unique personality structures that are relatively consistent across time and events.

Beck & Jackson
(2020, *JPSP*; 2021b; *EJP*)

Aim 2: Using longitudinal data to understand how well-being unfolds across contexts.

Beck, et al.
(revision submitted, *NHB*)

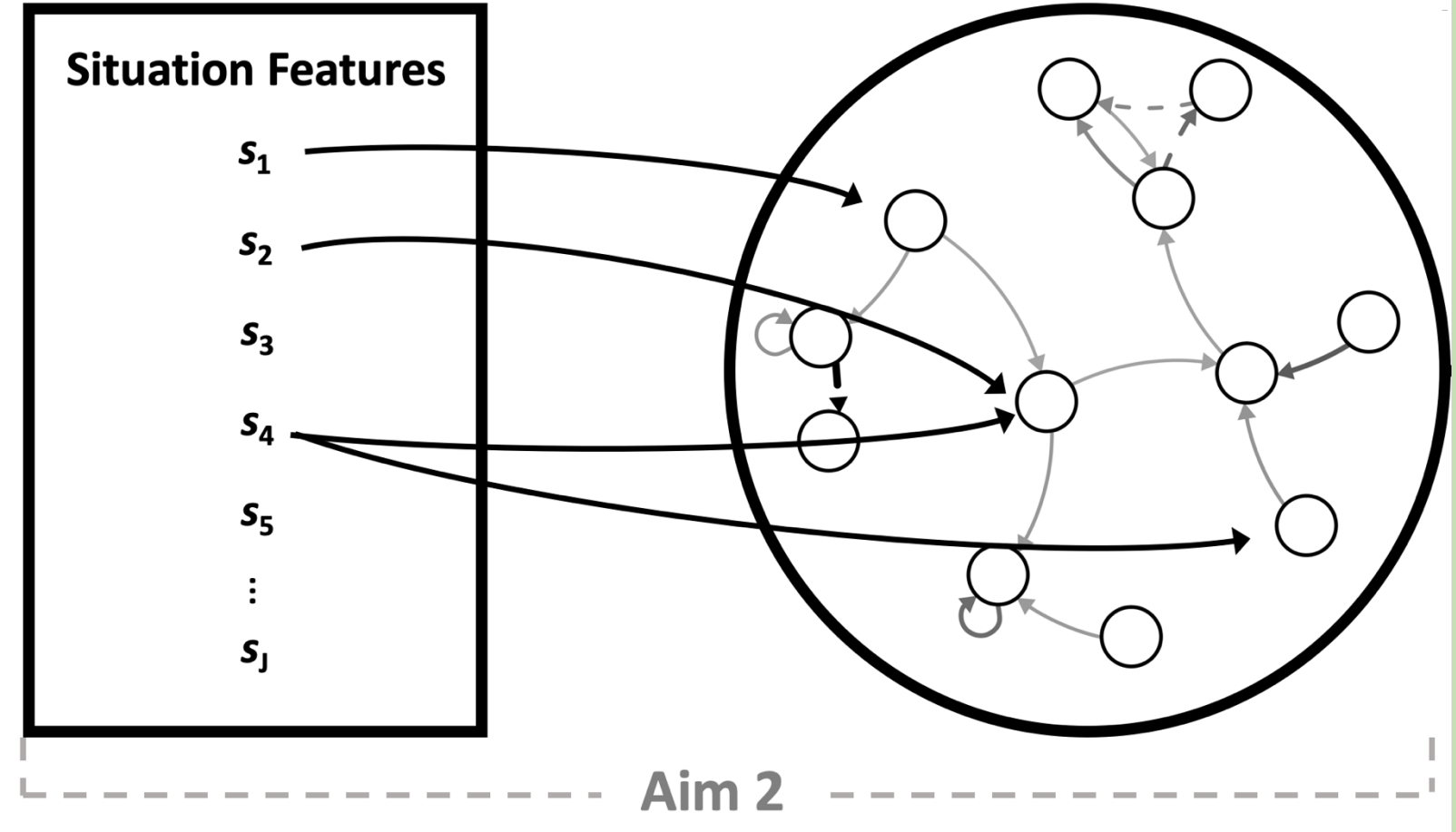


Aim 1: Individuals show unique personality structures that are relatively consistent across time and events.

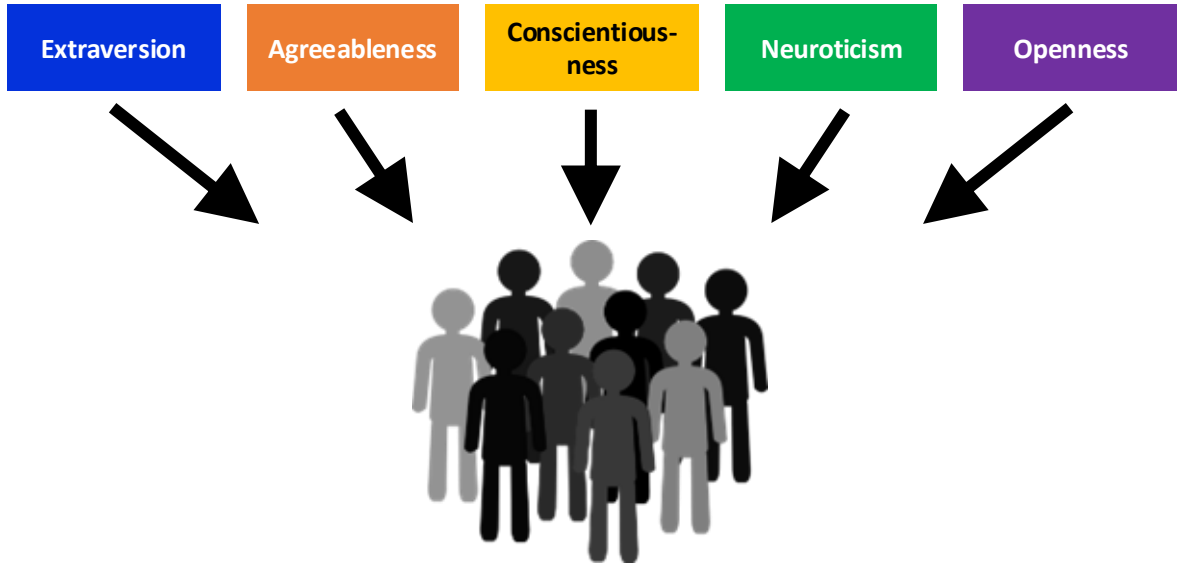
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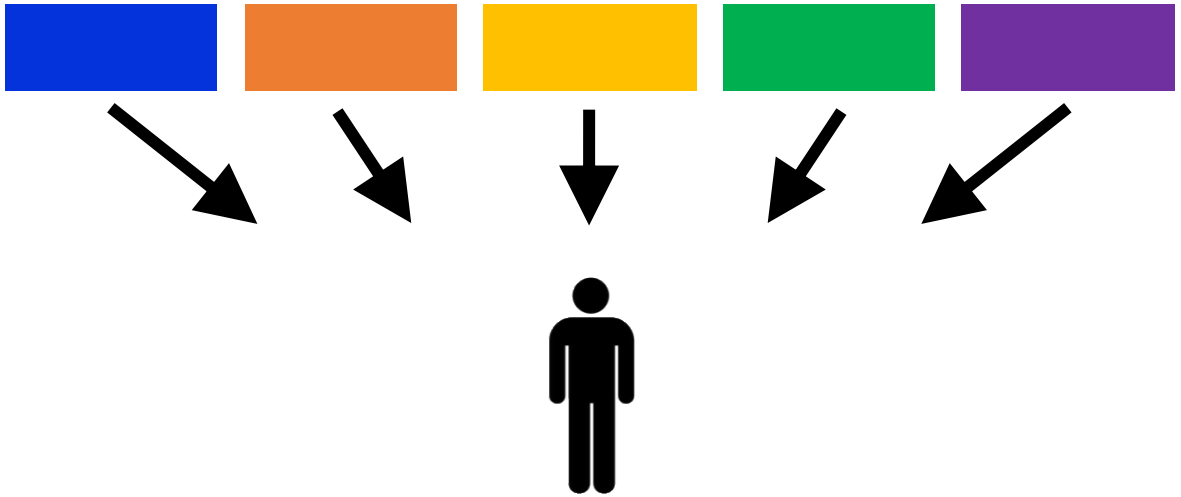
Beck, et al.
(revision submitted, *NHB*)



* = shared first authorship.



Nomothetic
Variable-Centered



Idiographic
Person-Specific

Life Satisfaction

Strong Theoretical Models

Rich longitudinal data

Unique patterns ignored

Domain Satisfaction

Health

Income

Housing

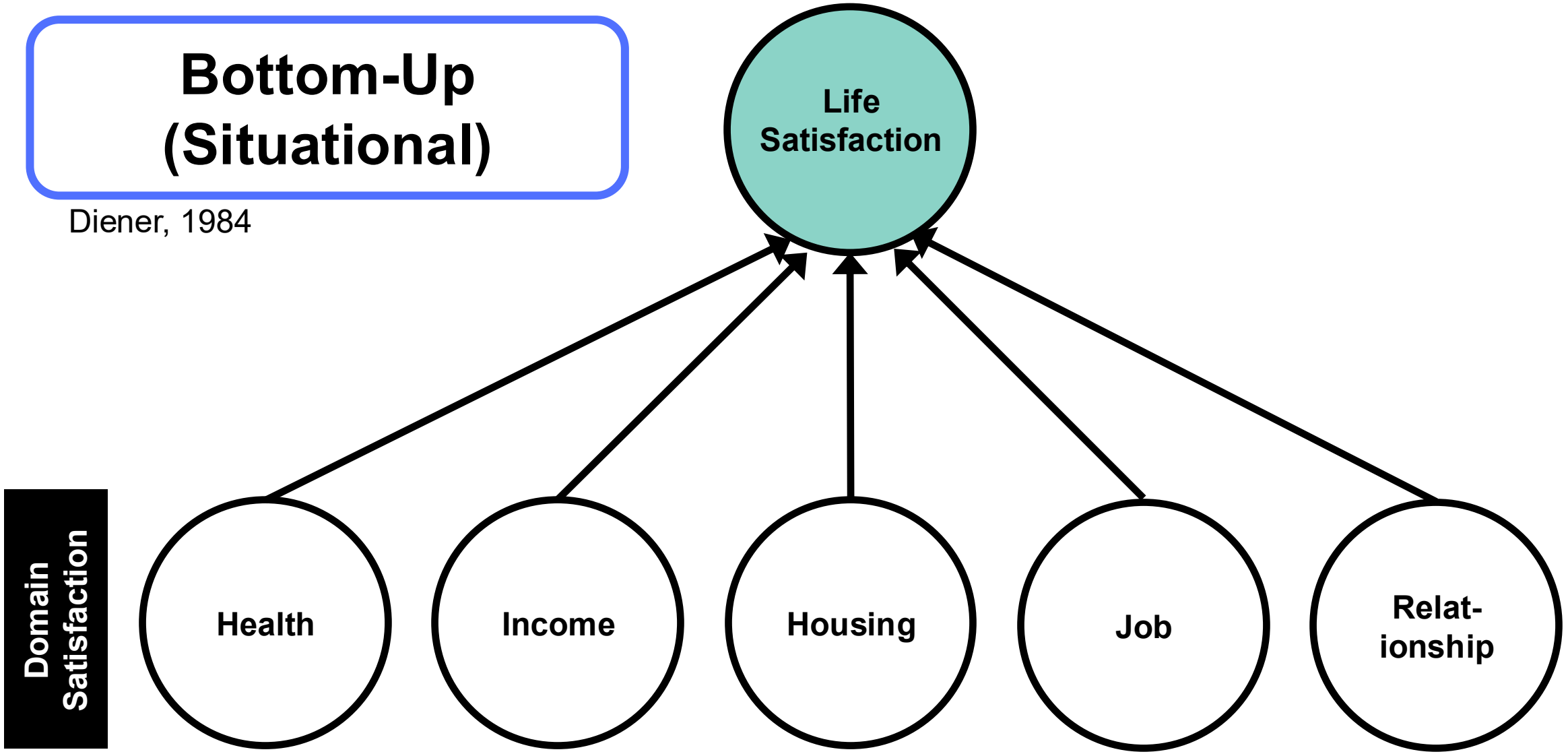
Job

Relat-ionship

Beck et al. (2025, *NHB*)

Bottom-Up (Situational)

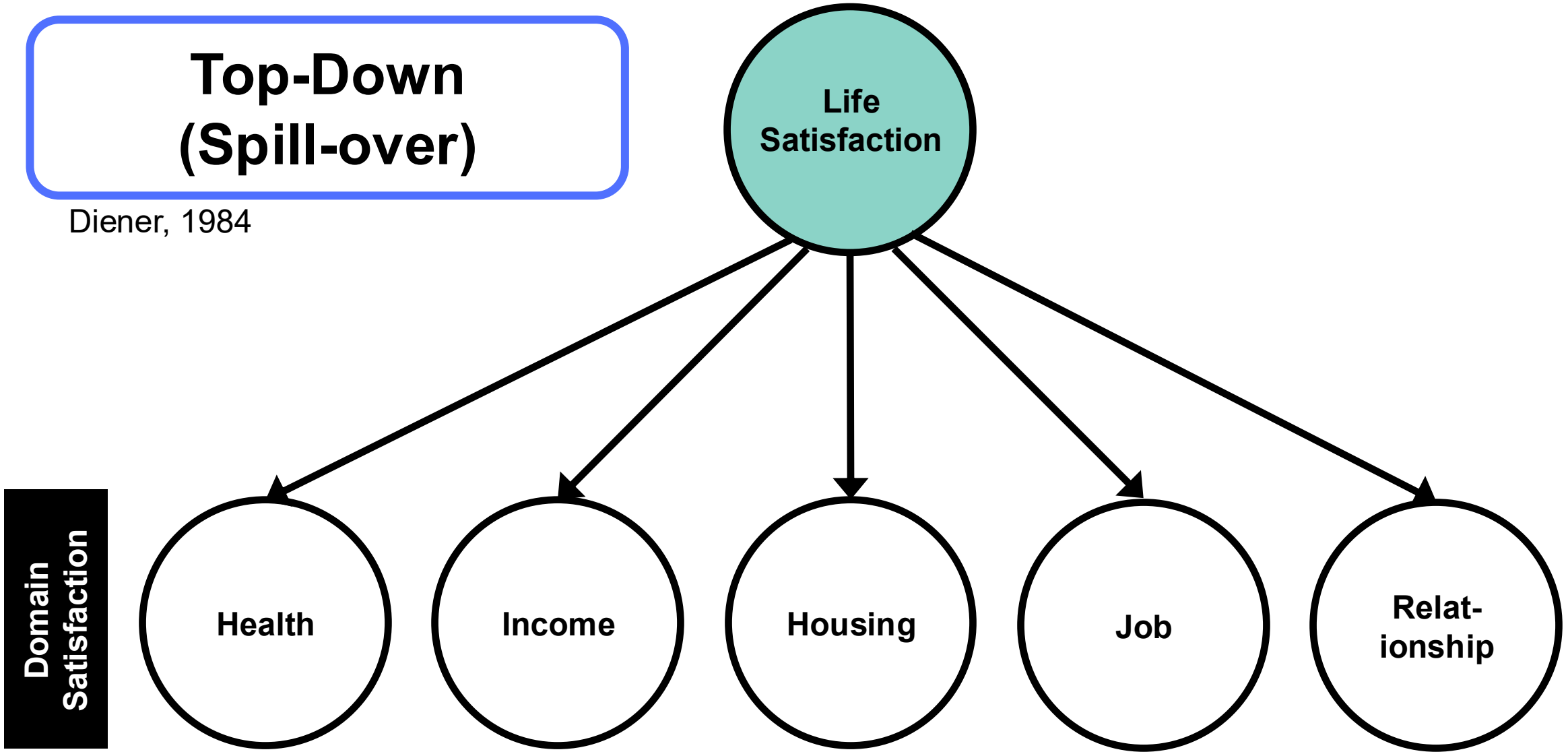
Diener, 1984



Beck et al. (2025, *NHB*)

Top-Down (Spill-over)

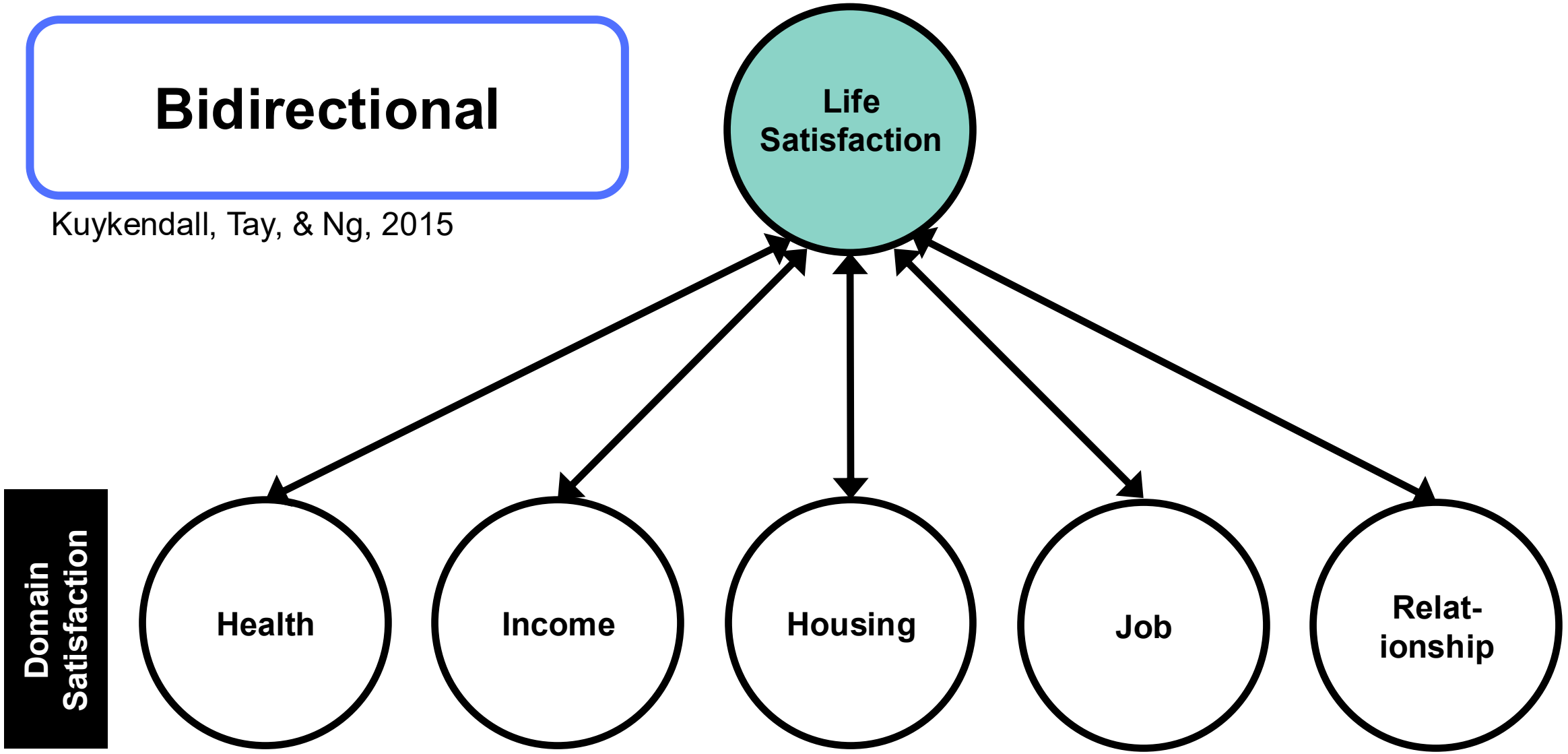
Diener, 1984



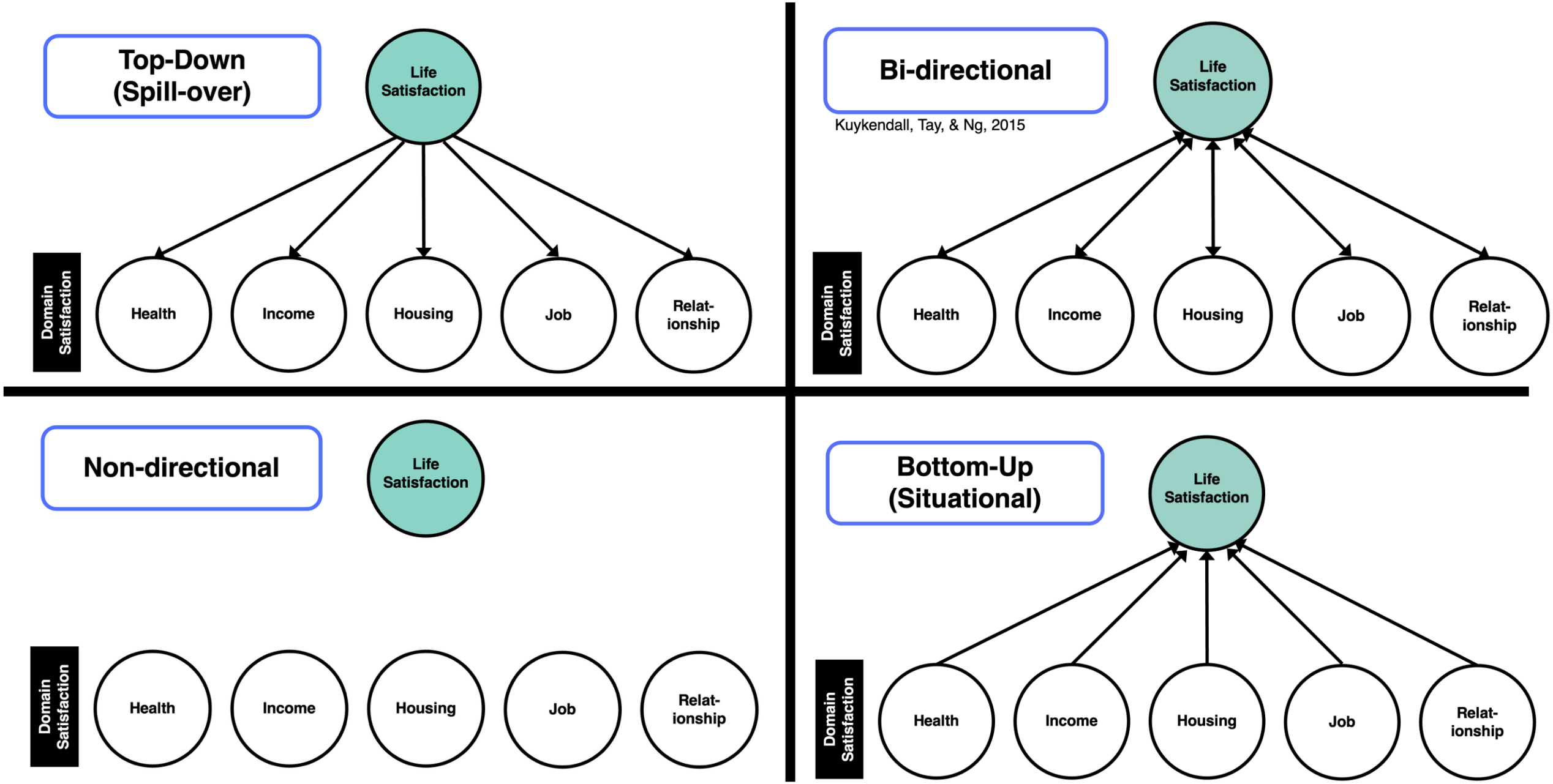
Beck et al. (2025, *NHB*)

Bidirectional

Kuykendall, Tay, & Ng, 2015

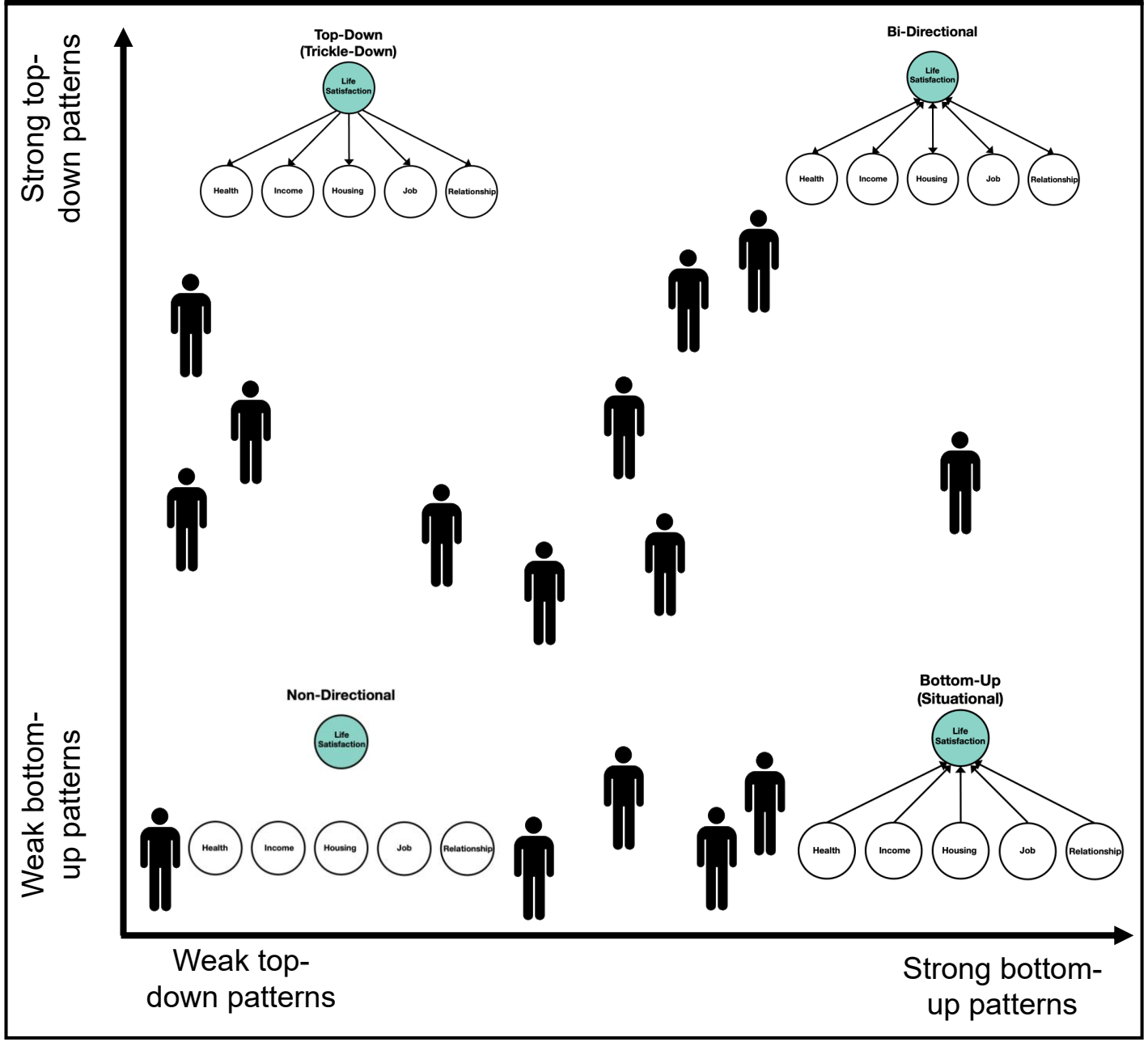


Beck et al. (2025, *NHB*)



Beck et al. (2025, *NHB*)

Personalised Perspectives on Happiness: For whom is each theory correct?



Beck et al. (2025, *NHB*)

GSOEP **HILDA** **SHP** **BHPS** **LISS**

Multilevel VAR(1) Models

Epskamp et al., 2018

Level 1:

$$Y_{ptj} = \beta_{0pj} + \beta_{1pj} * V_{1,p,t-1,} + \ldots + \beta_{kpj} * V_{k,p,t-1} + \varepsilon_{ptj}$$

Level 2:

$$\beta_{0pj} = \gamma_{00j} + \gamma_{01j} * \bar{V}_{1,p} + \ldots + \gamma_{01j} * \bar{V}_{kp} + r_{0pj}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$\beta_{kpj} = \gamma_{k0j} + \gamma_{k1j} * \bar{V}_{p,1} + \ldots + \gamma_{kkj} * \bar{V}_{pj} + r_{kpj}$$

Between-Person Effects (Level 2) $\gamma_{01j} - \gamma_{0kj}$

Within-Person Effects (Level 1) $\gamma_{10j} - \gamma_{kkj}$

Person-Specific Effects (Level 2 random effects) $r_{0pj} - r_{kpj}$

Beck et al. (2025, *NHB*)

GSOEP | HILDA | SHP | BHPS | LISS

Multilevel VAR(1) Models

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$$\beta_{kpj} = \gamma_{k0j} + \gamma_{k1j} * \bar{V}_{p,1} + \ldots + \gamma_{kkj} * \bar{V}_{pj} + r_{kpj}$$

Between-Person Effects (Level 2) $\gamma_{01j} - \gamma_{0kj}$

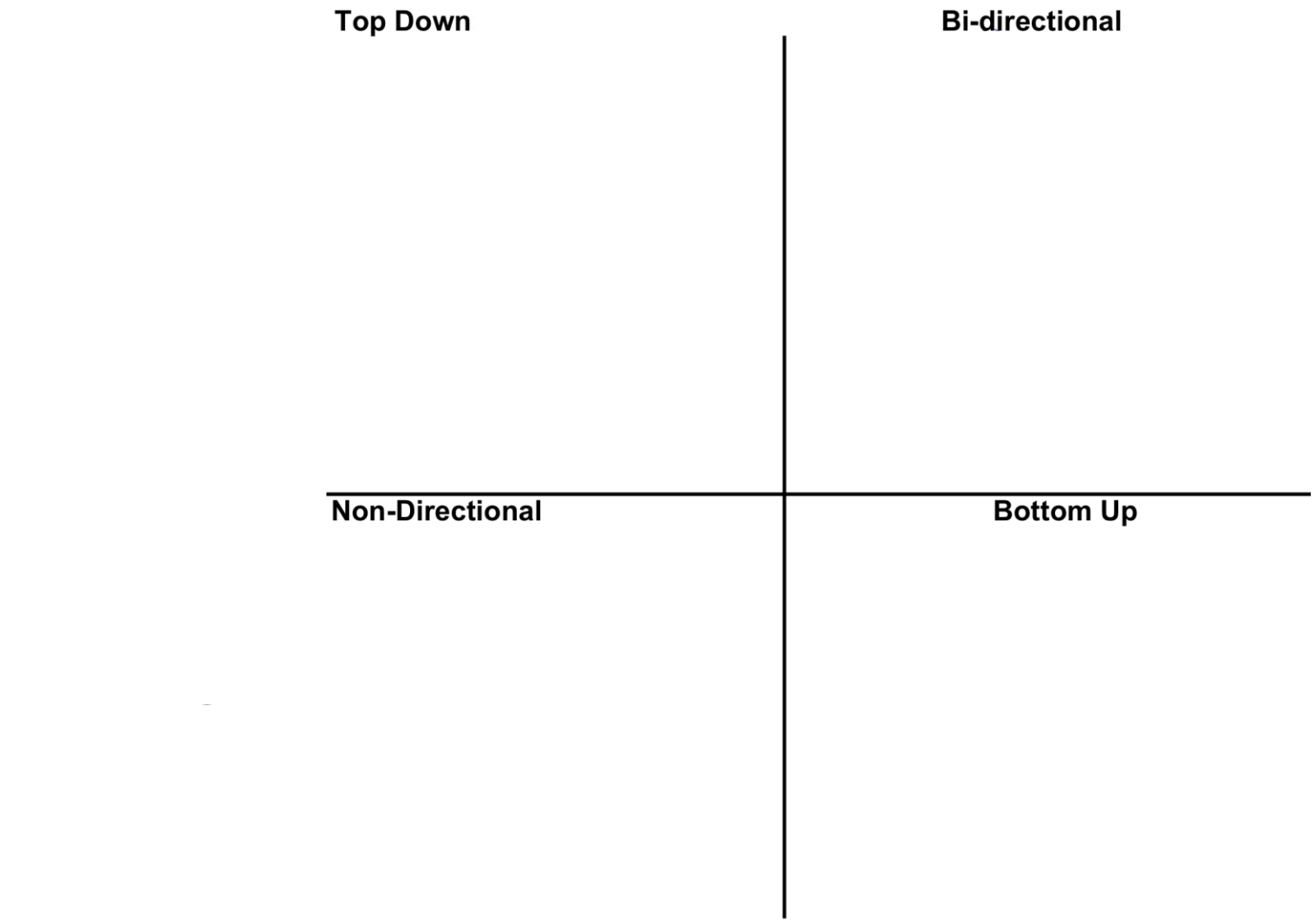
Within-Person Effects (Level 1) $\gamma_{10j} - \gamma_{kkj}$

Person-Specific Effects (Level 2 random effects) $r_{0pj} - r_{kpj}$

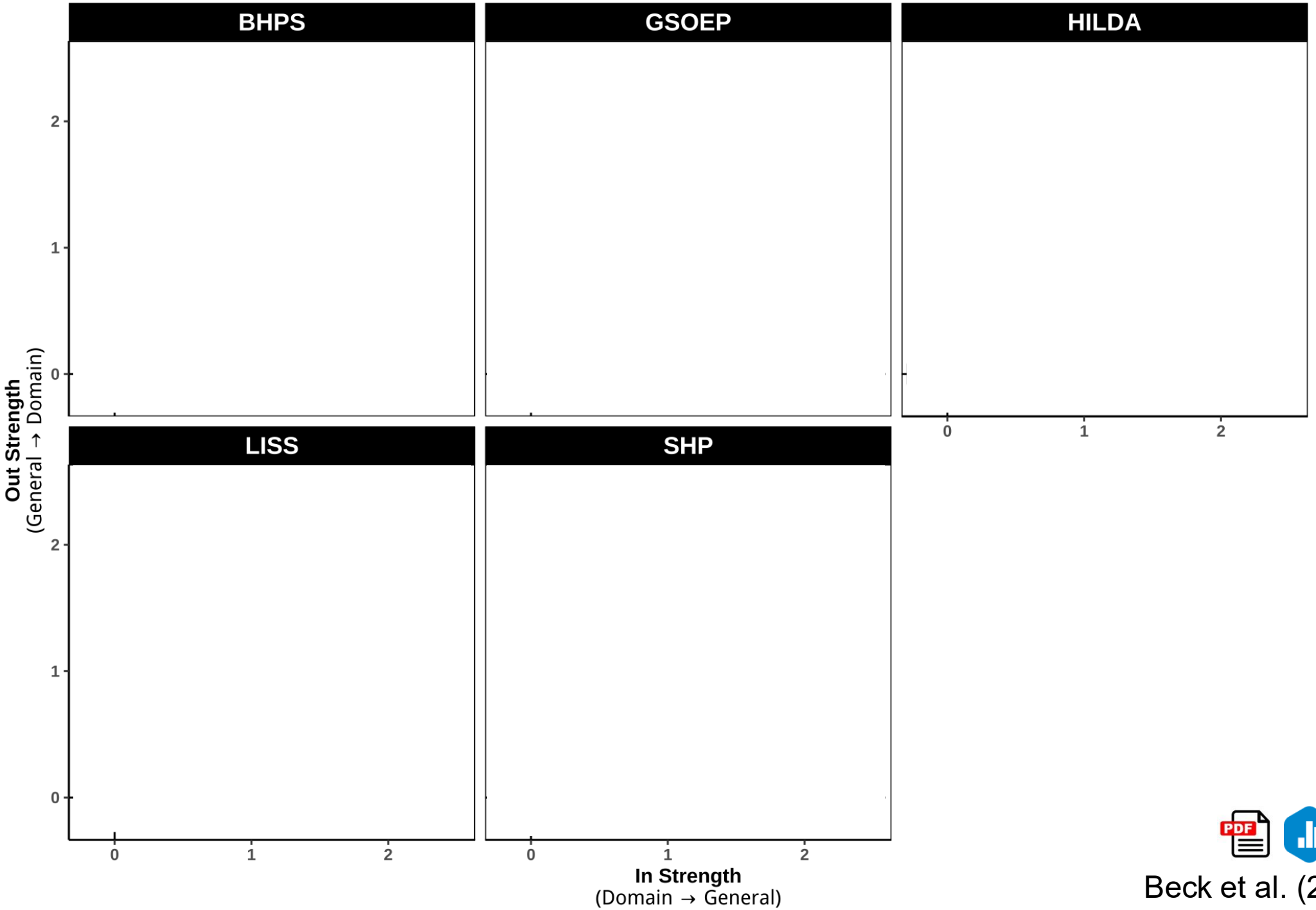
Beck et al. (2025, *NHB*)

RESULTS

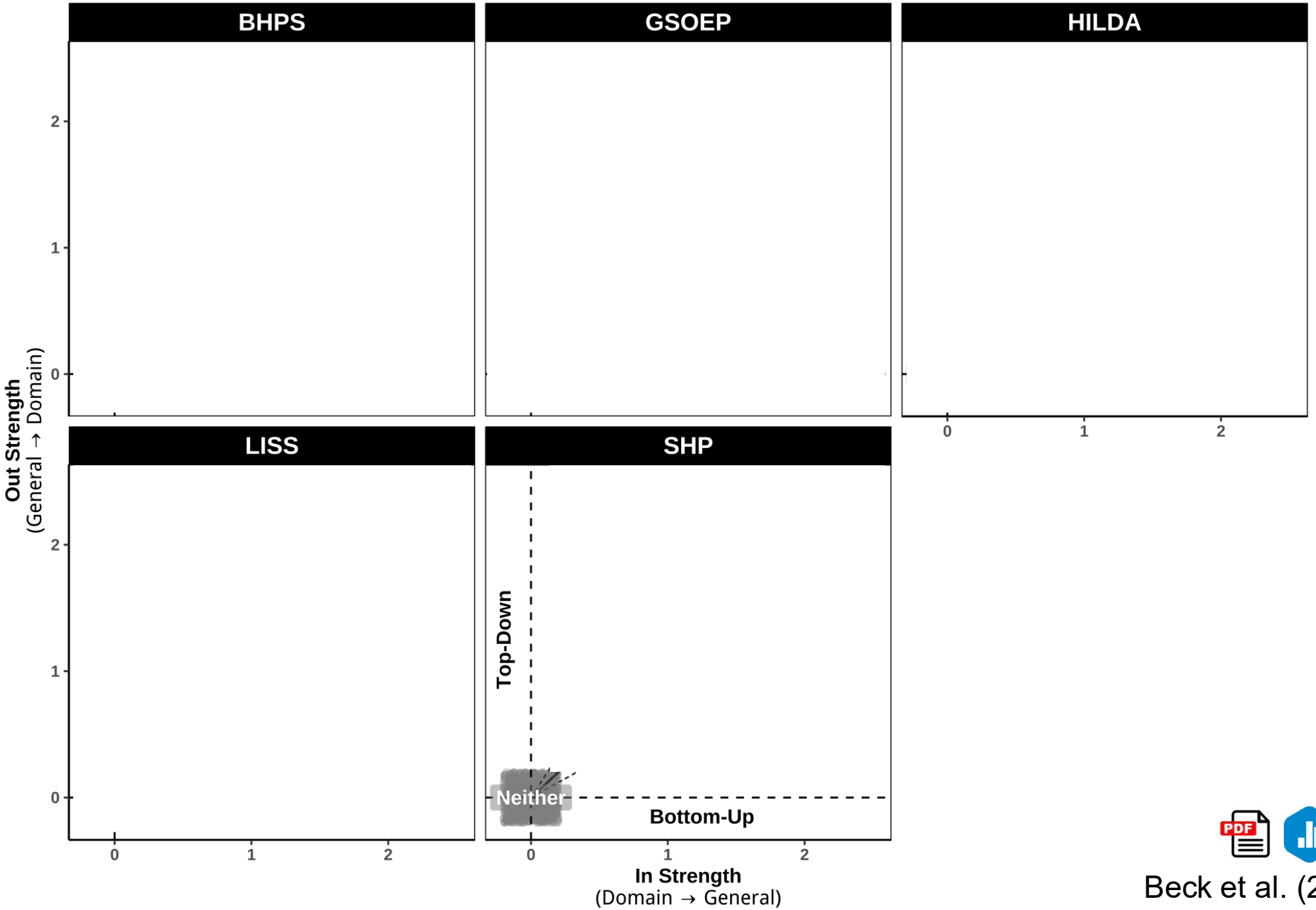
Aim 2



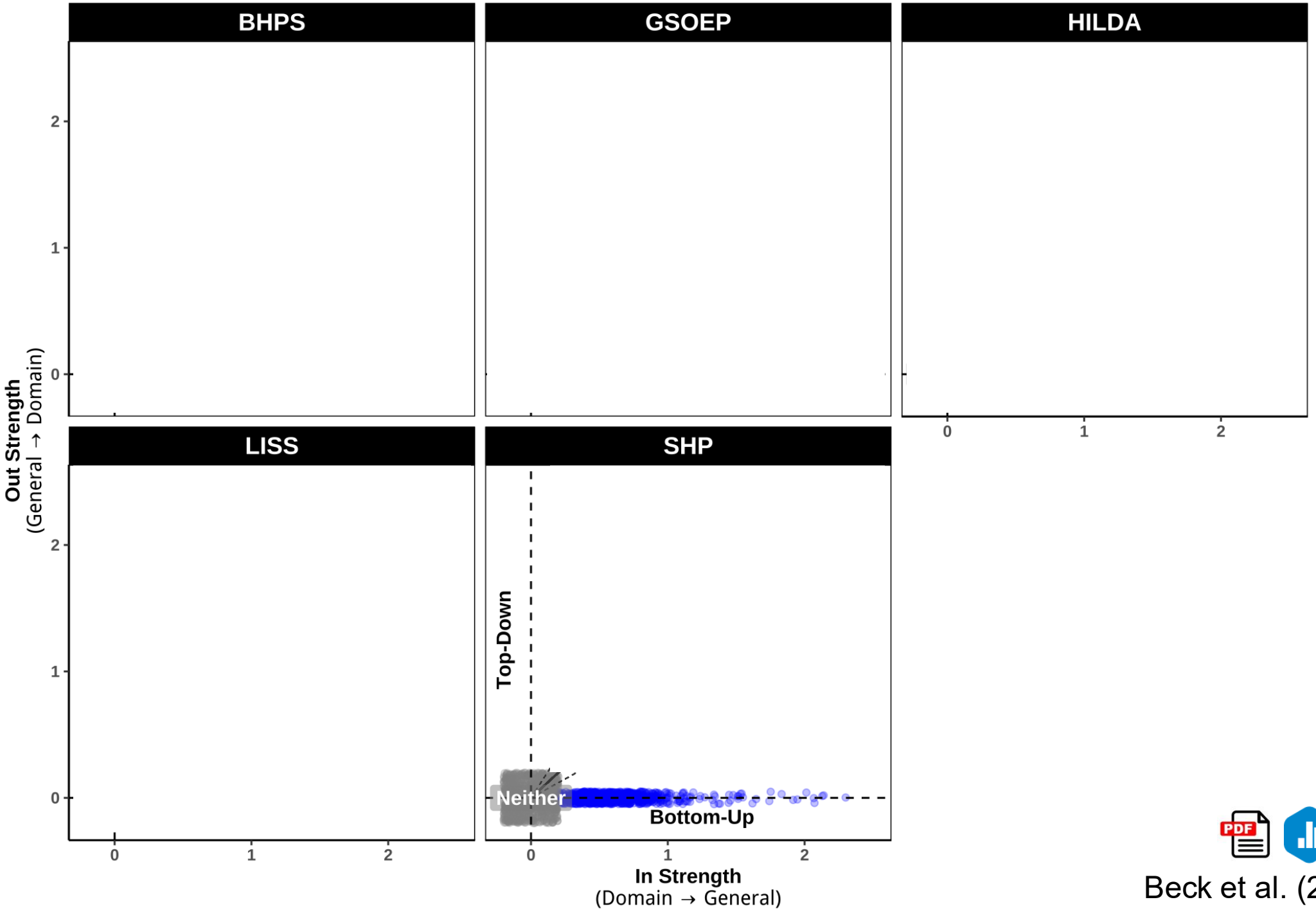
Beck et al. (2025, *NHB*)



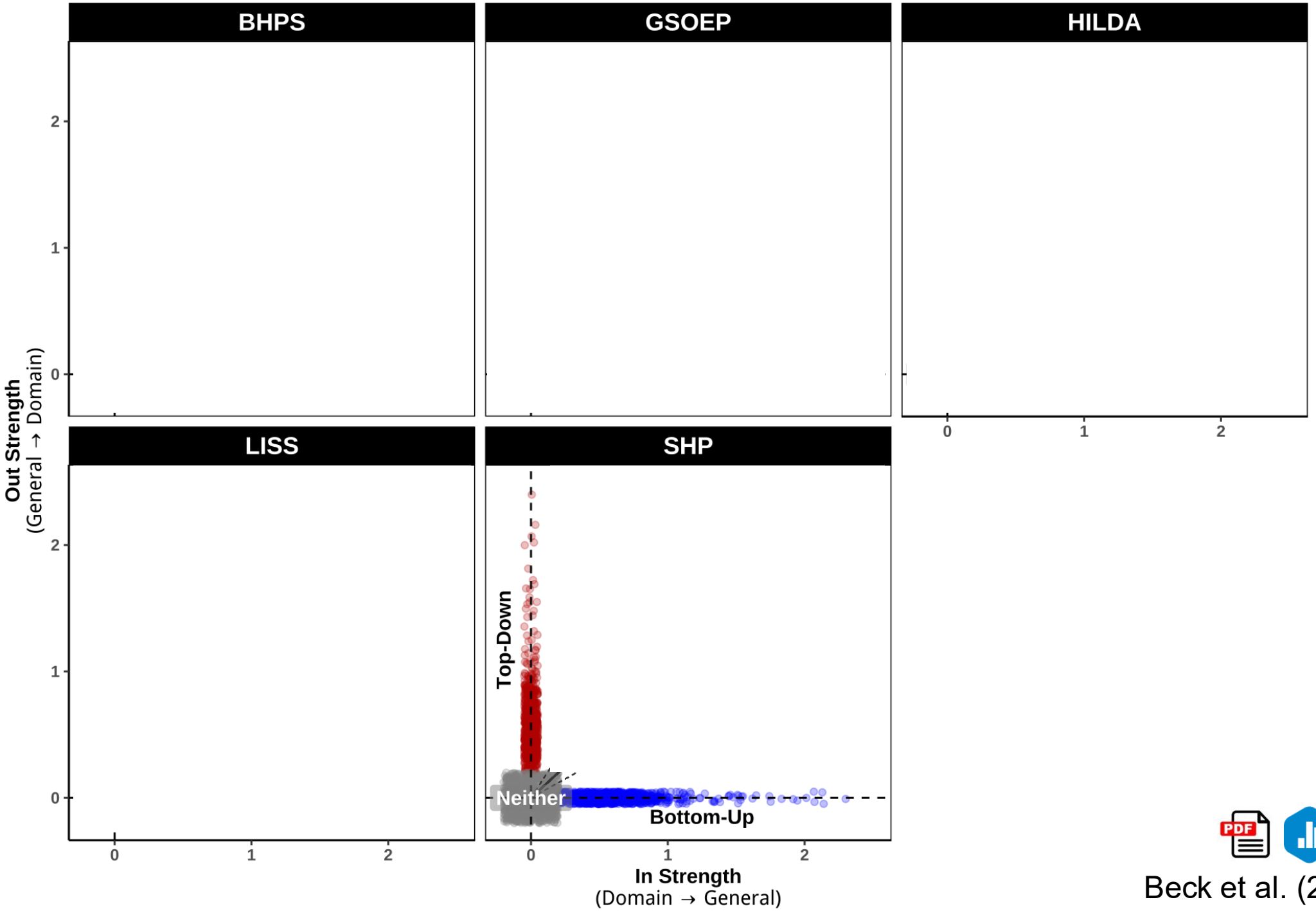
Beck et al. (2025, *NHB*)



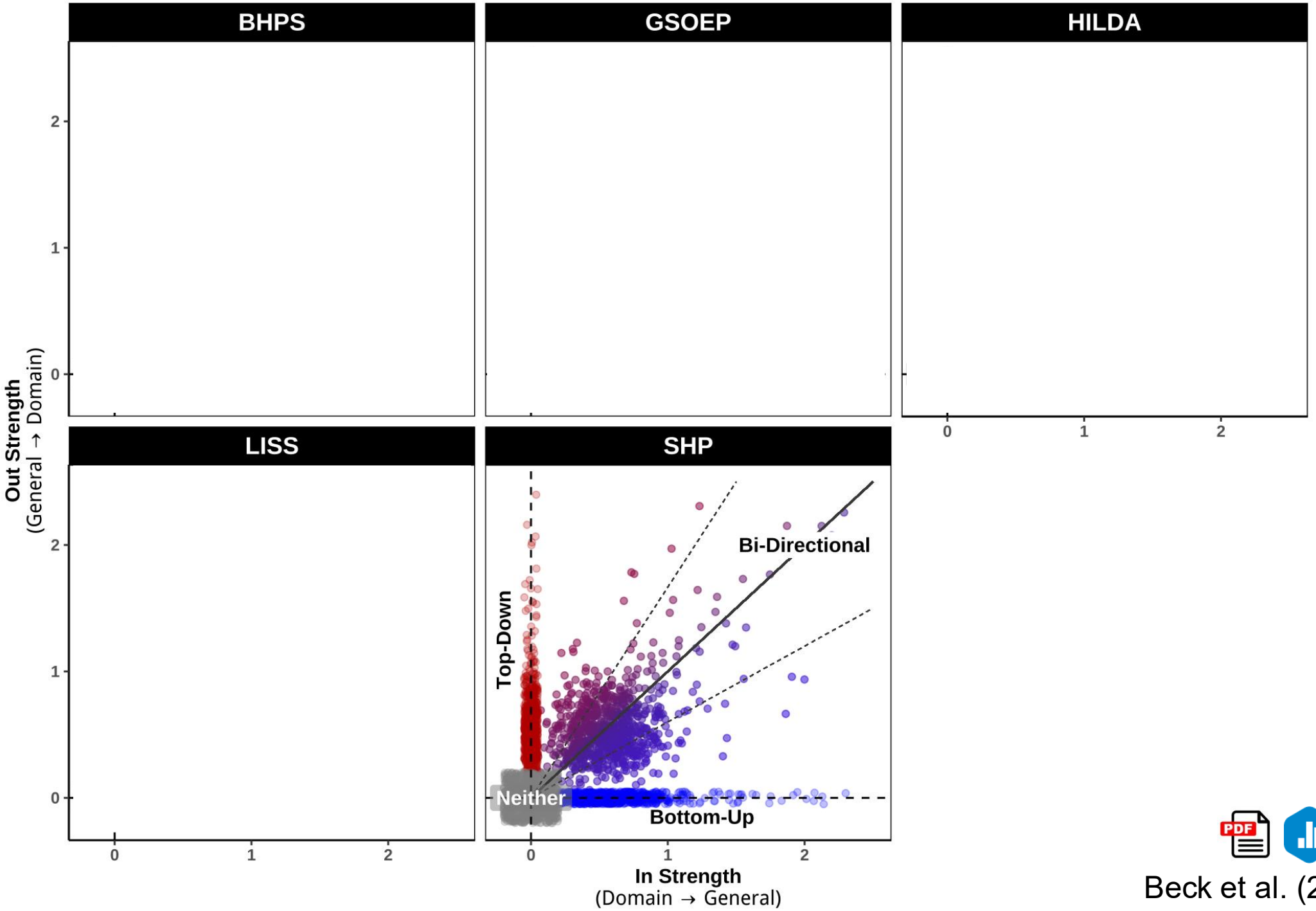
Beck et al. (2025, *NHB*)



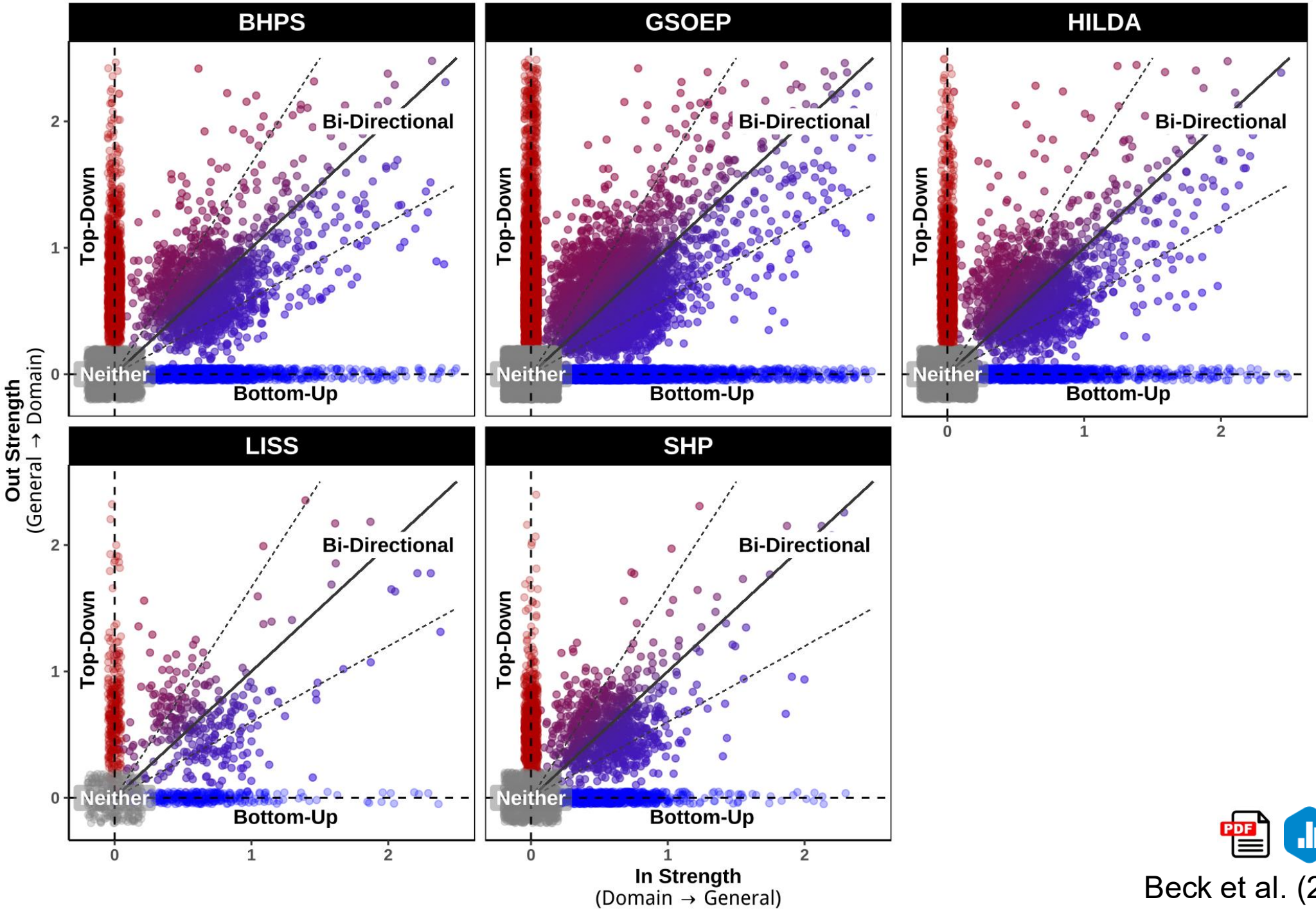
Beck et al. (2025, *NHB*)



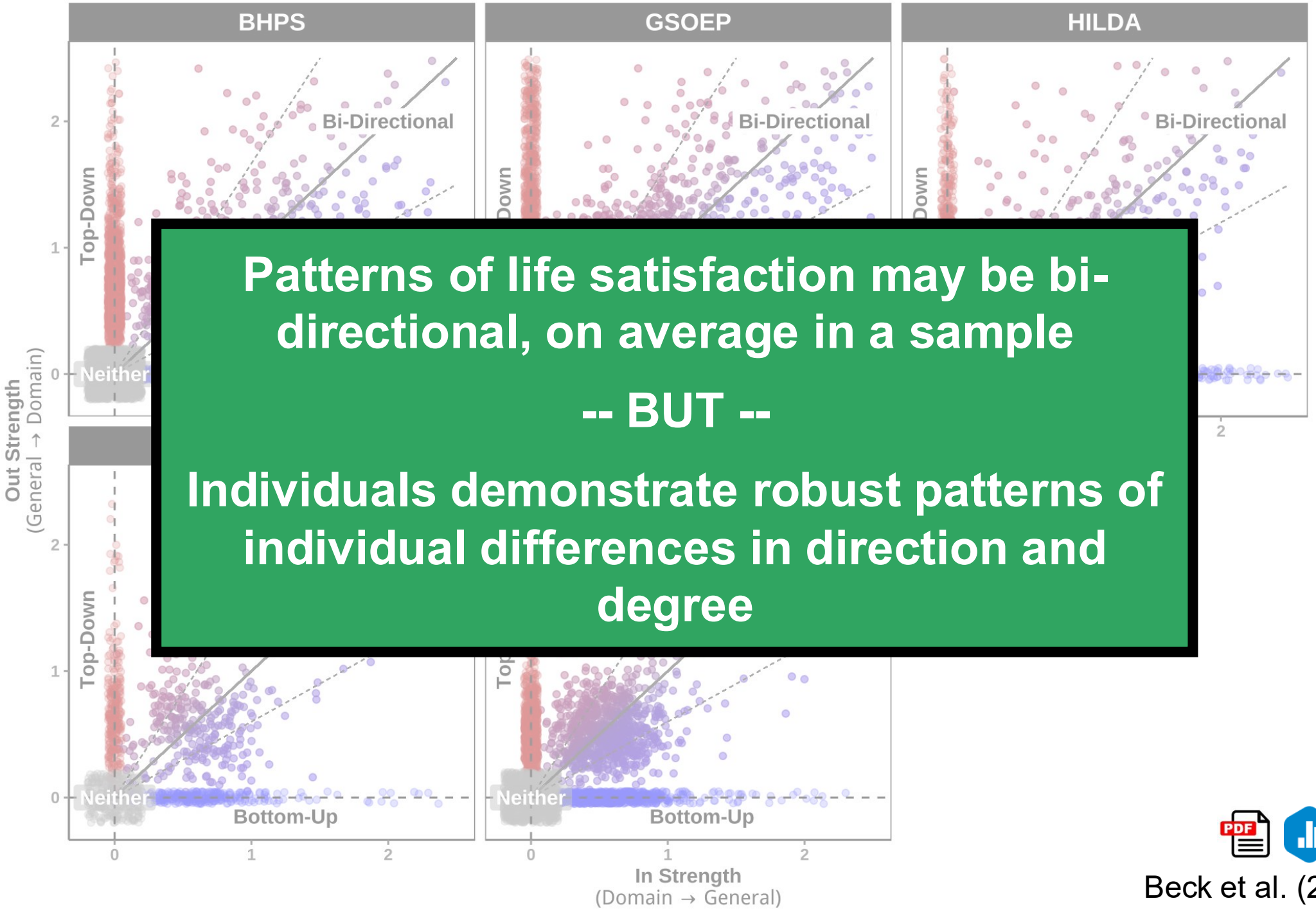
Beck et al. (2025, *NHB*)



Beck et al. (2025, *NHB*)



Beck et al. (2025, *NHB*)



Patterns of life satisfaction may be bi-directional, on average in a sample

-- BUT --

Individuals demonstrate robust patterns of individual differences in direction and degree

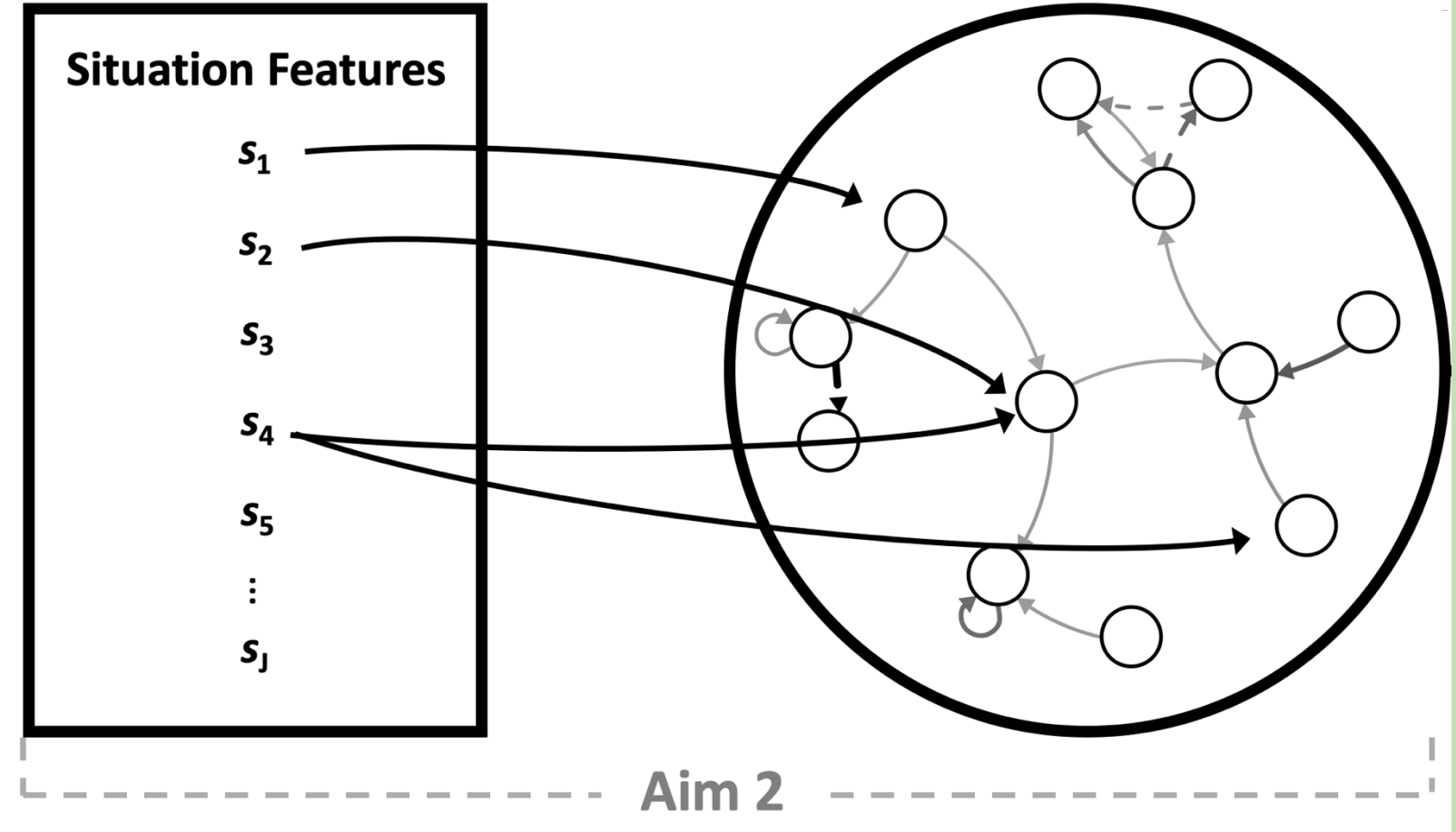
Beck et al. (2025, *NHB*)

Aim 1: Individuals show unique personality structures that are relatively consistent across time and events.

Beck & Jackson
(2020, *JPSP*; 2021b; *EJP*)

Aim 2: Using longitudinal data to understand how well-being unfolds across contexts.

Beck et al.
(revision submitted, *NHB*)



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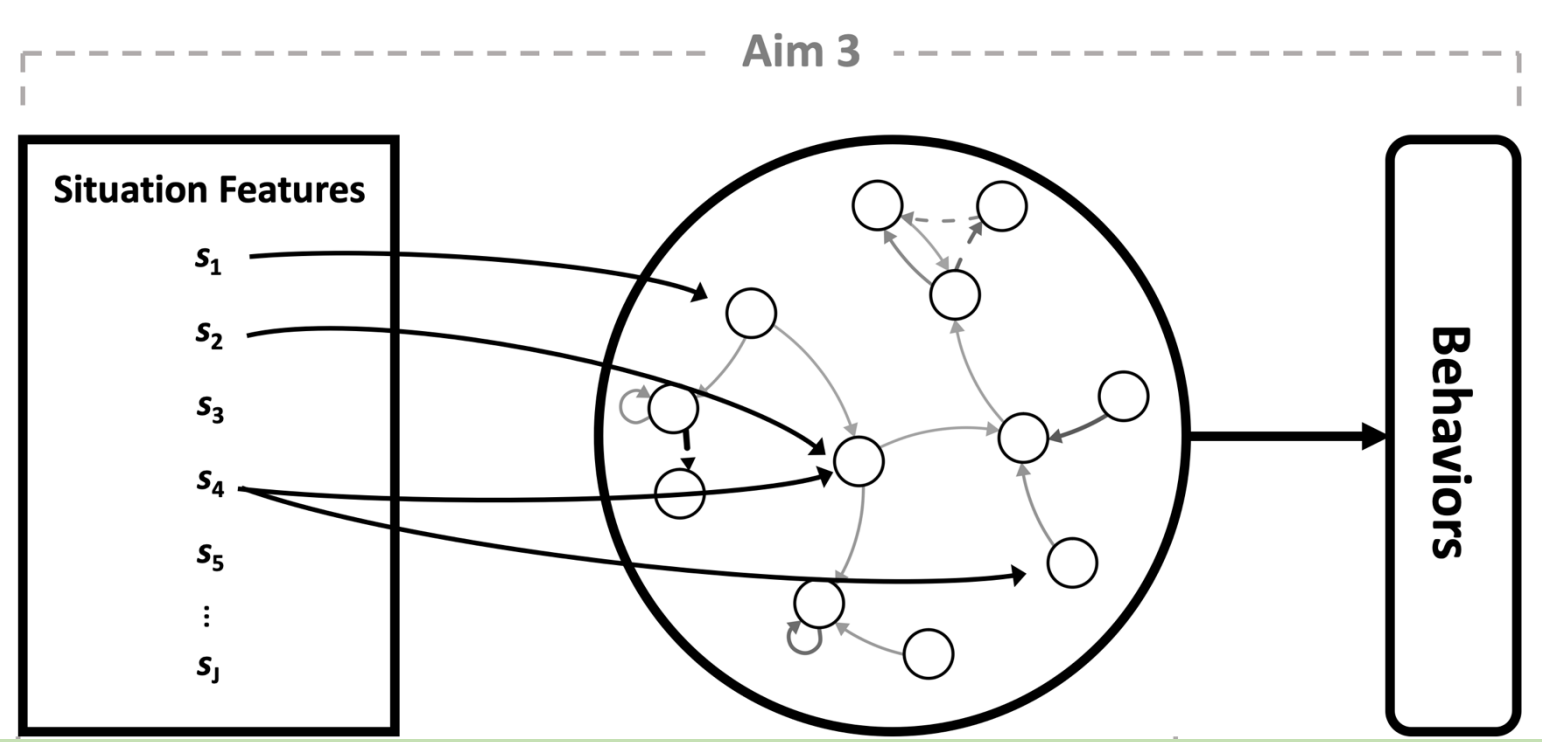
Aim 2: Using longitudinal data to understand how well-being unfolds across contexts.

Beck et al.
(revision submitted, *NHB*)

The dynamics of life and domain satisfaction:

- Unfold differently across people
- Do not appear well-captured by sample averages of idiographic dynamics
- Have similar patterns of individual differences across samples

Aim 1: Individuals show unique personality structures that are relatively consistent across time and events.	**Beck & Jackson** (2020, *JPSP*; 2021b; *EJP*)
Aim 2: The dynamics of well-being unfold differently across people.	**Beck et al.** (revision submitted, *NHB*)
Aim 3: Predicting behavior using machine learning.	**Beck & Jackson** (2023, *Psych Science*)




Psychological
Situations
Behavior / Experiences

Critical Assumption

Situations and experiences should have similar consequences across people.

People with similar levels of a personality characteristic should behave in similar ways.

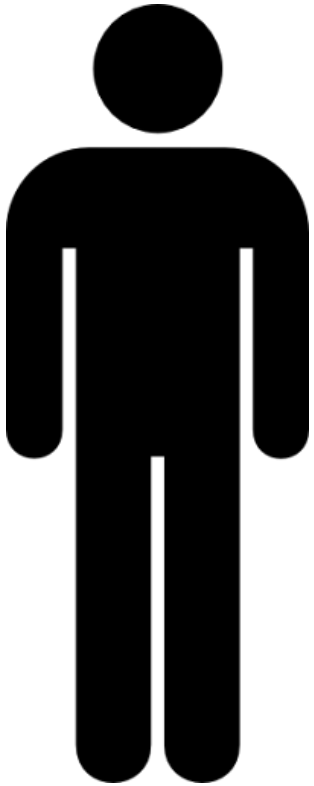
Beck & Jackson (2023; *Psychological Science*)

Alternative Assumption

Situations and experiences should have *different* consequences across people.

People with similar levels of a personality characteristic *may not* behave in similar ways.

Beck & Jackson (2023; *Psychological Science*)



Psychological

Situational

Time

Behaviors

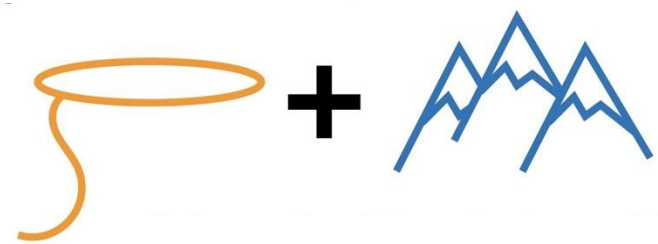
Experiences

Beck & Jackson (2023; *Psychological Science*)

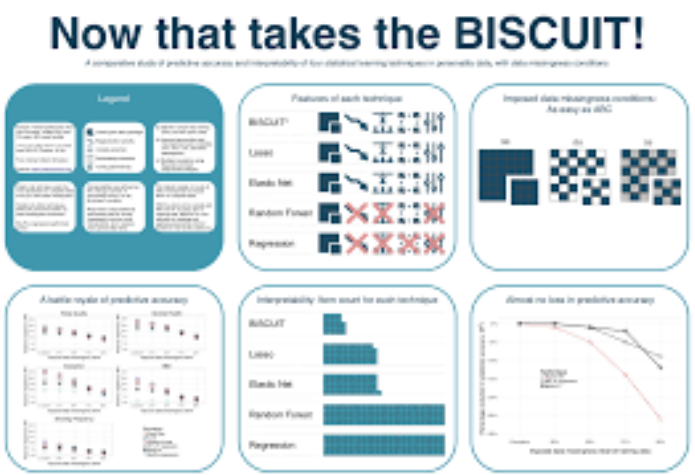
Analytic Plan

3 machine learning classification methods:

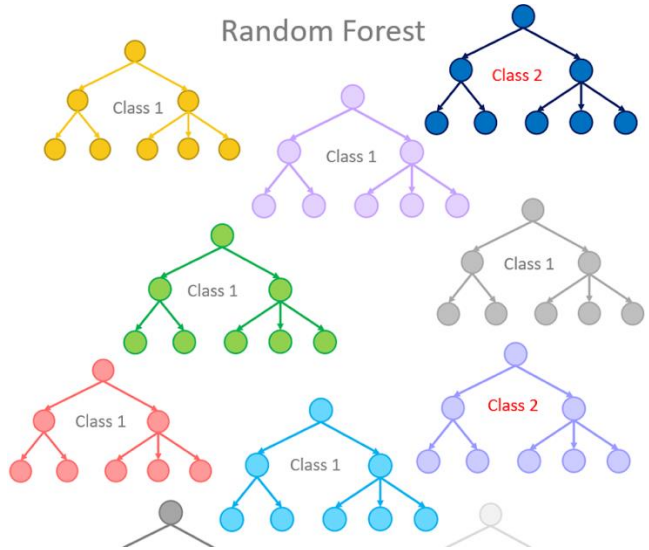
Elastic Net



BISCWIT



Random Forest



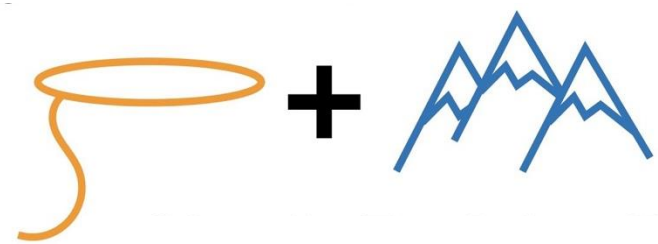
- Classification Accuracy
- Area under the receiver operating curve (AUC)

Beck & Jackson (2023; *Psychological Science*)

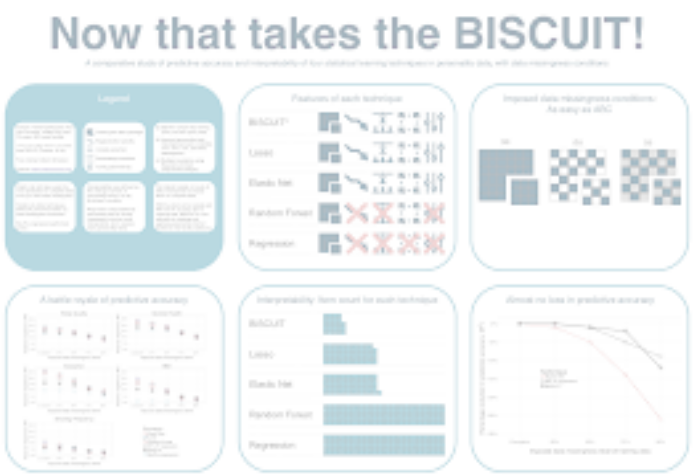
Analytic Plan

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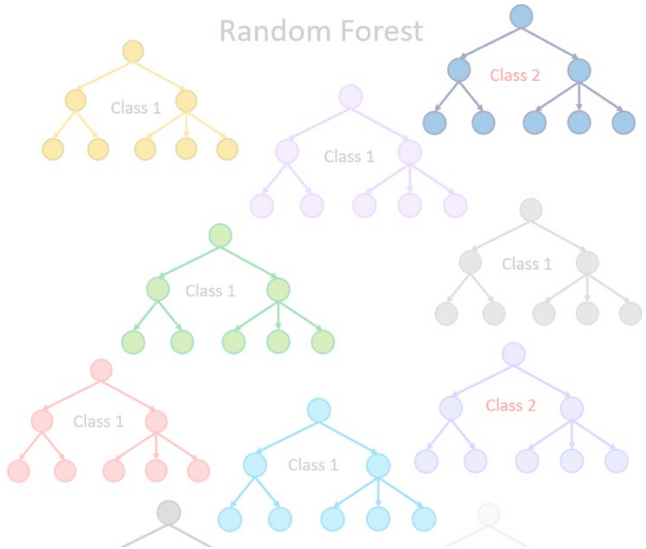
Elastic Net



BISCWIT



Random Forest



- Classification Accuracy
- Area under the receiver operating curve (AUC)

Beck & Jackson (2023; *Psychological Science*)

RESULTS

Aim 3

Classification Accuracy

	Elastic Net	
	Median (SD)	N
Procrastination		
Loneliness		
Argument		
Interacted		
Studying		
Sick		
Tired		

Beck & Jackson (2023; *Psychological Science*)

Classification Accuracy

Elastic Net

Classification accuracy was high, on average, with some variability across outcomes.

Sick

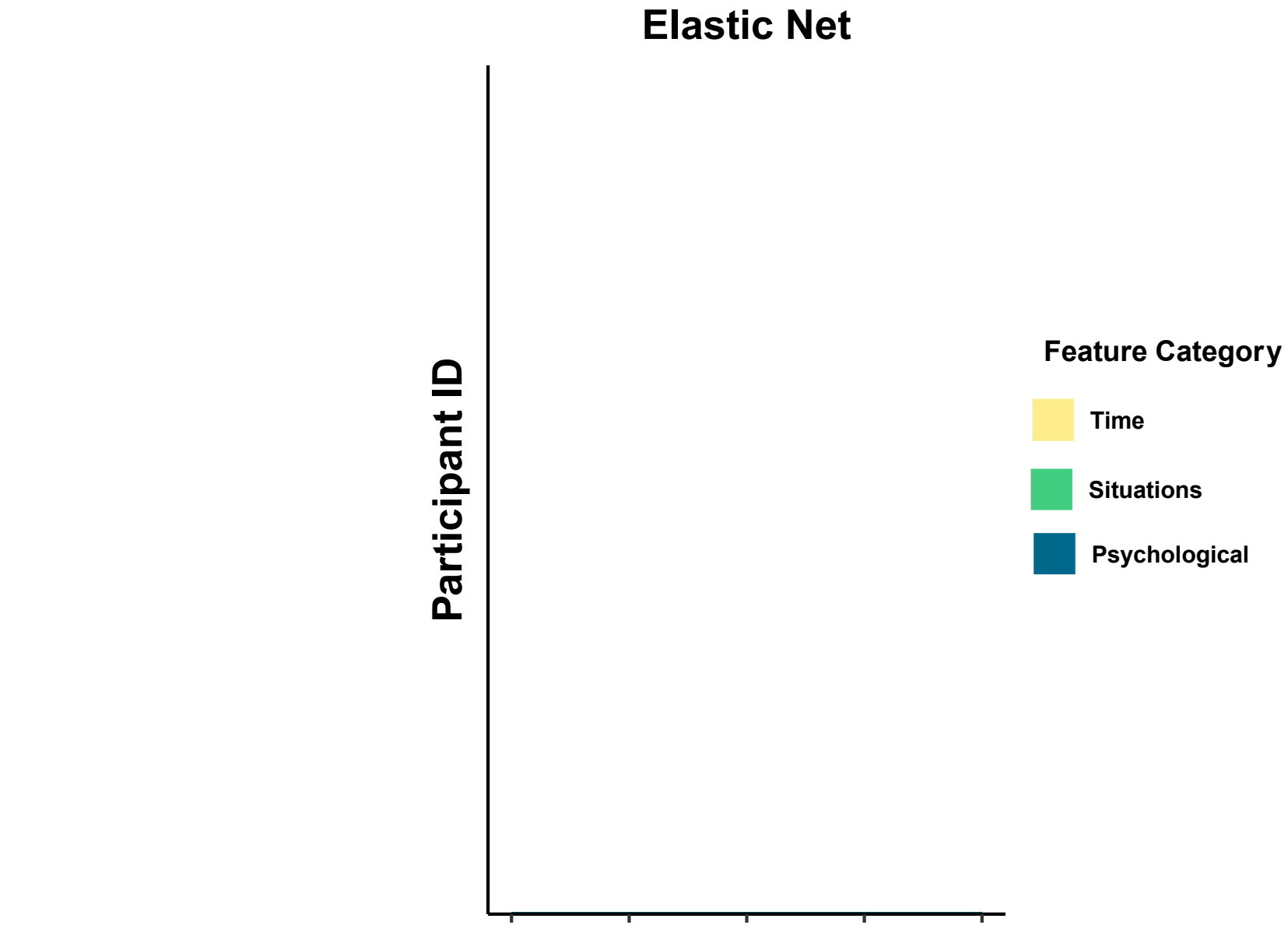
Tired

Beck & Jackson (2023; *Psychological Science*)

Do certain categories of features out-predict others?

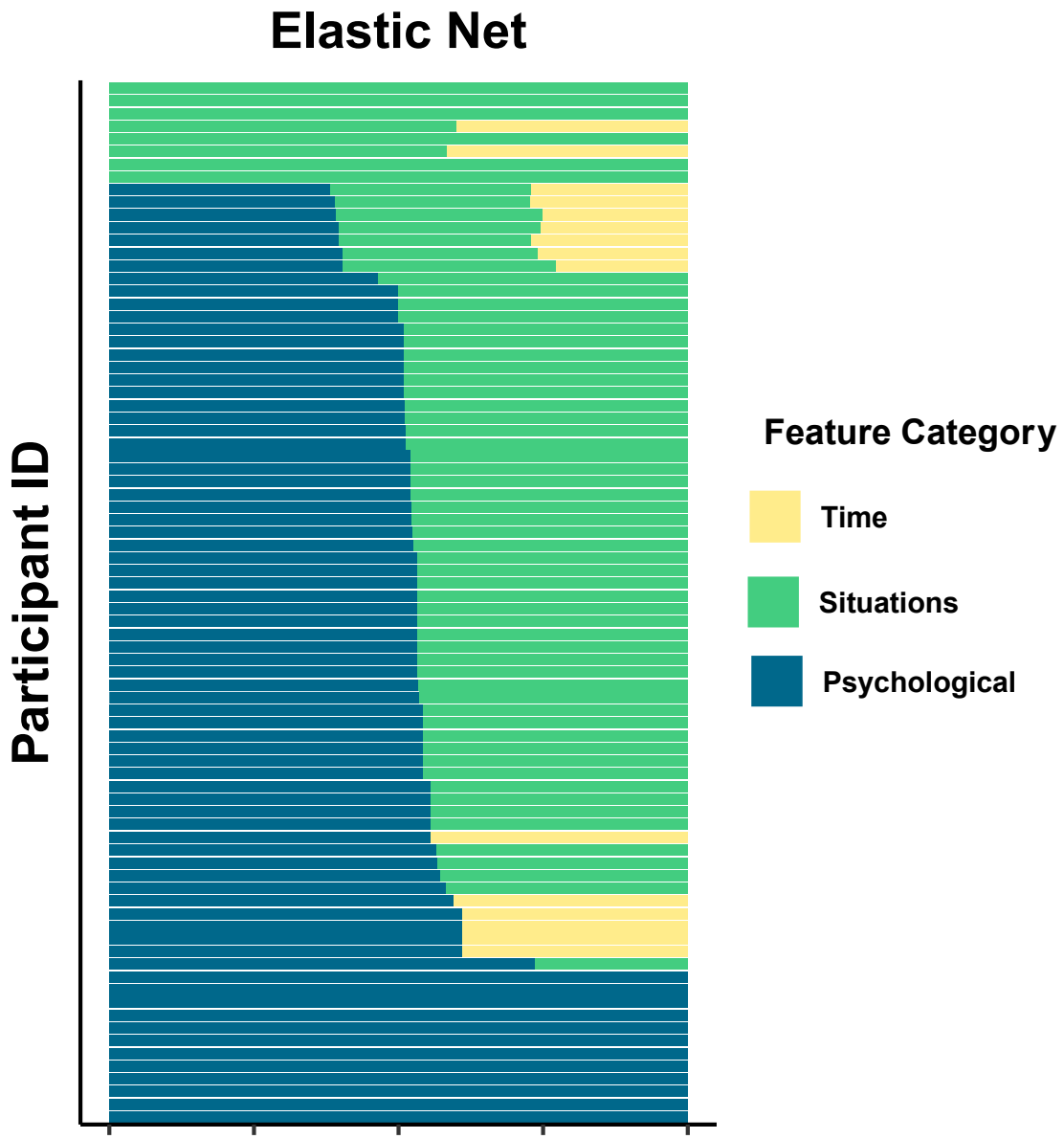
Beck & Jackson (2023; *Psychological Science*)

Do certain categories of features out-predict others?



Beck & Jackson (2023; *Psychological Science*)

Do certain categories of features out-predict others?



Beck & Jackson (2023; *Psychological Science*)

Do certain categories of features out-predict others?

Elastic Net

The relative contribution of person, situation, and timing features varies across people.

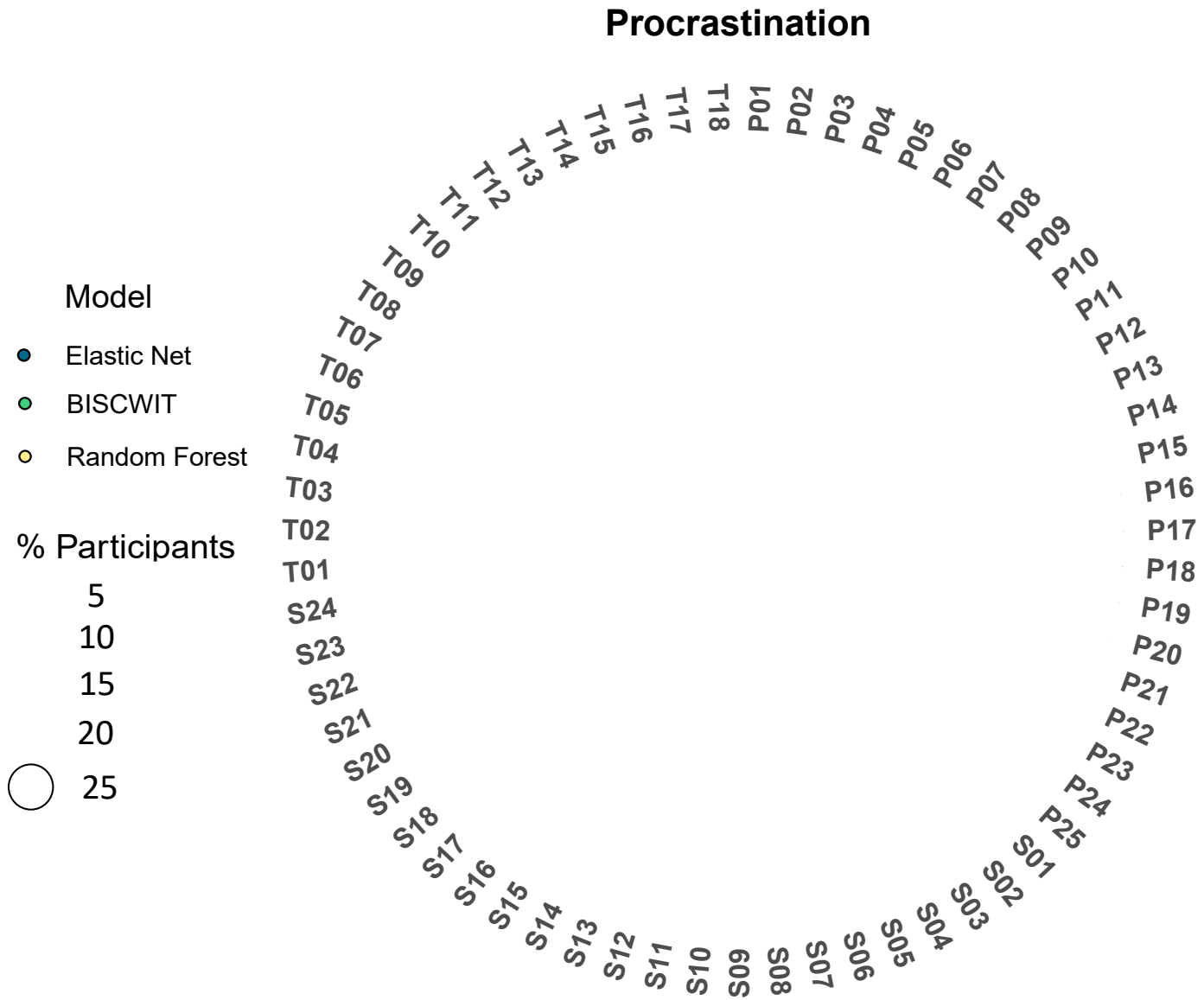


Beck & Jackson (2023; *Psychological Science*)

Which features play the strongest roles?

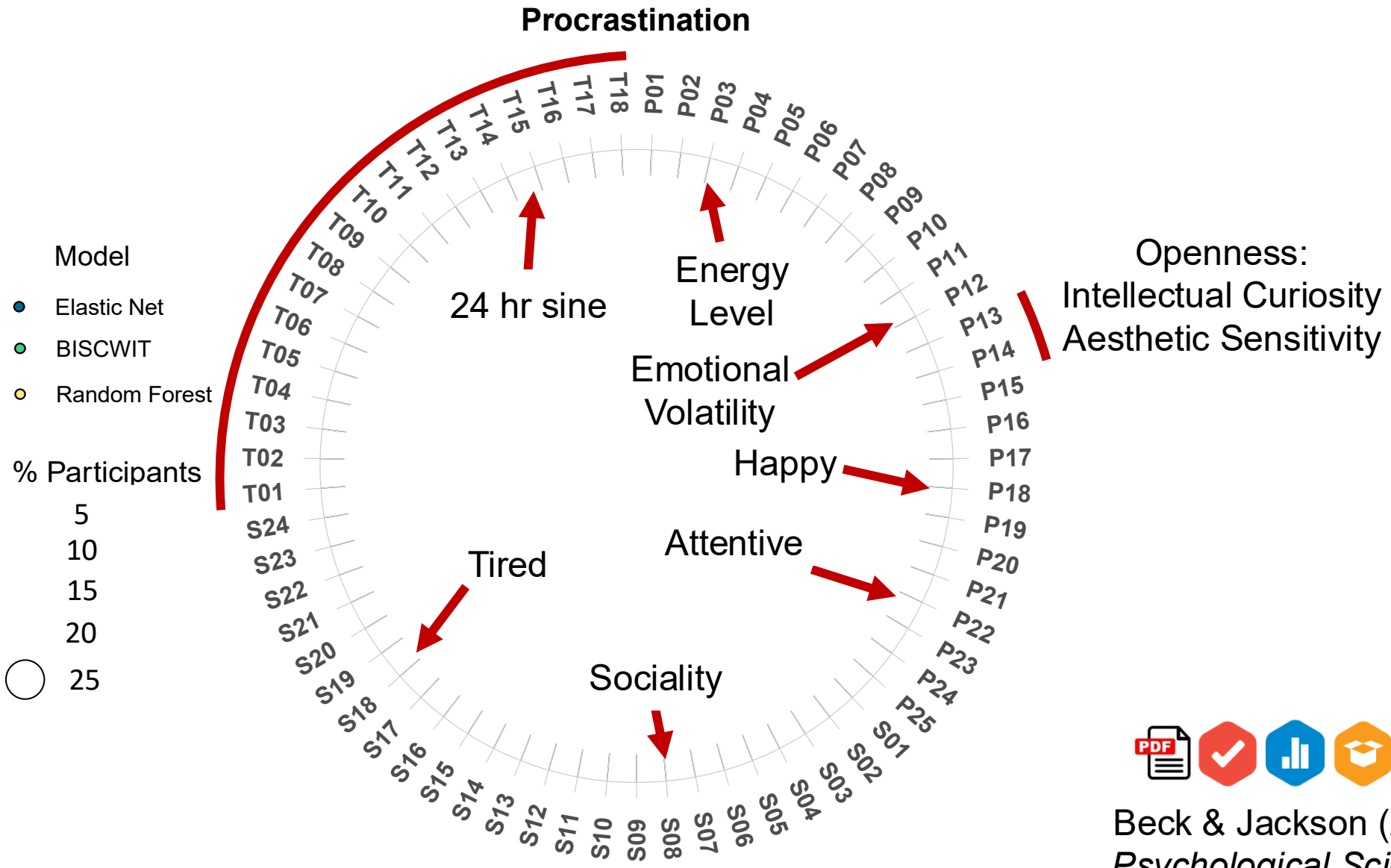
Beck & Jackson (2023; *Psychological Science*)

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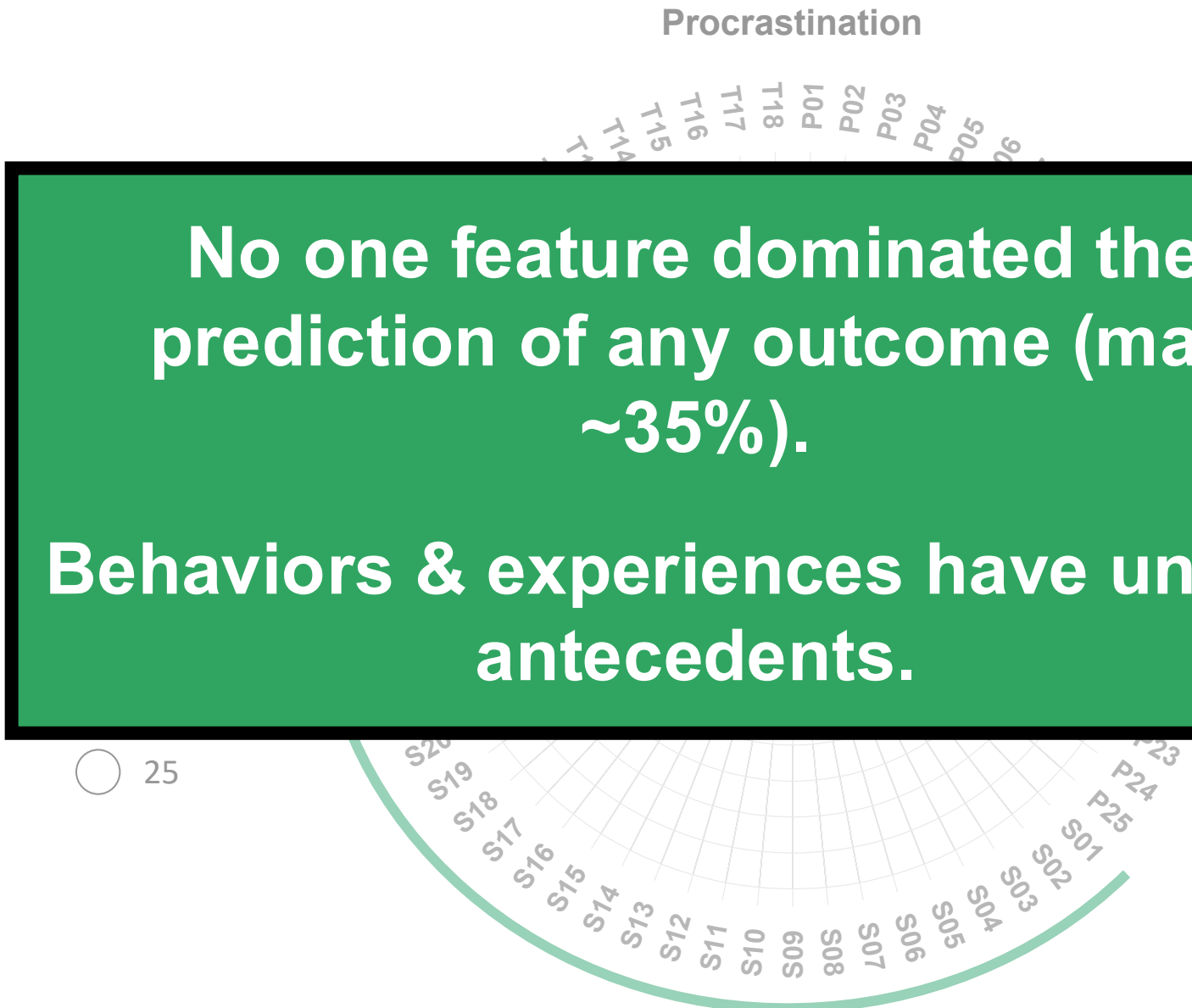
Beck & Jackson (2023; *Psychological Science*)

Which features play the strongest roles?



Beck & Jackson (2023; *Psychological Science*)

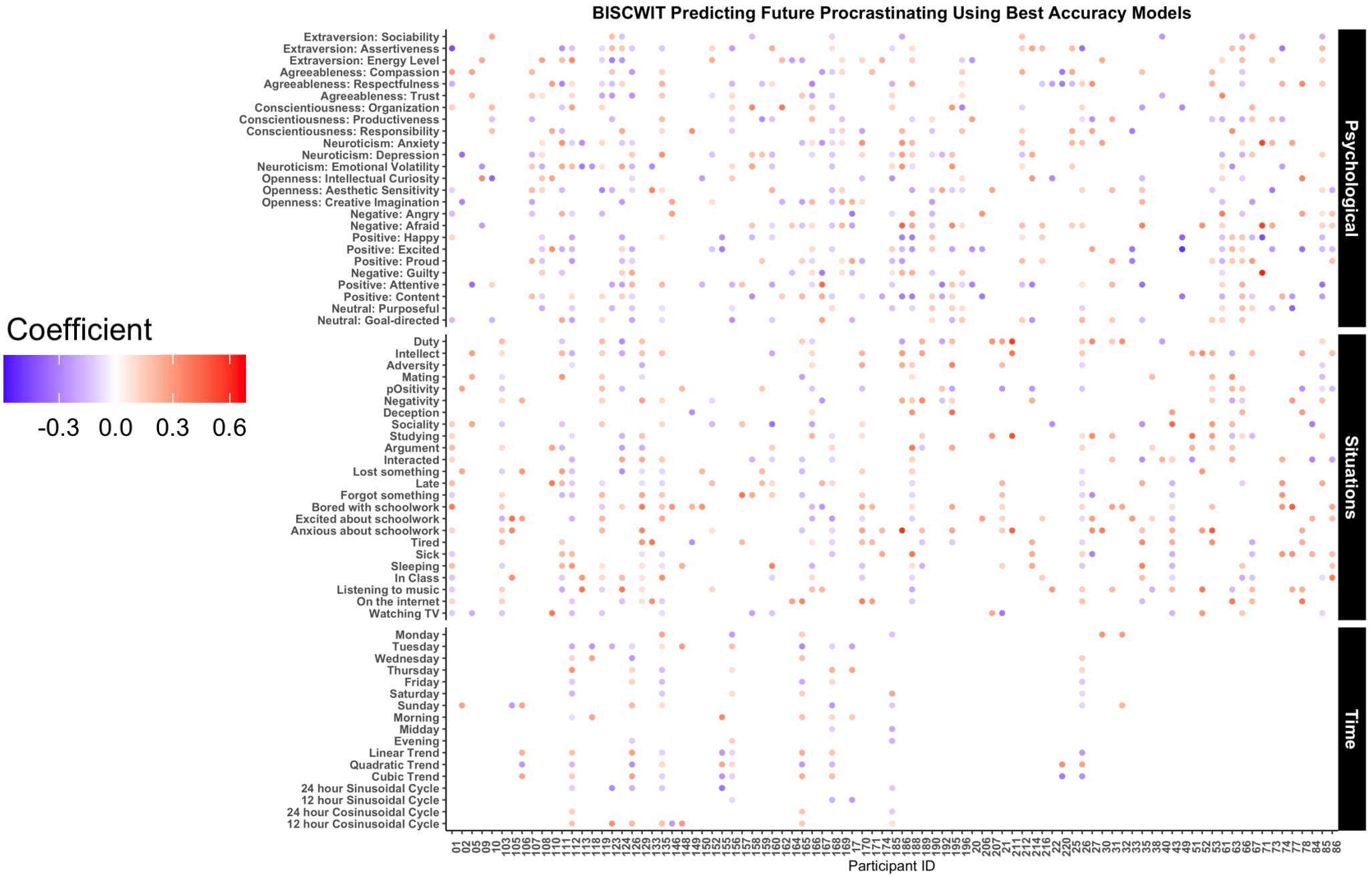
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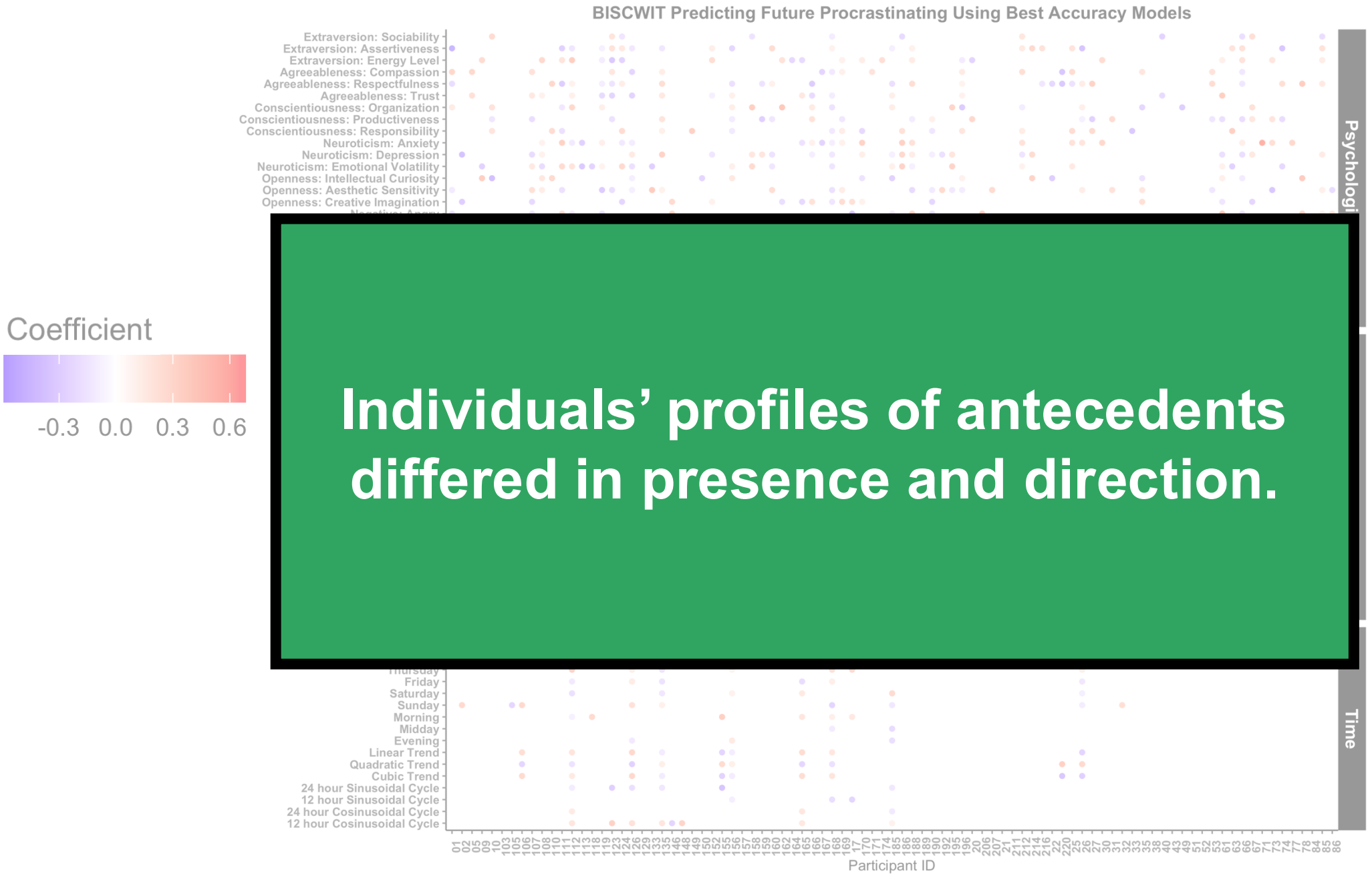


No one feature dominated the prediction of any outcome (max ~35%).

Behaviors & experiences have unique antecedents.

Beck & Jackson (2023; *Psychological Science*)

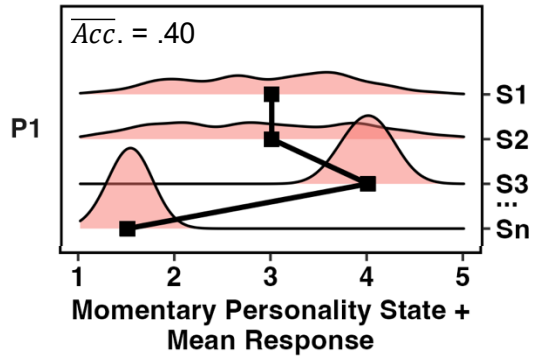
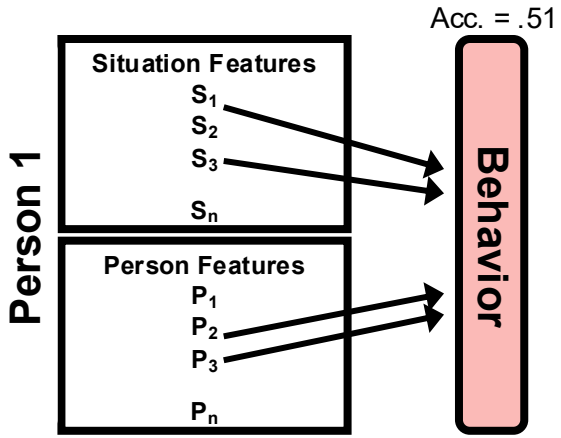
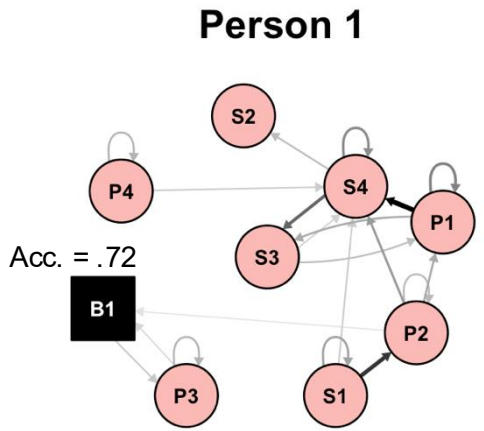

BISCWIT Predicting Future Procrastinating Using Best Accuracy Models
Coefficient
-0.3 0.0 0.3 0.6
Extraversion: Sociability
Extraversion: Assertiveness
Extraversion: Energy Level
Agreeableness: Compassion
Agreeableness: Respectfulness
Agreeableness: Trust
Conscientiousness: Organization
Conscientiousness: Productiveness
Conscientiousness: Responsibility
Neuroticism: Anxiety
Neuroticism: Depression
Neuroticism: Emotional Volatility
Openness: Intellectual Curiosity
Openness: Aesthetic Sensitivity
Openness: Creative Imagination
Negative: Angry
Negative: Afraid
Positive: Happy
Positive: Excited
Positive: Proud
Negative: Guilty
Positive: Attentive
Positive: Content
Neutral: Purposeful
Neutral: Goal-directed
Duty
Intellect
Adversity
Mating
pOsitivity
Negativity
Deception
Sociality
Studying
Argument
Interacted
Lost something
Late
Forgot something
Bored with schoolwork
Excited about schoolwork
Anxious about schoolwork
Tired
Sick
Sleeping
In Class
Listening to music
On the internet
Watching TV
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Morning
Midday
Evening
Linear Trend
Quadratic Trend
Cubic Trend
24 hour Sinusoidal Cycle
12 hour Sinusoidal Cycle
24 hour Cosinusoidal Cycle
12 hour Cosinusoidal Cycle
Psychological
Situations
Time
Participant ID


BISCWIT Predicting Future Procrastinating Using Best Accuracy Models
Extraversion: Sociability
Extraversion: Assertiveness
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Agreeableness: Compassion
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Conscientiousness: Responsibility
Neuroticism: Anxiety
Neuroticism: Depression
Neuroticism: Emotional Volatility
Openness: Intellectual Curiosity
Openness: Aesthetic Sensitivity
Openness: Creative Imagination
Psychologi
Coefficient
-0.3 0.0 0.3 0.6
Individuals' profiles of antecedents differed in presence and direction.
Friday
Saturday
Sunday
Morning
Midday
Evening
Linear Trend
Quadratic Trend
Cubic Trend
24 hour Sinusoidal Cycle
12 hour Sinusoidal Cycle
24 hour Cosinusoidal Cycle
12 hour Cosinusoidal Cycle
Time
Participant ID

AIM 3: Which assessment + conceptual framework is the most accurate model of behavior predictio

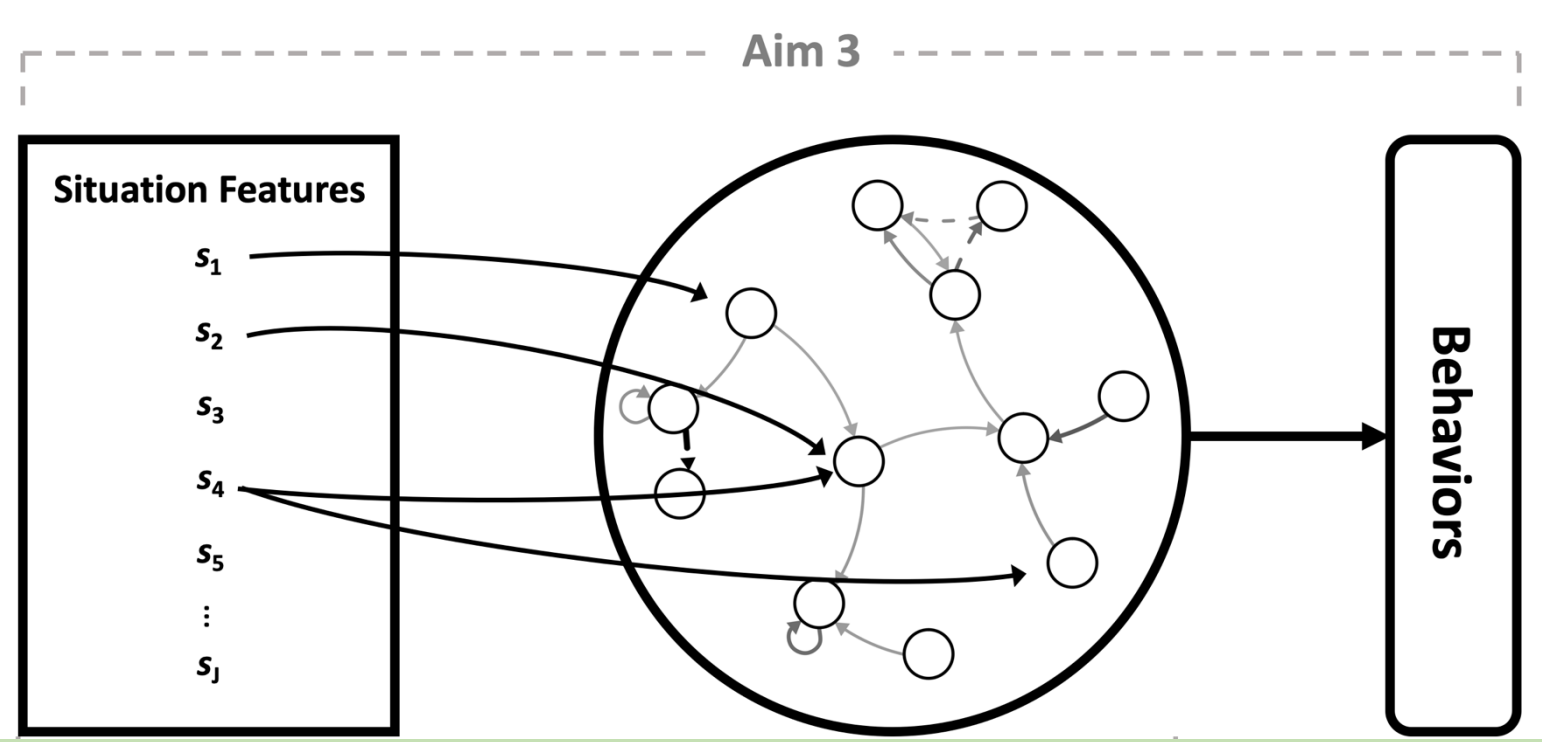
Unique Behavior & Experiences Study

UBE

	AIM 2A	AIM 2B	AIM 2C
Conceptual Frameworks	*Conditional Frameworks* (e.g., Wright & Mischel, 1987) 	*Person x Situation Interaction Frameworks* (e.g., Funder, 2006) 	*Systems Frameworks* (Mischel & Shoda, 1995) 
Traditional Model	**Predicting *mean levels*** Mixed Effects Models with Situations Predicting Behavior (e.g., Sherman et al., 2015)	***Univariate* PxS** Mixed Effects Models w/ Trait x Situation Interactions Predicting Behavior (e.g., Sherman et al., 2015)	***Static* Dynamics** Vector Autoregressive Models (VAR) (e.g., Beck & Jackson, 2020a)
Model Innovation	**Predicting *variability*** Bayesian Univariate Generalized Mixed Effects Location Scale Models (e.g., Williams et al., 2021)	***Multivariate* PxS** 2 Machine Learning Classification Models: (1) Elastic Net Regression and (2) BISCWIT (e.g., Beck & Jackson, 2022)	***Time-Varying* Dynamics** Time-Varying Mixed Vector Autoregressive Models (mVAR) (e.g., Haselbeck et al., 2021)

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Aim 1: Individuals show unique personality structures that are relatively consistent across time and events.	**Beck & Jackson** (2020, *JPSP*; 2021b; *EJP*)
Aim 2: The dynamics of well-being unfold differently across people.	**Beck et al.** (revision submitted, *NHB*)
Aim 3: Predicting behavior using machine learning.	**Beck & Jackson** (2023, *Psych Science*)



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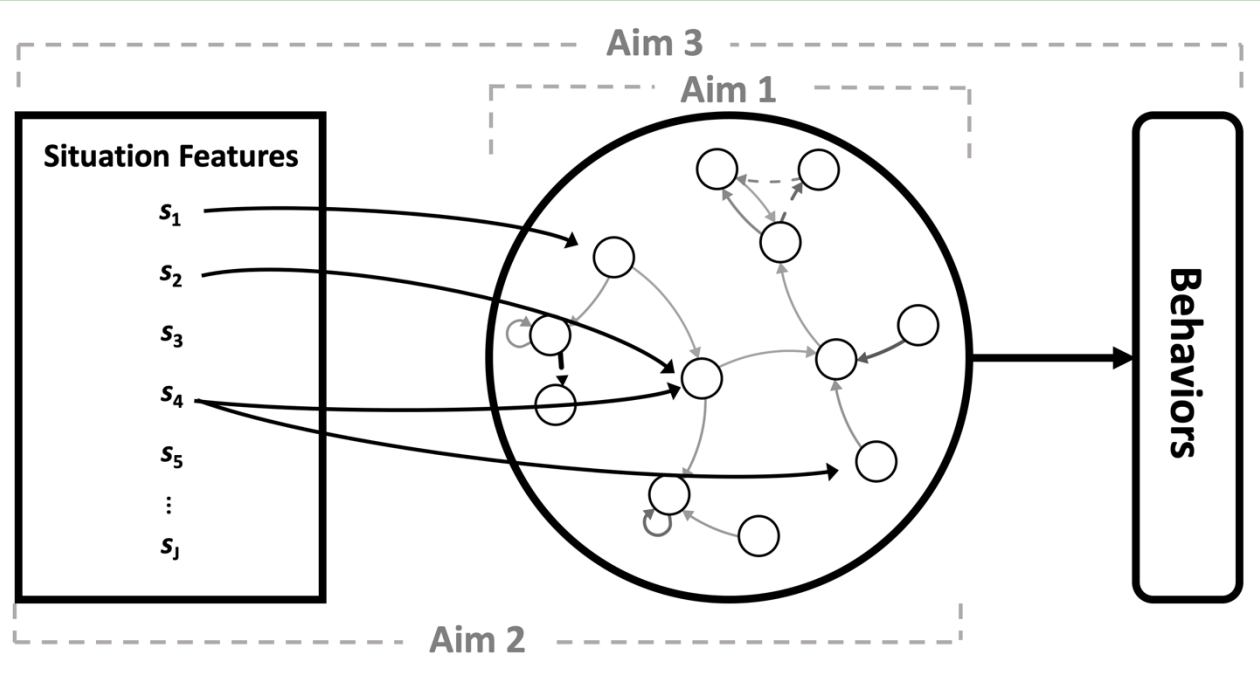
The antecedents of behaviors and experiences:

- Can be used to accurately predict those behaviors and experiences
- Differ across people
- Vary in the degree of psychological, situational, and timing antecedents

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Ongoing Directions



Other Ongoing Directions

From Personalized Prediction to Personalized Intervention

Personality as a Dynamic System

Personalized Assessment

Bottom-up Integration

Social-Cognitive Dynamics in Aging

Person-Specific Systems as a Scientific Hub

Personalized Assessment

What are we missing when we use the same measures for everyone?

We have no idea. That’s an empirical question!

What are we missing when we use the same measures for everyone?

Content	Variability	Predictive Utility
Hypothesis: People will generate content that doesn't overlap with the Big Five or other typical shared indicators we measure.	**Hypothesis**: People will show more variability and use the full scale more for unique items than shared items, on average.	**Hypothesis:** Unique items will improve predictive utility and play important roles in personalized prediction models.

Unique Behaviors & Experience Study

Unique Behavior & Experiences Study UBE

Study Design

UC DAVIS — N = 300

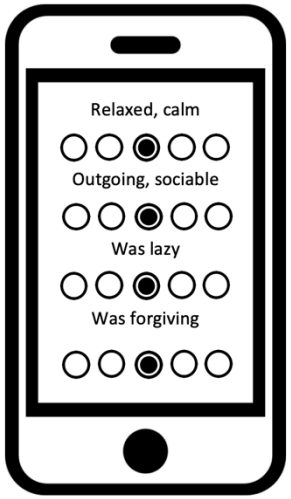
Washington University in St. Louis — N = 300

PREREGISTERED | OPEN DATA | OPEN MATERIALS



Baseline Surveys

- Big Five Personality
- Cardinal Traits
- Demographics
- **Unique Item Generation**
- …
- etc.



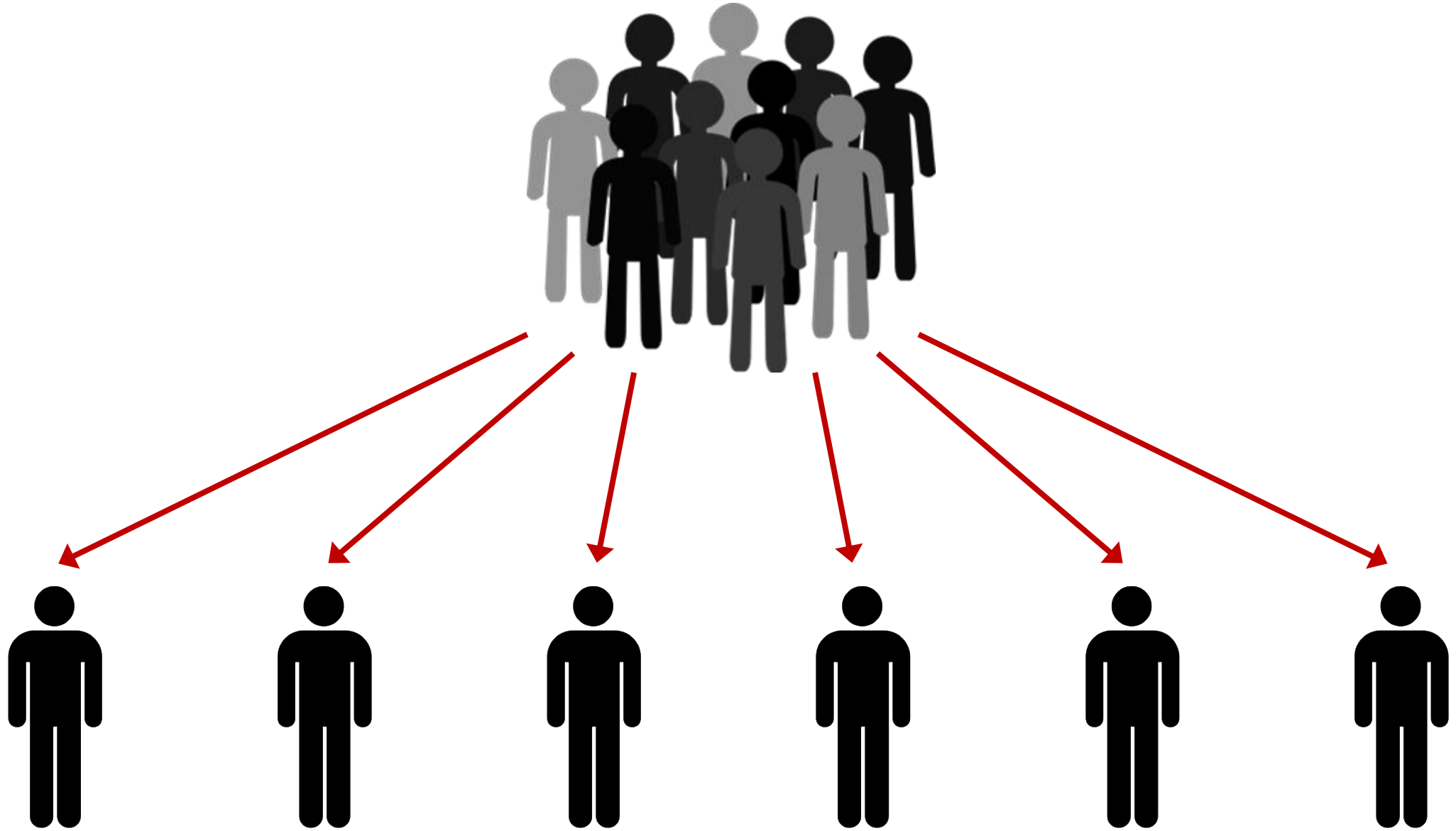
Experience Sampling Method (ESM)

5 x / day for 3 weeks (max n = 105)

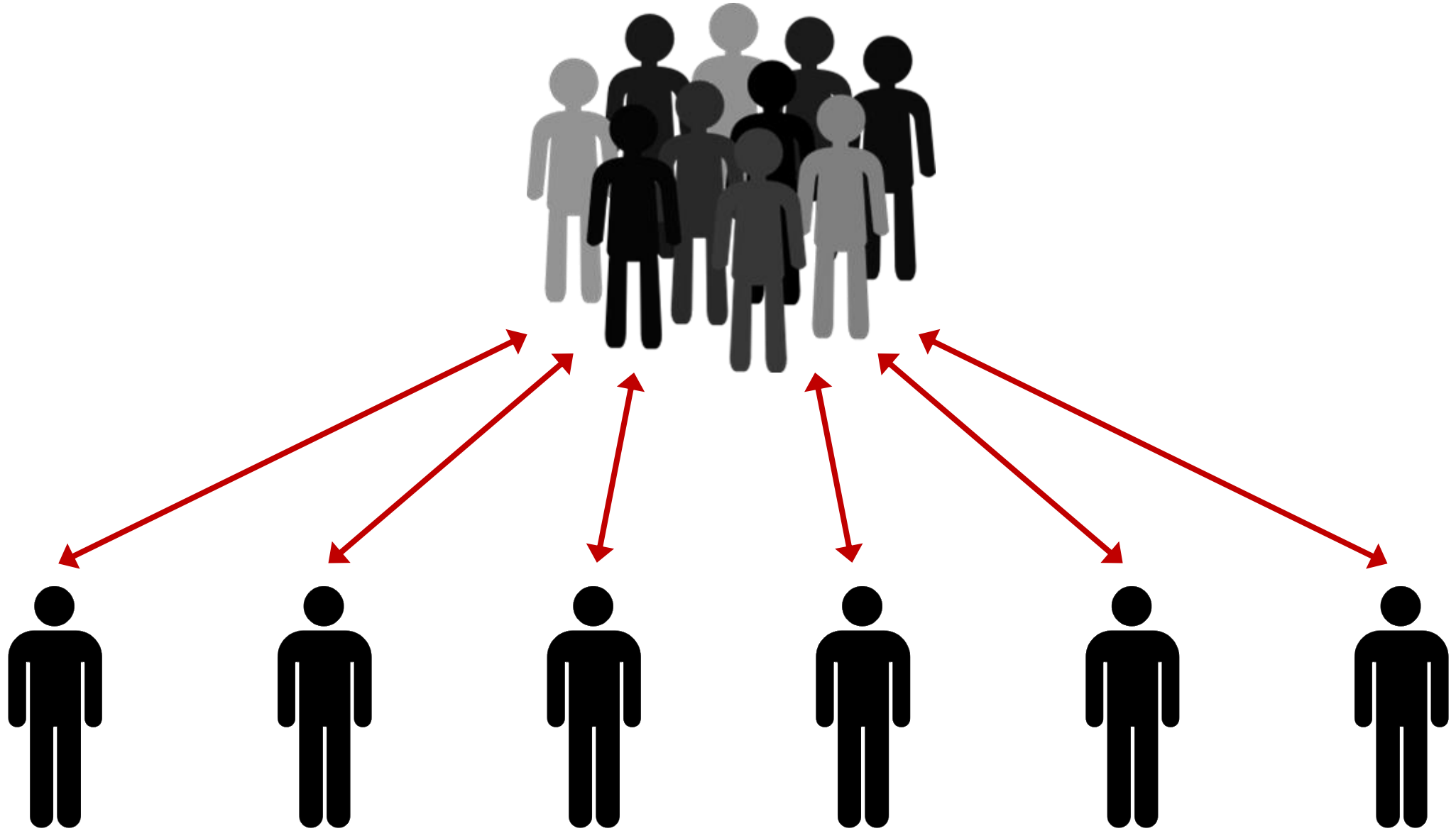
- Big Five Personality States
- Unique, participant generated "Cardinal States"
- DIAMONDS Situation Characteristics
- Binary Behavior Indicators
- Passive Sensing

NSF **BCS-2336406**

Top-Down Psychological Science



Bottom-Up Psychological Science



Roadmap to Bottom-Up Integration

From person-specific dynamics to population-level structure

"Instead of growing impatient with the single case and hastening on to generalization, why should we not grow impatient with our generalizations and hasten to the internal pattern?"

(Allport, 1961, p. 84)



Roadmap to Bottom-Up Integration

From person-specific dynamics to population-level structure

High integration

High specificity

5  **Translate these insights into scalable population-level assessment** — POPULATION TRANSLATION

4  **Validate system-level regularities, stability, and consequences** — VALIDATION

3  **Identify structured regularities ("lumps in the batter")** — EMERGENCE ZONE

2  **Extract system features for comparison** — SYSTEM FEATURES

1  **Define and model person-specific (lowest-level) dynamics** — PERSON-SPECIFIC MODELING

What is personality?

"Personality is the **dynamic organization** within the individual of those **psychophysical systems** that determine his **unique adjustments to the environment**."

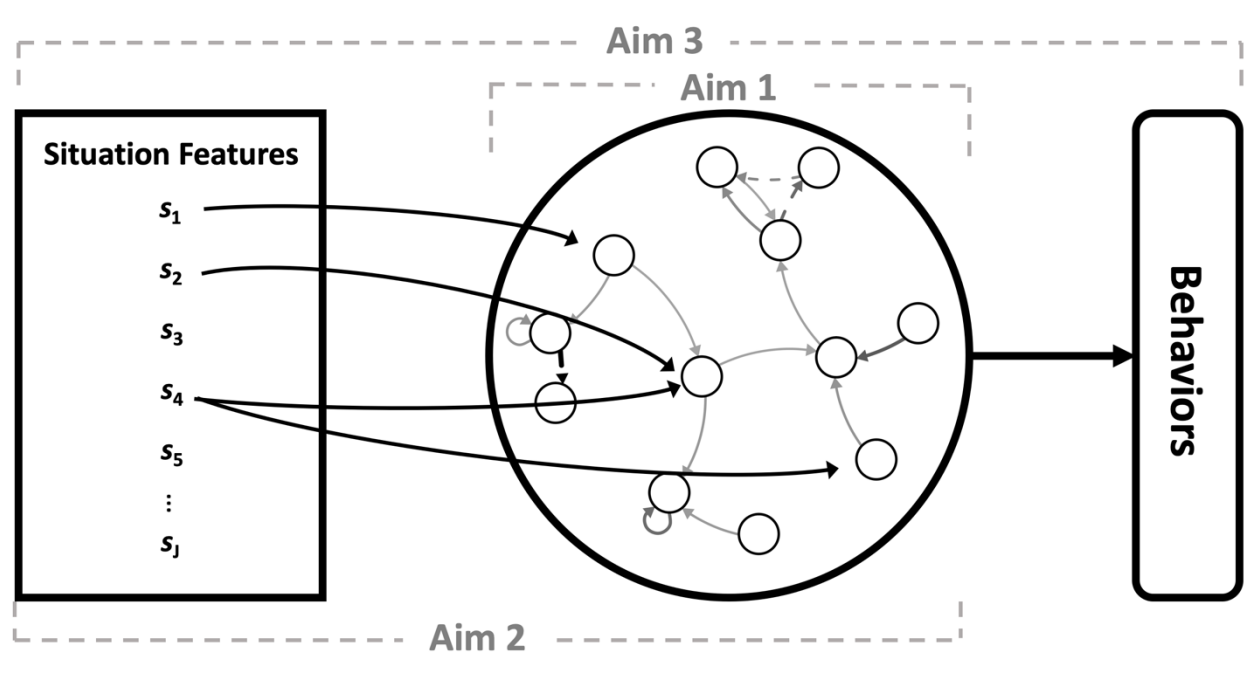
(Allport, 1937, p. 32)



e.g., Allport, 1937, 1961, 1968

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Ongoing Directions



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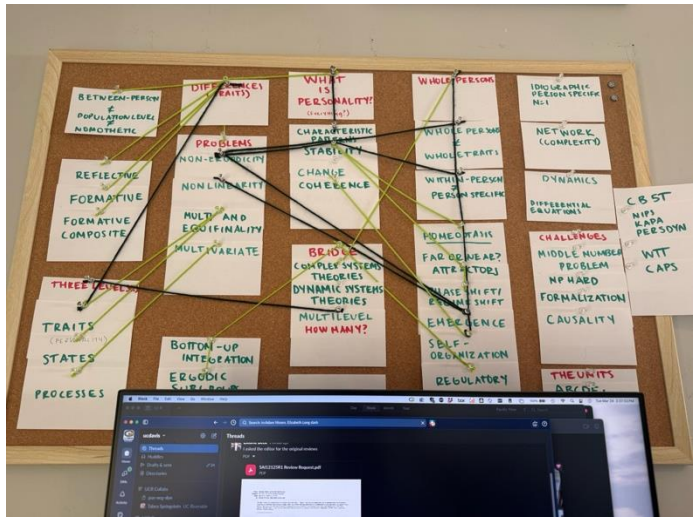
NIH National Institute on Aging

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